



Kandinsky - Abstract Interpretation, 1925

What Could Possibly Go Right?

Analisi Statica e Verifica del Software
Università di Parma

Caterina Urban

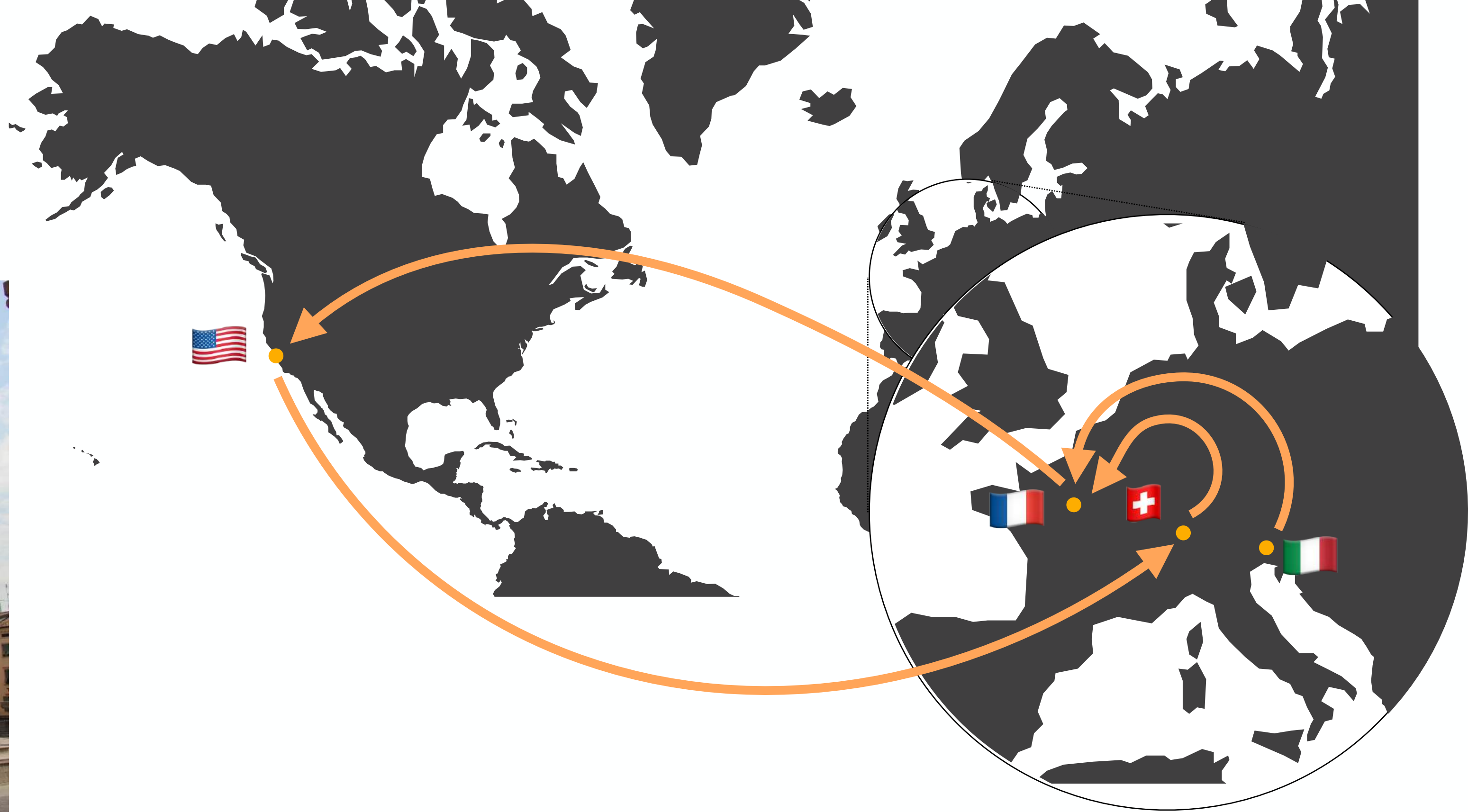
Inria & École Normale Supérieure | Université PSL

Who am I?



1987	Udine, Italie
2006 - 2011	Università degli Studi di Udine
2011 - 2015	École Normale Supérieure
2015	NASA & Carnegie Mellon University
2015 - 2019	ETH Zurich
Since 2019	Inria

BSc, MSc
PhD
Internship
Postdoc
Research Scientist



Lecture(s) Plan

The Art of Losing Precision

No Surprises, Please

**What Could
Possibly Go Right?**

It's Complicated

Principles of Abstract Interpretation

The Art of Losing Precision

No Surprises, Please

**What Could
Possibly Go Right?**

It's Complicated

Static Analysis by Abstract Interpretation



Language Syntax

Numeric Expressions

$expr ::=$	X	(variable, $X \in \mathbb{X}$)
	c	(constant, $c \in \mathbb{Z}$)
	$[c_1, c_2]$	(non-deterministic input, $c_1, c_2 \in \mathbb{Z} \cup \{-\infty, +\infty\}$)
	$-expr$	(negation)
	$expr \diamond expr$	(binary operation, $\diamond \in \{+, -, \dots\}$)

Statements

$stmt ::=$	${}^\ell X \leftarrow expr^\ell$	(assignment, $\ell \in \mathcal{L}$)
	if ${}^\ell expr \bowtie 0$ then $stmt$ end ${}^\ell$	(conditional, $\bowtie \in \{=, \leq, \dots\}$)
	while ${}^\ell expr \bowtie 0$ do $stmt$ done ${}^\ell$	(loop)
	$stmt; stmt$	(sequence)

Example

```
1
a ← [0, +∞]
2
b ← [0, +∞]
3
q ← 0
4
r ← a
5
while 6(r ≥ b) do
7
  r ← r − b
8
  q ← q + 1
9
done
10
```

Static Analysis by Abstract Interpretation

3-Step Recipe

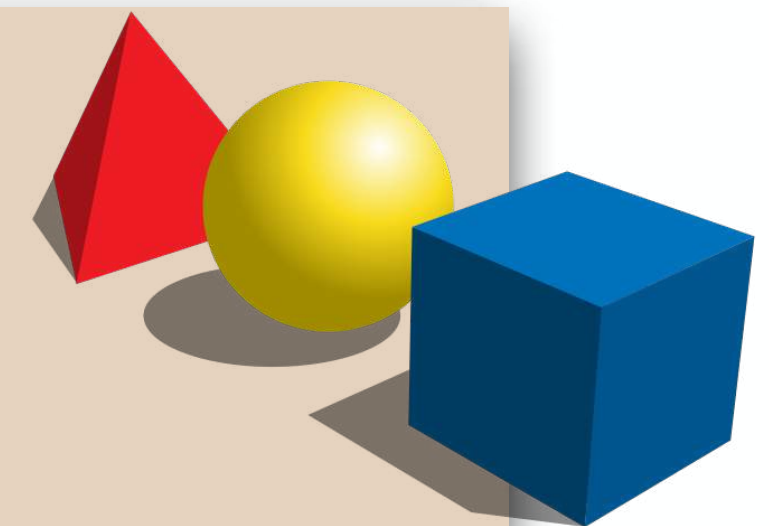
practical tools

targeting specific programs



abstract semantics, abstract domains

algorithmic approaches to decide program properties



concrete semantics

mathematical models of the program behavior



Static Analysis by Abstract Interpretation

Program Semantics

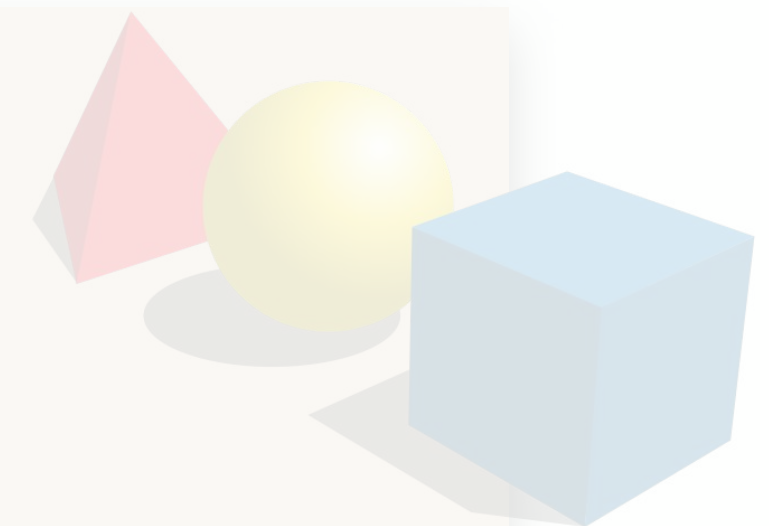
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Static Analysis by Abstract Interpretation



Expression Semantics

- $E[\![expr]\!]: \underbrace{(\mathbb{X} \rightarrow \mathbb{Z})}_{\rho \in \mathcal{E} \text{ memory state}} \rightarrow \mathcal{P}(\mathbb{Z})$

$$E[\![X]\!]\rho \stackrel{\text{def}}{=} \{\rho(X)\}$$

$$E[\![c]\!]\rho \stackrel{\text{def}}{=} \{c\}$$

$$E[\![c_1, c_2]\!]\rho \stackrel{\text{def}}{=} \{x \in \mathbb{Z} \mid c_1 \leq x \leq c_2\}$$

$$E[\![-e]\!]\rho \stackrel{\text{def}}{=} \{-x \mid x \in E[\![e]\!]\rho\}$$

$$E[\![e_1 + e_2]\!]\rho \stackrel{\text{def}}{=} \{x_1 + x_2 \mid x_1 \in E[\![e]\!]\rho, x_2 \in E[\![e]\!]\rho\}$$

$$E[\![e_1 - e_2]\!]\rho \stackrel{\text{def}}{=} \{x_1 - x_2 \mid x_1 \in E[\![e]\!]\rho, x_2 \in E[\![e]\!]\rho\}$$

$$\vdots$$

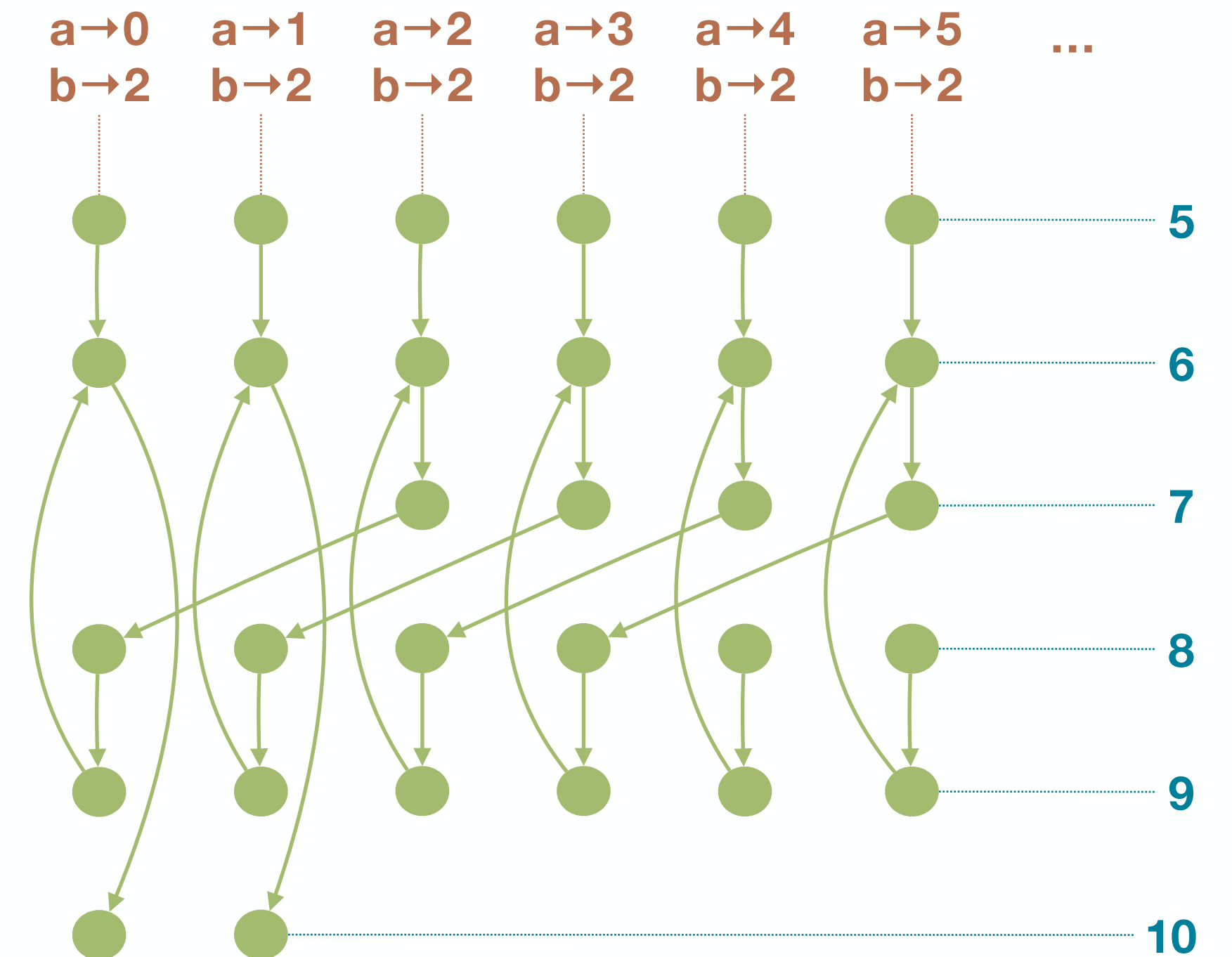
$$\begin{array}{l} expr ::= X \\ \quad | c \\ \quad | [c_1, c_2] \\ \quad | -expr \\ \quad | expr \diamond expr \end{array}$$

Transition Semantics

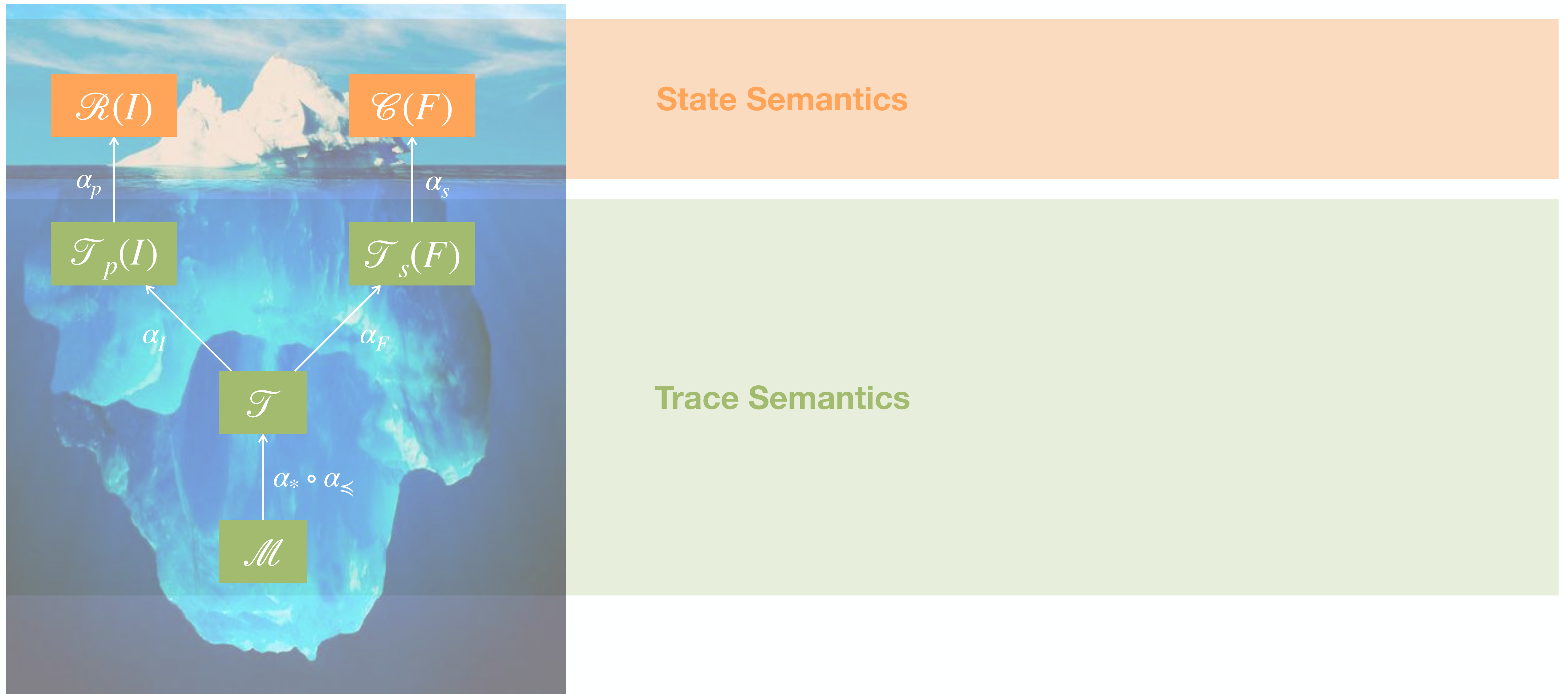
Program Executions as Discrete Transitions between States

- states: $\Sigma \stackrel{\text{def}}{=} \mathcal{L} \times (\mathbb{X} \rightarrow \mathbb{Z})$
- transition relation: $\tau \subseteq \Sigma \times \Sigma$

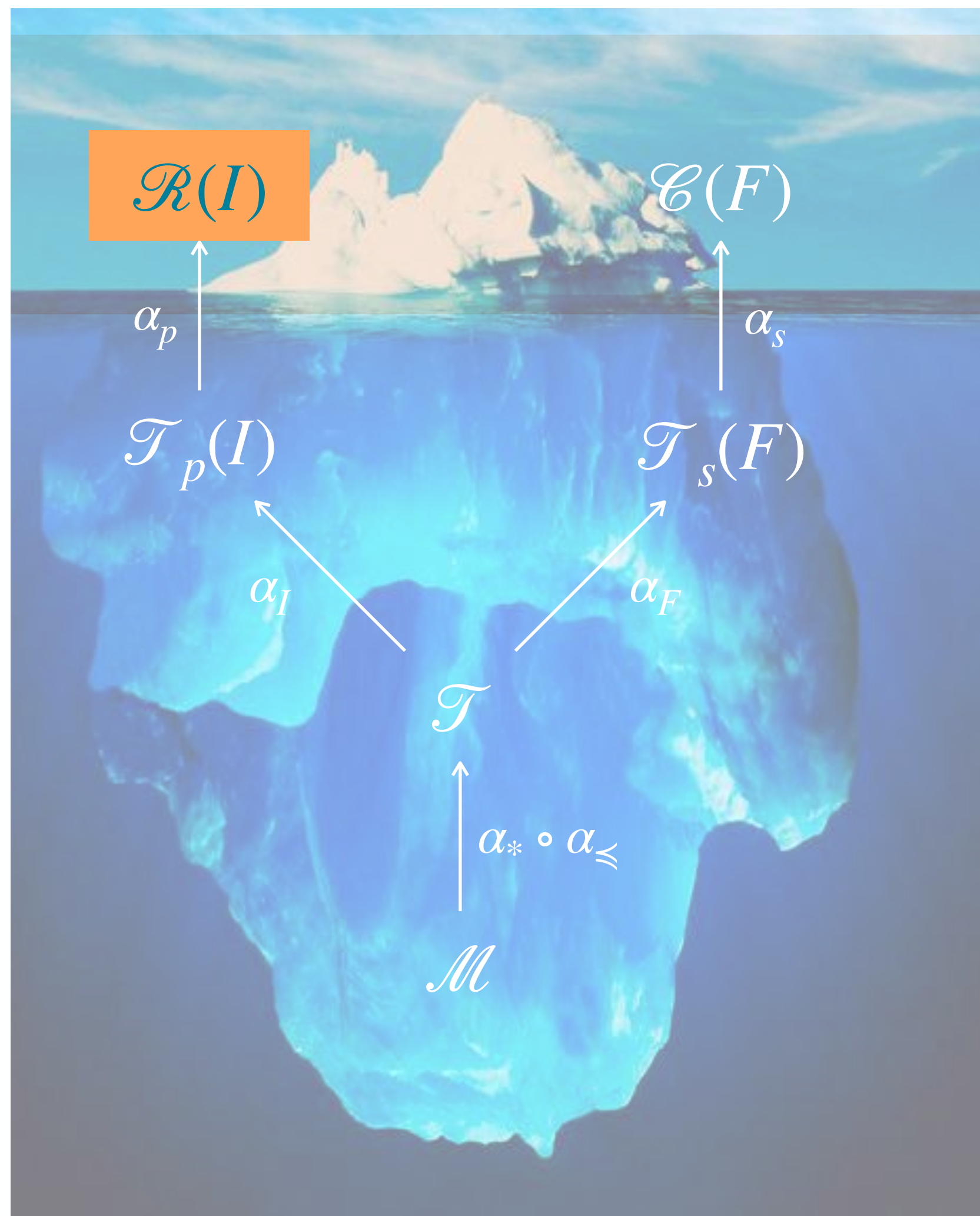
```
3  
q ← 0  
4  
r ← a  
5  
while 6(r ≥ b) do  
7  
  r ← r - b  
8  
  q ← q + 1  
9  
od  
10
```



Hierarchy of Semantics



Forward Reachability Semantics



State Semantics

Forward Reachability Semantics

Program States Reachable From $I \in \mathcal{P}(\Sigma)$

- $\mathcal{R}(I) \in \mathcal{P}(\Sigma)$

computational order

$$\mathcal{R}(I) = \text{lfp}_{\subseteq}^{\emptyset} F_r$$

$$F_r(S) \stackrel{\text{def}}{=} I \cup \text{post}(S)$$

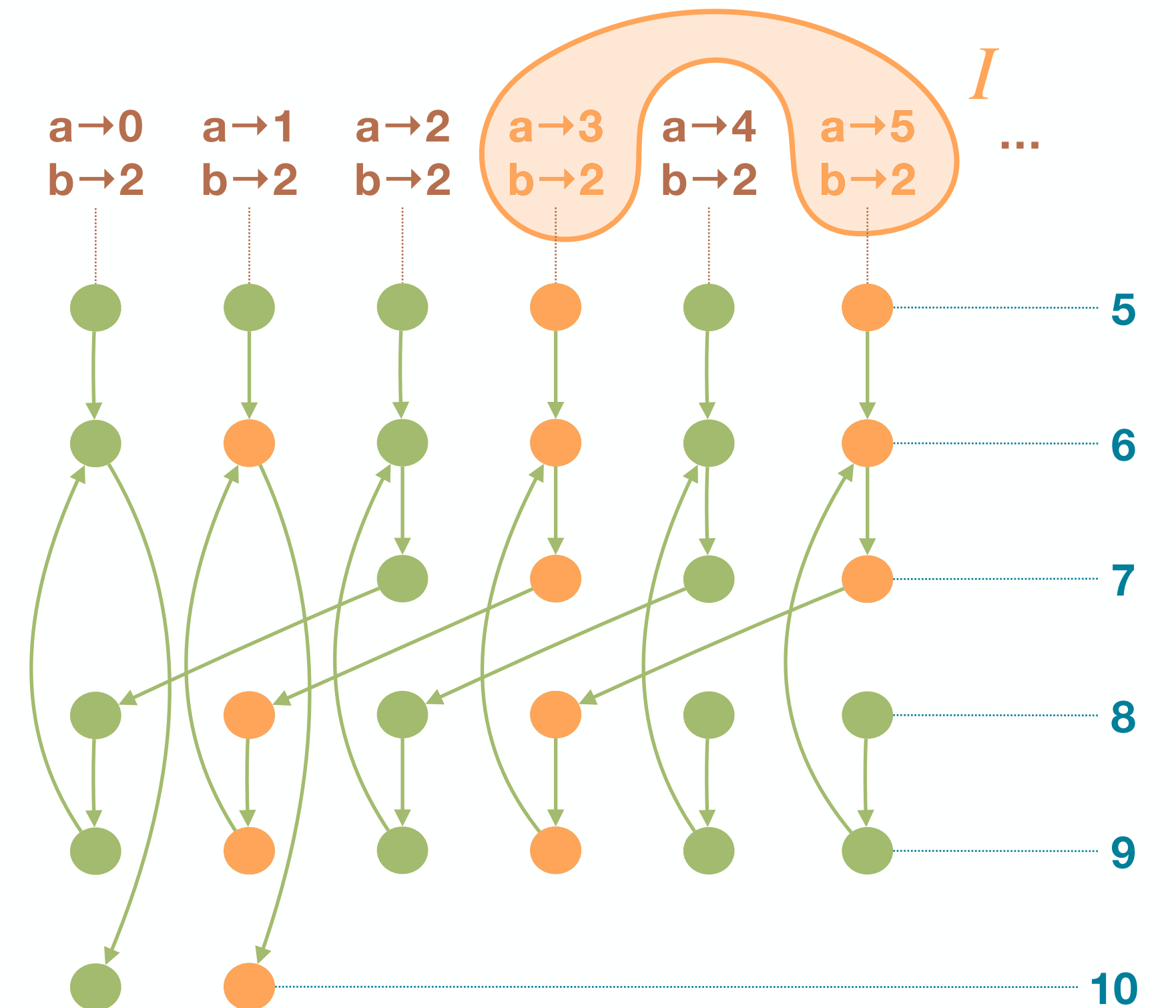
Definition

Given a transition system $\langle \Sigma, \tau \rangle$, the **image function** $\text{post}: \mathcal{P}(\Sigma) \rightarrow \mathcal{P}(\Sigma)$ maps a set of program states $X \in \mathcal{P}(\Sigma)$ to the set of their successors with respect to the transition relation τ :

$$\text{post}(X) \stackrel{\text{def}}{=} \{s' \in \Sigma \mid \exists s \in X: \langle s, s' \rangle \in \tau\}$$

```

3
q ← 0
4
r ← a
5
while 6(r ≥ b) do
7
  r ← r - b
8
  q ← q + 1
9
od
10
    
```



Forward Reachability

Denotational Formulation

$stmt ::= {}^{\ell}X \leftarrow expr^{\ell}$
| $\text{if } {}^{\ell}expr \bowtie 0 \text{ then } stmt \text{ end}^{\ell}$
| $\text{while } {}^{\ell}expr \bowtie 0 \text{ do } stmt \text{ done}^{\ell}$
| $stmt; stmt$

$$\mathcal{R}[\![{}^{\ell_1}X \leftarrow e^{\ell_2}]\!]S \stackrel{\text{def}}{=} \{(\ell_2, \rho[X \mapsto v]) \mid (\ell_1, \rho) \in S, v \in E[e]\rho\}$$

$$\begin{aligned} \mathcal{R}[\![\text{if } {}^{\ell_1}e \bowtie 0 \text{ then } {}^{\ell_2}s^{\ell_3} \text{ end}^{\ell_4}]\!]S &\stackrel{\text{def}}{=} \\ &\{(\ell_4, \rho) \mid (\ell_3, \rho) \in \mathcal{R}[\![{}^{\ell_2}s^{\ell_3}]\!]\{(\ell_2, \rho) \mid (\ell_1, \rho) \in S, \exists v \in E[e]\rho: v \bowtie 0\}\} \cup \\ &\{(\ell_4, \rho) \mid (\ell_1, \rho) \in S, \exists v \in E[e]\rho: v \not\bowtie 0\} \end{aligned}$$

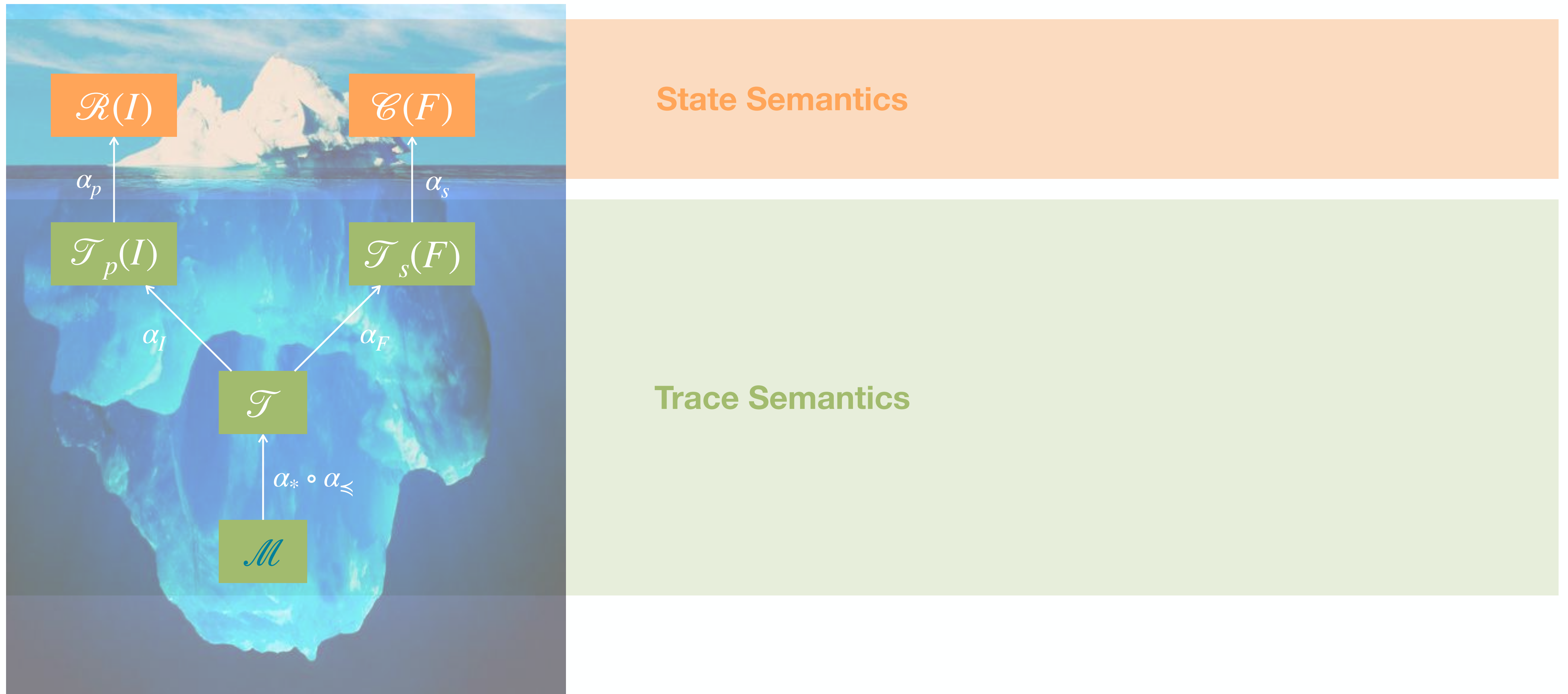
$$\begin{aligned} \mathcal{R}[\![\text{while } {}^{\ell_1}e \bowtie 0 \text{ then } {}^{\ell_2}s^{\ell_3} \text{ done}^{\ell_4}]\!]S &\stackrel{\text{def}}{=} \\ &\{(\ell_4, \rho) \mid (\ell_1, \rho) \in \text{lfp}_{\emptyset}^{\subseteq} F_r, \exists v \in E[e]\rho: v \not\bowtie 0\} \end{aligned}$$

where

$$F_r(Y) \stackrel{\text{def}}{=} S \cup \{(\ell_1, \rho) \mid (\ell_3, \rho) \in \mathcal{R}[\![{}^{\ell_2}s^{\ell_3}]\!]\{(\ell_2, \rho) \mid (\ell_1, \rho) \in Y, \exists v \in E[e]\rho: v \not\bowtie 0\}\}$$

$$\mathcal{R}[\![s_1; s_2]\!]S \stackrel{\text{def}}{=} \mathcal{R}[\![s_2]\!](\mathcal{R}[\![s_1]\!]S)$$

Maximal Trace Semantics



Program State Sequences

- ϵ **empty** sequence
- $s_0, \dots, s_{n-1}$ sequence of **length n**
- Σ^n set of **sequences of length n**
- $\Sigma^* \stackrel{\text{def}}{=} \bigcup_{i \in \mathbb{N}} \Sigma^i$ set of **all finite sequences**
- Σ^ω set of **all infinite sequences**
- $\Sigma^\infty \stackrel{\text{def}}{=} \Sigma^* \cup \Sigma^\omega$ set of **all sequences**

Operations on Sequences

- **concatenation:** $(s_0, \dots, s_n) \cdot (s'_0, \dots, s'_n) \stackrel{\text{def}}{=} s_0, \dots, s_n s'_0, \dots, s'_n$
 $A \cdot B \stackrel{\text{def}}{=} \{a \cdot b \mid a \in A \wedge b \in B\}$
- **merging:** $(s_0, \dots, s) ; (s, s'_1, \dots, s'_n) \stackrel{\text{def}}{=} s_0, \dots, s s'_1, \dots, s'_n$
 $A ; B \stackrel{\text{def}}{=} \{a ; b \mid a \in A \wedge b \in B\}$

Maximal Trace Semantics

Finite and Infinite Program Traces

- $\mathcal{M} \in \mathcal{P}(\Sigma^\infty)$

$$\mathcal{M} \stackrel{\text{def}}{=} \{s_0, \dots, s_n \in \Sigma^* \mid s_n \in \mathcal{B} \wedge \forall i: \langle s_i, s_{i+1} \rangle \in \tau\} \cup \{s_0, \dots \in \Sigma^\omega \mid \forall i: \langle s_i, s_{i+1} \rangle \in \tau\}$$

$$\mathcal{B} \stackrel{\text{def}}{=} \{s \in \Sigma \mid \forall s' \in \Sigma: \langle s, s' \rangle \notin \tau\}$$

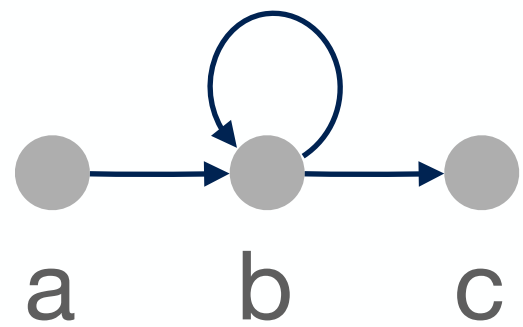
Maximal Trace Semantics

Least Fixpoint Formulation

computational order

$$\mathcal{M} = \text{lfp}_{\Sigma^\omega} F$$

$$F(T) \stackrel{\text{def}}{=} \mathcal{B} \cup \tau; T$$



- $F^0(\emptyset) = \Sigma^\omega$
- $F^1(F^0) = \{c\} \cup \{ab\Sigma^\omega, bb\Sigma^\omega, bc\Sigma^\omega\}$
- $F^2(F^1) = \{bc, c\} \cup \{abb\Sigma^\omega, bbb\Sigma^\omega, abc\Sigma^\omega, bbc\Sigma^\omega\}$
- $F_p^3(F_p^2) = \{abc, bbc, bc, c\} \cup \{abbb\Sigma^\omega, bbbb\Sigma^\omega, abbc\Sigma^\omega, bbbc\Sigma^\omega\}$

$$\mathcal{M} = \{ab^i c, b^i c, c \mid i \geq 1\} \cup \{ab^\omega, b^\omega\}$$

Maximal Trace Semantics

Denotational Formulation

```

stmt ::=  ${}^{\ell}X \leftarrow \text{expr}^{\ell}$ 
        |  $\text{if } {}^{\ell}\text{expr} \bowtie 0 \text{ then } \text{stmt} \text{ end}^{\ell}$ 
        |  $\text{while } {}^{\ell}\text{expr} \bowtie 0 \text{ do } \text{stmt} \text{ done}^{\ell}$ 
        |  $\text{stmt}; \text{stmt}$ 

```

$$\mathcal{M}[\![{}^{\ell_1}X \leftarrow e^{\ell_2}]\!]T \stackrel{\text{def}}{=} \{(\ell_1, \rho)(\ell_2, \rho[X \mapsto v])\sigma \mid \sigma \in \Sigma^{\infty}, (\ell_2, \rho[X \mapsto v])\sigma \in T, v \in E[e]\rho\}$$

$$\begin{aligned} \mathcal{M}[\![\text{if } {}^{\ell_1}e \bowtie 0 \text{ then } {}^{\ell_2}s^{\ell_3} \text{ end}^{\ell_4}]\!]T &\stackrel{\text{def}}{=} \\ &\{(\ell_1, \rho)(\ell_2, \rho)\sigma \mid \sigma \in \Sigma^{\infty}, (\ell_2, \rho)\sigma \in \mathcal{C}[\![{}^{\ell_2}s^{\ell_3}]\!]\{(\ell_3, \rho')\sigma' \mid \sigma' \in T\}, \exists v \in E[e]\rho: v \bowtie 0\} \cup \\ &\{(\ell_1, \rho)(\ell_4, \rho)\sigma \mid \sigma \in \Sigma^{\infty}, (\ell_4, \rho)\sigma \in T, \exists v \in E[e]\rho: v \not\bowtie 0\} \end{aligned}$$

$$\mathcal{M}[\![\text{while } {}^{\ell_1}e \bowtie 0 \text{ then } {}^{\ell_2}s^{\ell_3} \text{ done}^{\ell_4}]\!]T \stackrel{\text{def}}{=} \text{lfp}_{\Sigma^{\omega}}^{\sqsubseteq} F$$

where

$$\begin{aligned} F(Y) &\stackrel{\text{def}}{=} \{(\ell_1, \rho)(\ell_4, \rho)\sigma \mid \sigma \in \Sigma^{\infty}, (\ell_4, \rho)\sigma \in T, \exists v \in E[e]\rho: v \not\bowtie 0\} \cup \\ &\quad \{(\ell_1, \rho)(\ell_2, \rho)\sigma \mid \sigma \in \Sigma^{\infty}, (\ell_2, \rho)\sigma \in \mathcal{M}[\![{}^{\ell_2}s^{\ell_3}]\!]\{(\ell_3, \rho')\sigma' \mid \sigma' \in Y\}, \exists v \in E[e]\rho: v \bowtie 0\} \end{aligned}$$

$$\mathcal{M}[\![s_1; s_2]\!]T \stackrel{\text{def}}{=} \mathcal{M}[\![s_1]\!](\mathcal{M}[\![s_2]\!]S)$$

Static Analysis by Abstract Interpretation

Program Properties

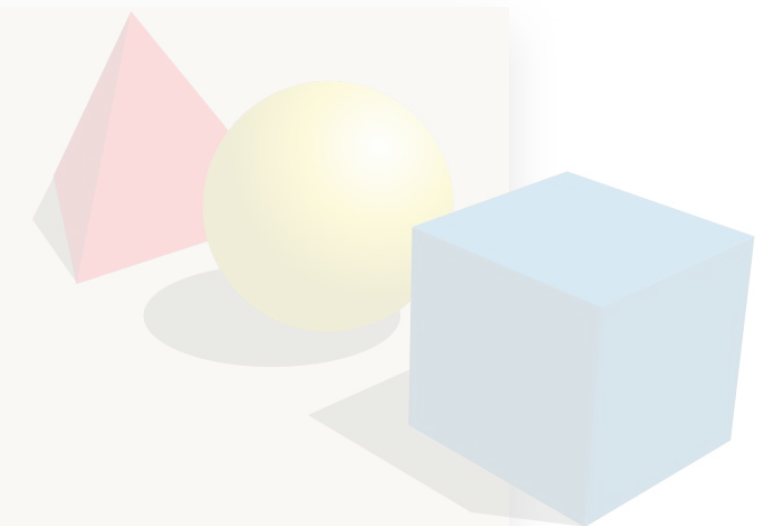
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Static Analysis by Abstract Interpretation



PROPERTY OF INTEREST



State Properties

$$S \in \mathcal{P}(\Sigma)$$

Example

$$\bullet S \stackrel{\text{def}}{=} \{ \langle \ell, \rho \rangle \in \Sigma \mid \ell \in \mathcal{L}, \rho \in \mathcal{E}, \rho(r) \geq 0 \}$$

```
1
a ← [0, +∞]
2
b ← [0, +∞]
3
q ← 0
4
r ← a
5
while 6(r ≥ b) do
7
  r ← r - b
8
  q ← q + 1
9
done
10
```

State Properties

$$S \in \mathcal{P}(\Sigma)$$

```
1  
a ← [0, +∞]  
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9  
done  
10
```

Example

$$\bullet S \stackrel{\text{def}}{=} \{ \langle \ell, \rho \rangle \in \Sigma \mid \ell \in \mathcal{L}, \rho \in \mathcal{E}, \rho(r) \geq 0 \}$$

State Property Verification



$$\mathcal{R}(I) \subseteq S$$

$$\mathcal{C}(F) \subseteq S$$

Trace Properties

$$T \in \mathcal{P}(\Sigma^\infty)$$

Example

- Termination: $T \stackrel{\text{def}}{=} \Sigma^*$
- Non-Termination: $T \stackrel{\text{def}}{=} \Sigma^\omega$
- Any State Property $S \in \mathcal{P}(\Sigma)$: $T \stackrel{\text{def}}{=} S^\infty$

Trace Property Verification



$$\mathcal{M} \subseteq T$$

Program Properties

$$P \in \mathcal{P}(\mathcal{P}(\Sigma^\infty))$$

Example

- Determinism: $P \stackrel{\text{def}}{=} \{ \{ \sigma \} \mid \sigma \in \Sigma^\infty \}$

Program Property Verification



$$\mathcal{M} \in P \Leftrightarrow \underbrace{\{ \mathcal{M} \}}_{\text{Collecting Semantics}} \subseteq P$$

Collecting Semantics

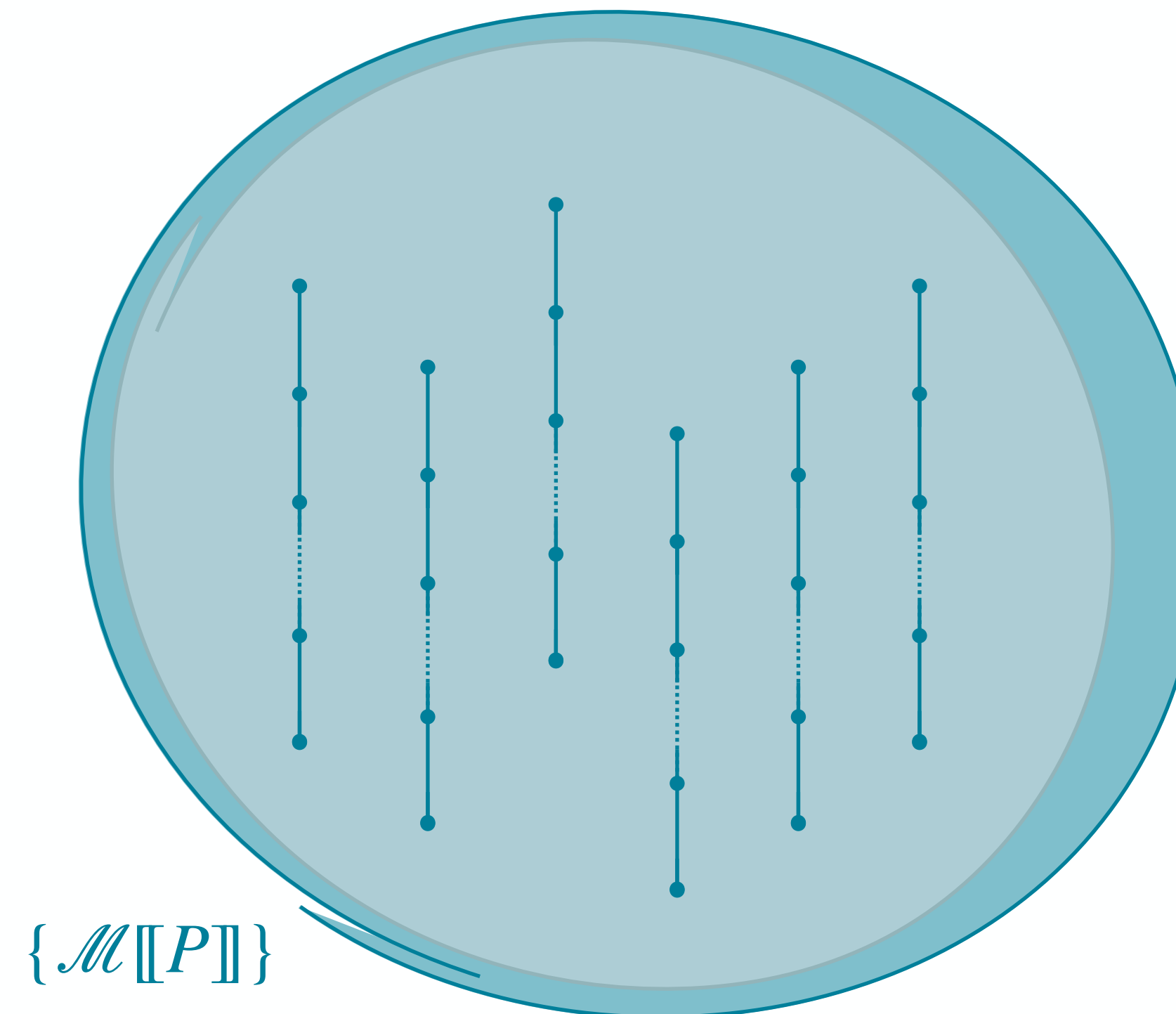
Intuition

Property (by extension): set of elements that have that property

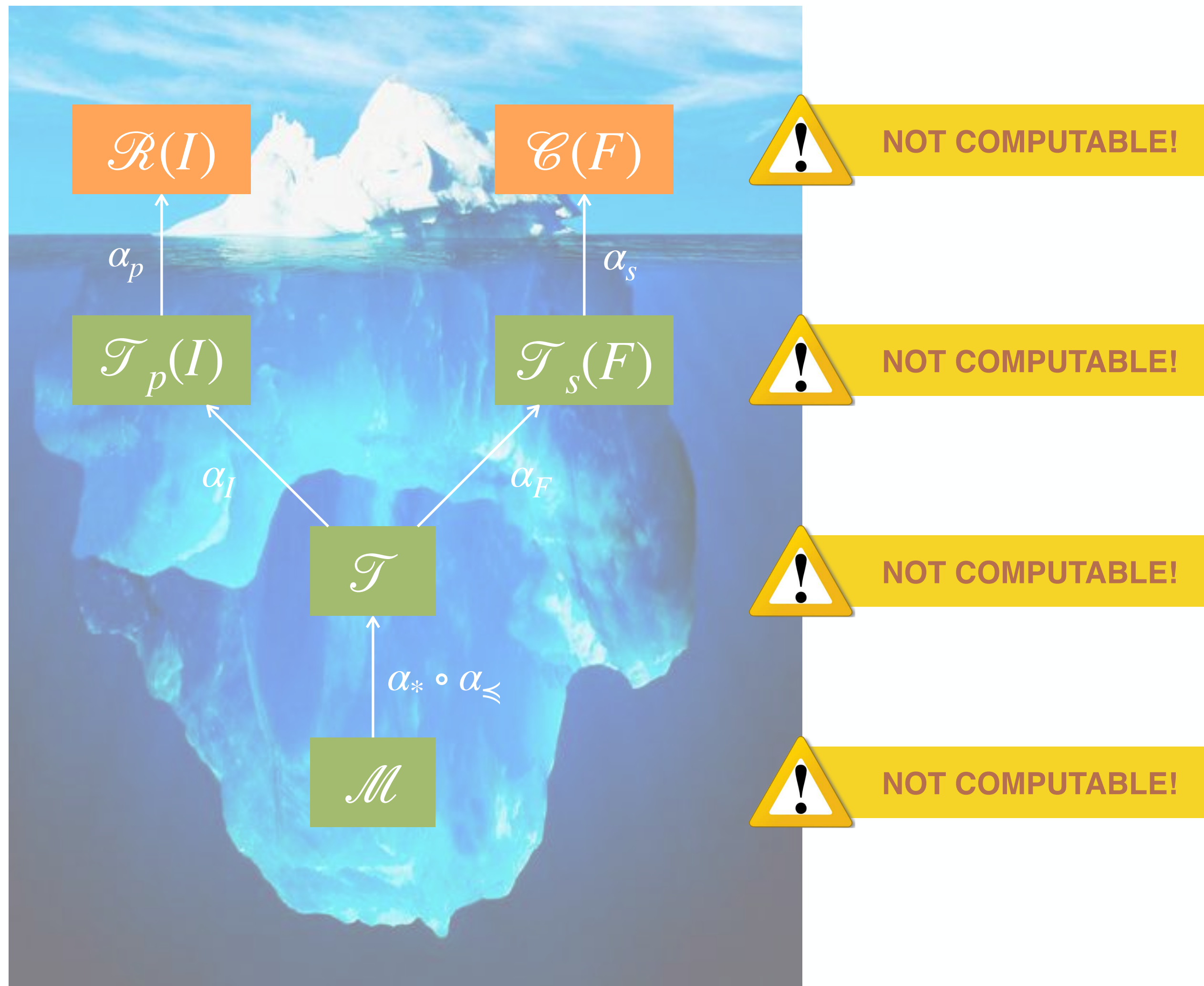
Property “being Vincenzo”



Property “being program P”



Hierarchy of Mathematical Objects



Static Analysis by Abstract Interpretation

Abstract Program Semantics

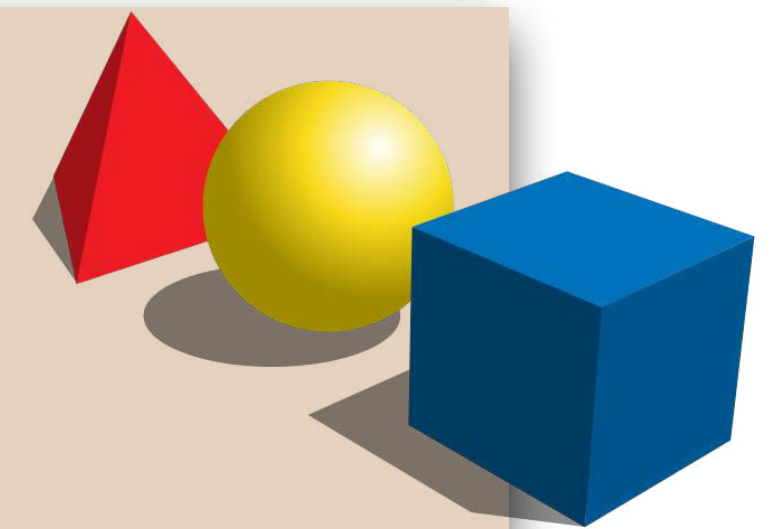
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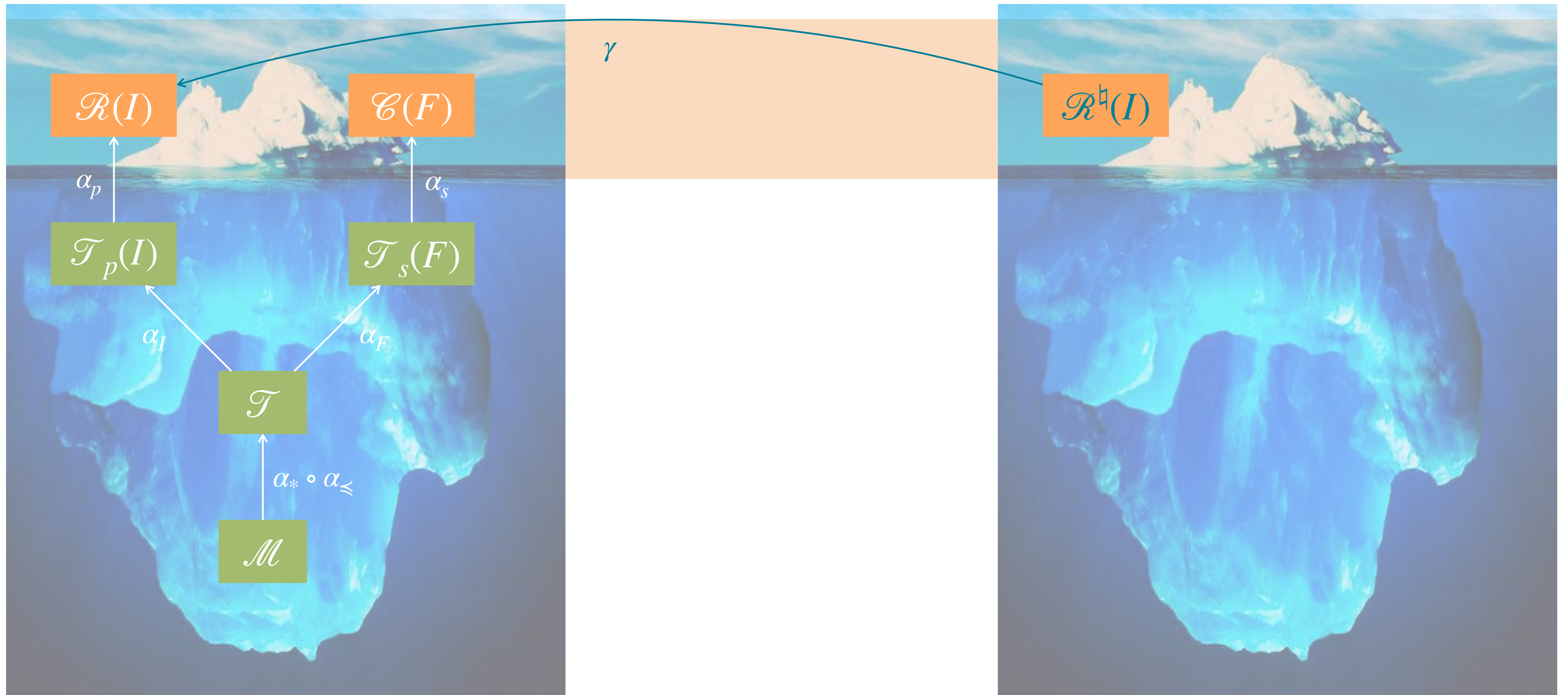
mathematical models of the program behavior



Static Analysis by Abstract Interpretation



Abstract Forward Reachability Semantics



Abstract Forward Reachability Semantics

Over-Approximation of Program States Reachable From $I \in \mathcal{P}(\Sigma)$

- $\mathcal{R}(I) \subseteq \gamma(\mathcal{R}^\#(I)) \in \mathcal{P}(\Sigma)$

Abstract Forward Reachability Semantics

Denotational Formulation

$$\mathcal{R}^\#[\ell_1 X \leftarrow e^{\ell_2}](\ell_1, a) \stackrel{\text{def}}{=} (\ell_2, \text{ASSIGN}_A[[X \leftarrow e]]a)$$

$$\mathcal{R}^\#[\text{if } \ell_1 e \bowtie 0 \text{ then } \ell_2 s^{\ell_3} \text{ end } \ell_4](\ell_1, a) \stackrel{\text{def}}{=} (\ell_4, a') \sqcup_A (\ell_4, \text{FILTER}_A[[e \bowtie 0]]a)$$

where $(\ell_3, a') = \mathcal{R}^\#[\ell_2 s^{\ell_3}](\ell_2, \text{FILTER}_A[[e \bowtie 0]]a)$

$$\mathcal{R}^\#[\text{while } \ell_1 e \bowtie 0 \text{ then } \ell_2 s^{\ell_3} \text{ done } \ell_4](\ell_1, a) \stackrel{\text{def}}{=} (\ell_4, \text{FILTER}_A[[e \bowtie 0]]a')$$

where $(\ell_1, a') = \text{lfp}_{(\ell_1, \perp_A)}^{\# \sqsubseteq_A} F_r^\#$

$$F_r^\#((\ell_1, y)) \stackrel{\text{def}}{=} (\ell_1, a) \sqcup_A (\ell_1, a'')$$

$$(\ell_3, a'') = \mathcal{R}^\#[\ell_2 s^{\ell_3}](\ell_2, \text{FILTER}_A[[e \bowtie 0]]y)$$

$$\mathcal{R}^\# [s_1; s_2](\ell_1, a) \stackrel{\text{def}}{=} \mathcal{R}^\# [s_2](\mathcal{R}^\# [s_1](\ell_1, a))$$

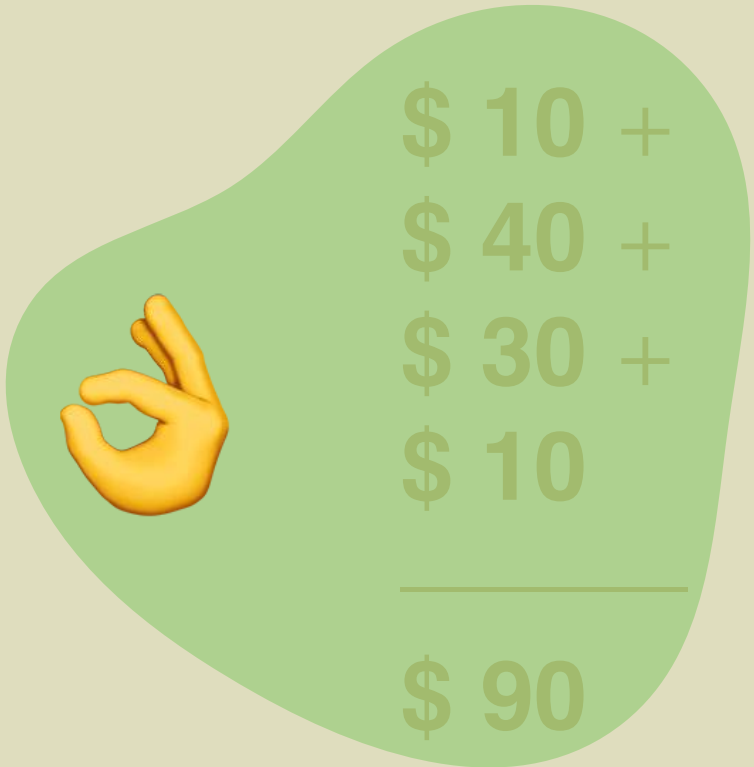
```

stmt ::=  $\ell X \leftarrow expr^\ell$ 
       | if  $\ell expr \bowtie 0$  then stmt end $^\ell$ 
       | while  $\ell expr \bowtie 0$  do stmt done $^\ell$ 
       | stmt; stmt
    
```

Static Analysis by Abstract Interpretation



SOUNDNESS



COMPLETENESS



State Properties

$$S \in \mathcal{P}(\Sigma)$$

```
1  
a ← [0, +∞]  
2  
b ← [0, +∞]  
3  
q ← 0  
4  
r ← a  
5  
while 6(r ≥ b) do  
7  
  r ← r − b  
8  
  q ← q + 1  
9  
done  
10
```

Example

$$\bullet S \stackrel{\text{def}}{=} \{ \langle \ell, \rho \rangle \in \Sigma \mid \ell \in \mathcal{L}, \rho \in \mathcal{E}, \rho(r) \geq 0 \}$$

Sound State Property Verification



$$\mathcal{R}(I) \subseteq \gamma(\mathcal{R}^\sharp(I)) \subseteq S$$

Static Analysis by Abstract Interpretation

Abstract Domains

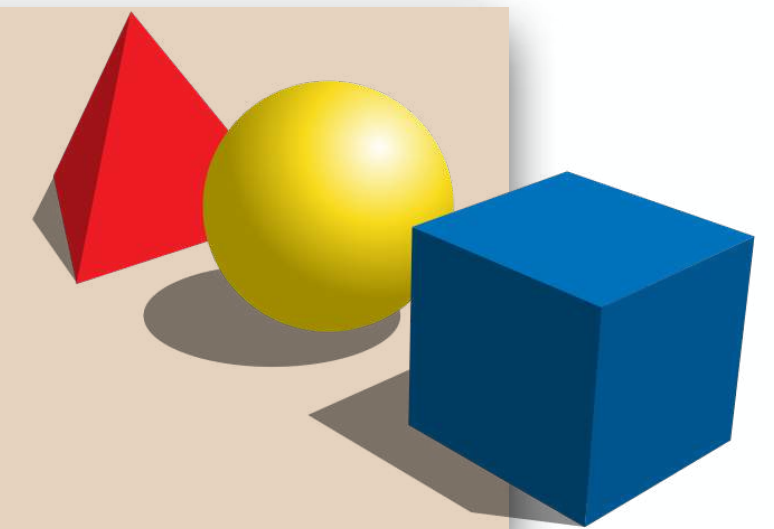
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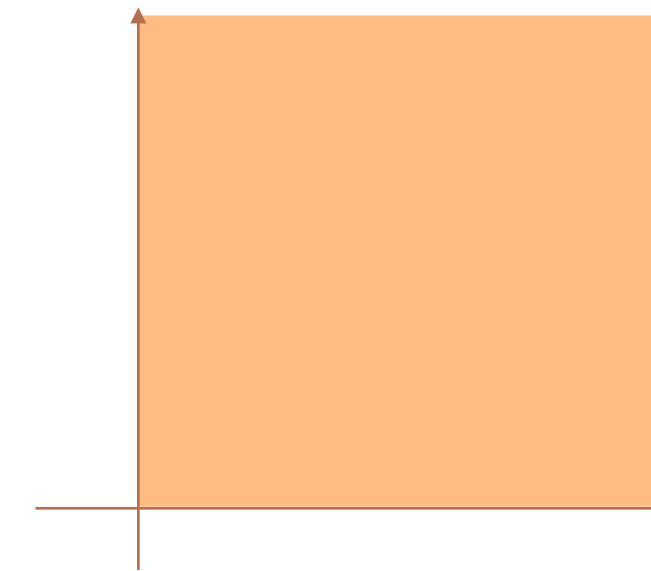
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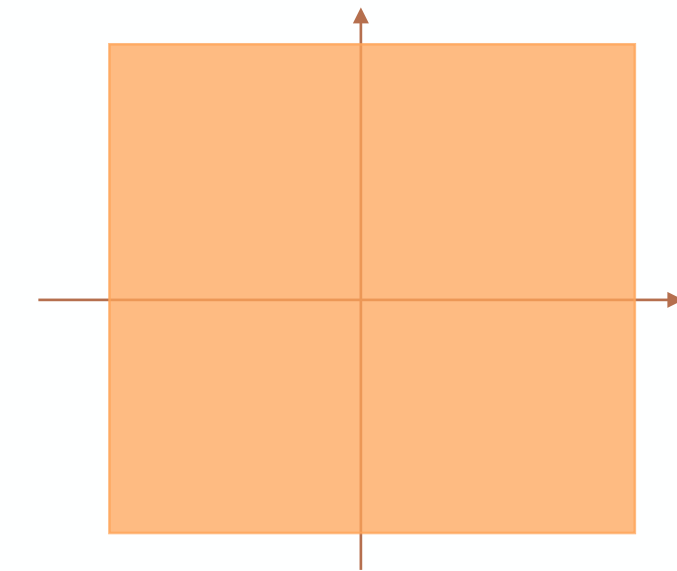


Numerical Abstract Domains

Non-Relational Domains

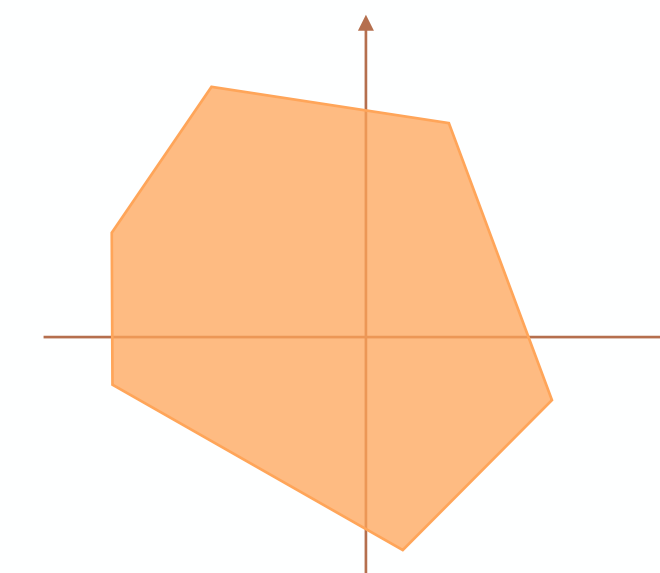


Sign Domain



Interval Domain

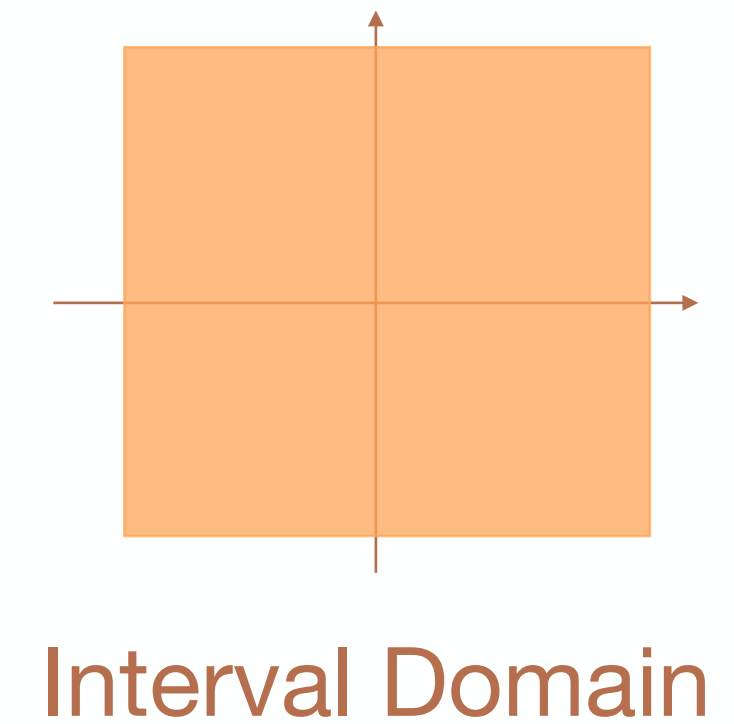
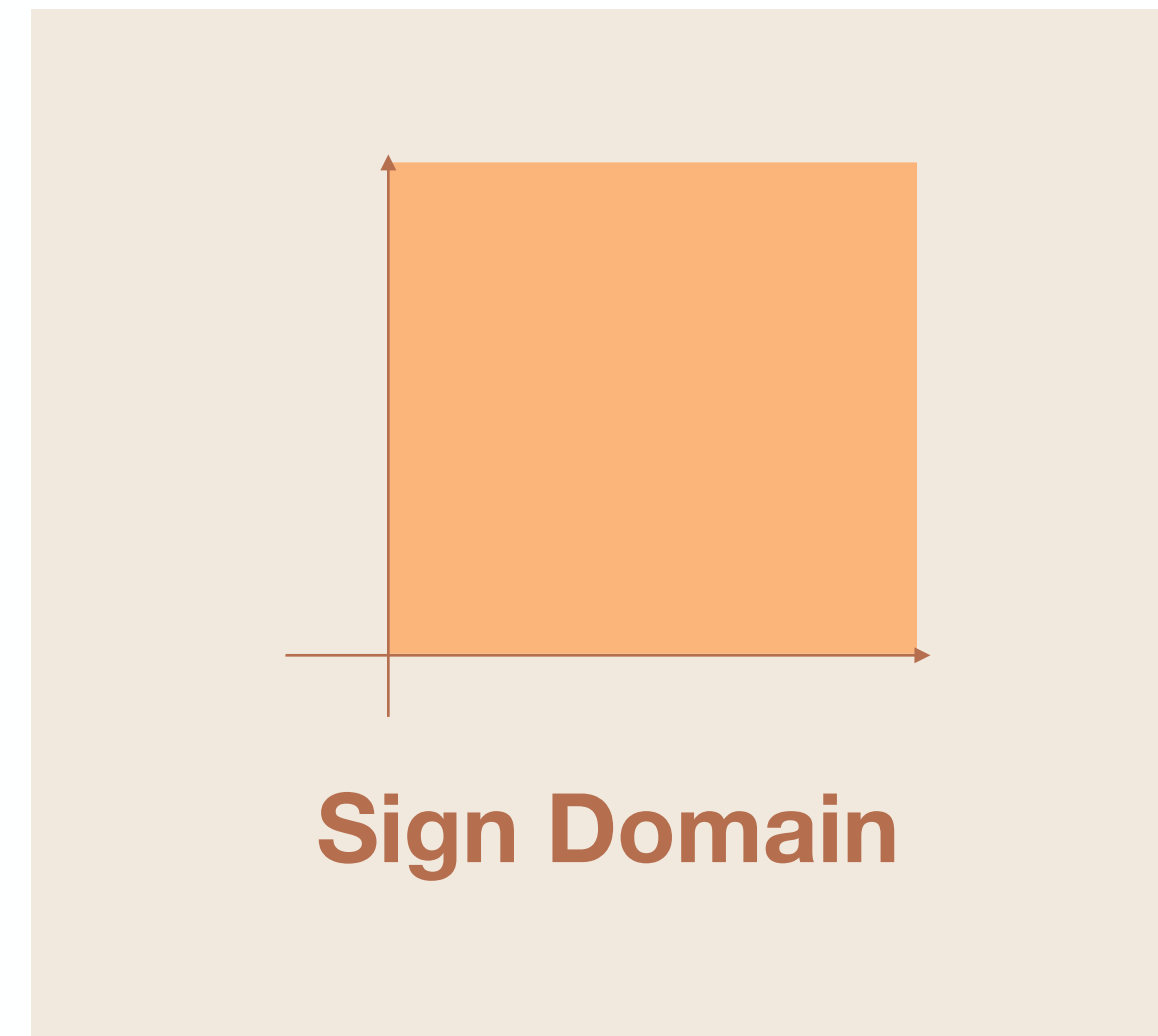
Relational Domains



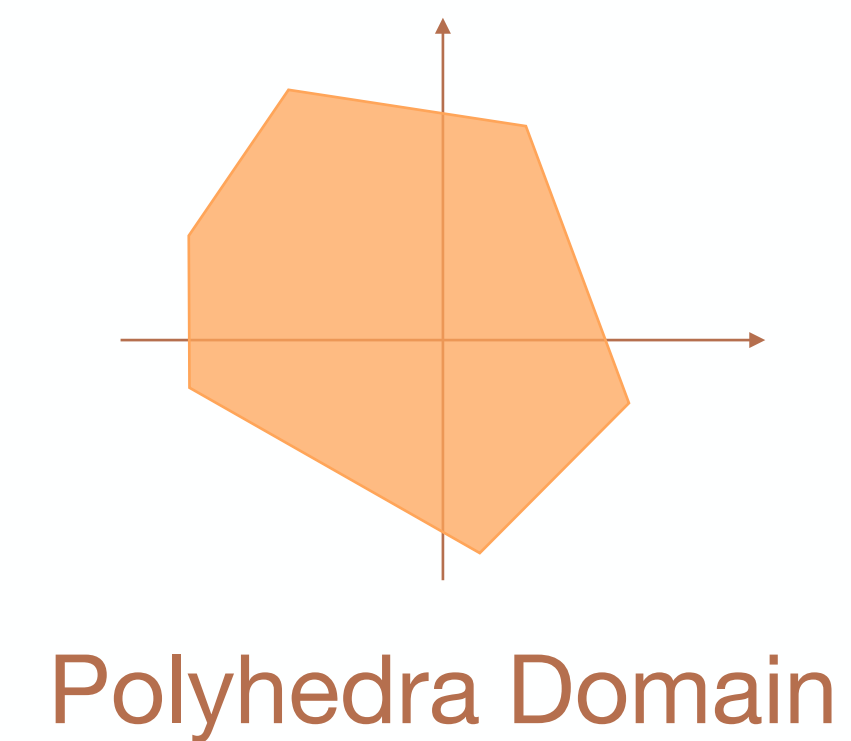
Polyhedra Domain

Numerical Abstract Domains

Non-Relational Domains

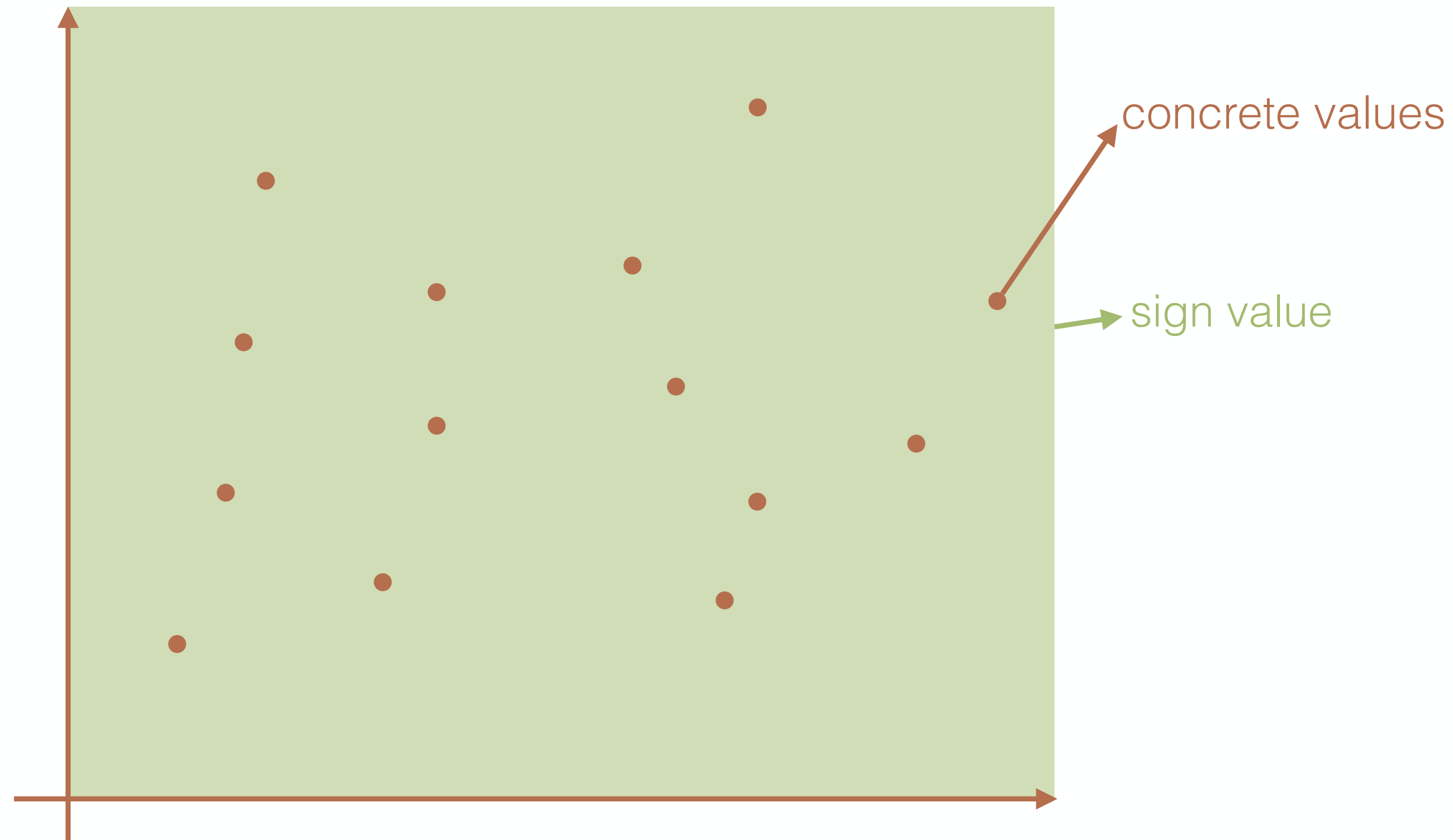


Relational Domains



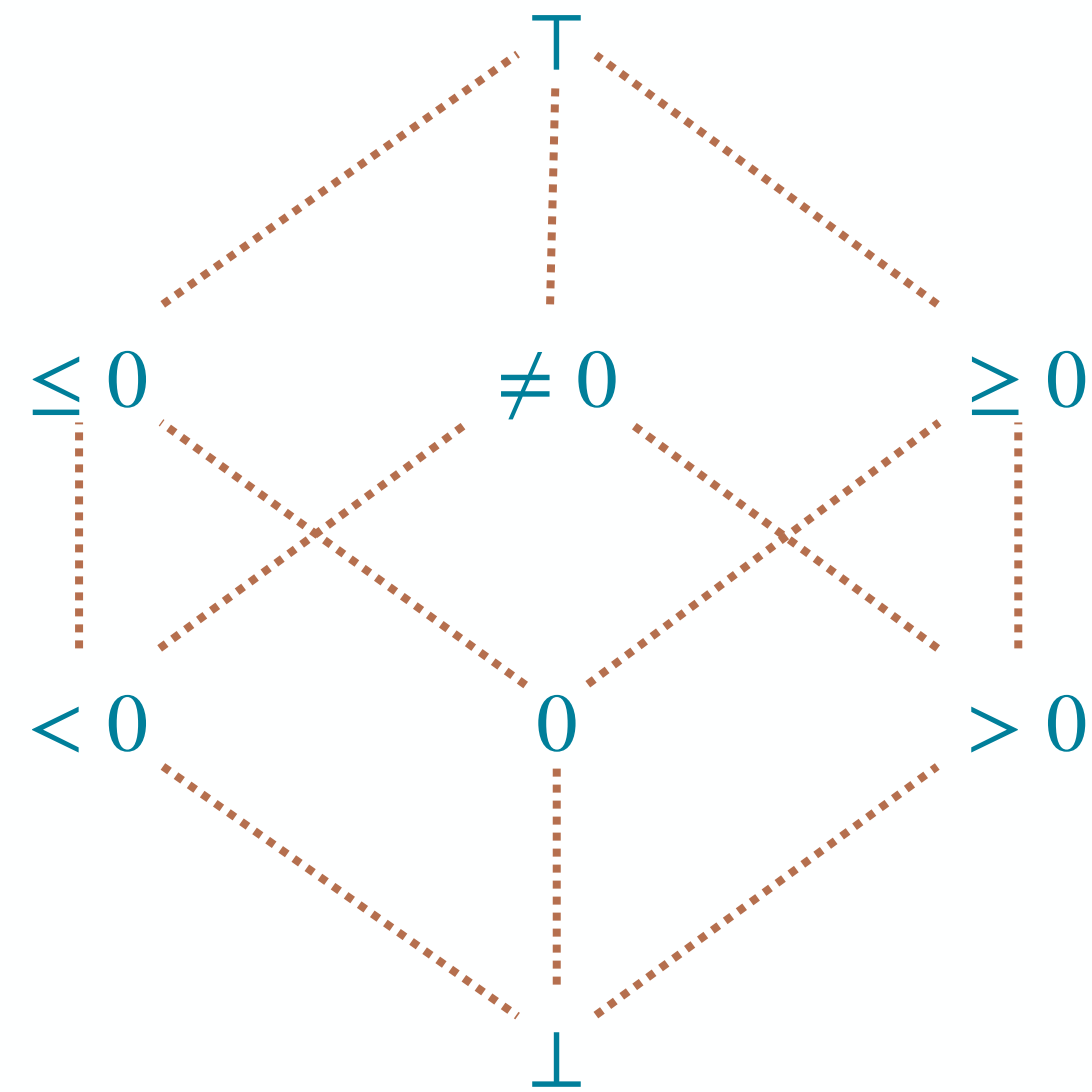
Sign Abstract Domain

Concrete Value are Replaced with Sign Values



Sign Abstract Domain

$A : \mathbb{X} \rightarrow \text{Sign}$ maps **variables** to their **sign**



$\gamma_{\text{Sign}} : \text{Sign} \rightarrow \mathcal{P}(\mathbb{Z})$

$$\gamma_{\text{Sign}}(\perp) \stackrel{\text{def}}{=} \emptyset$$

$$\gamma_{\text{Sign}}(0) \stackrel{\text{def}}{=} \{0\}$$

$\vdots$

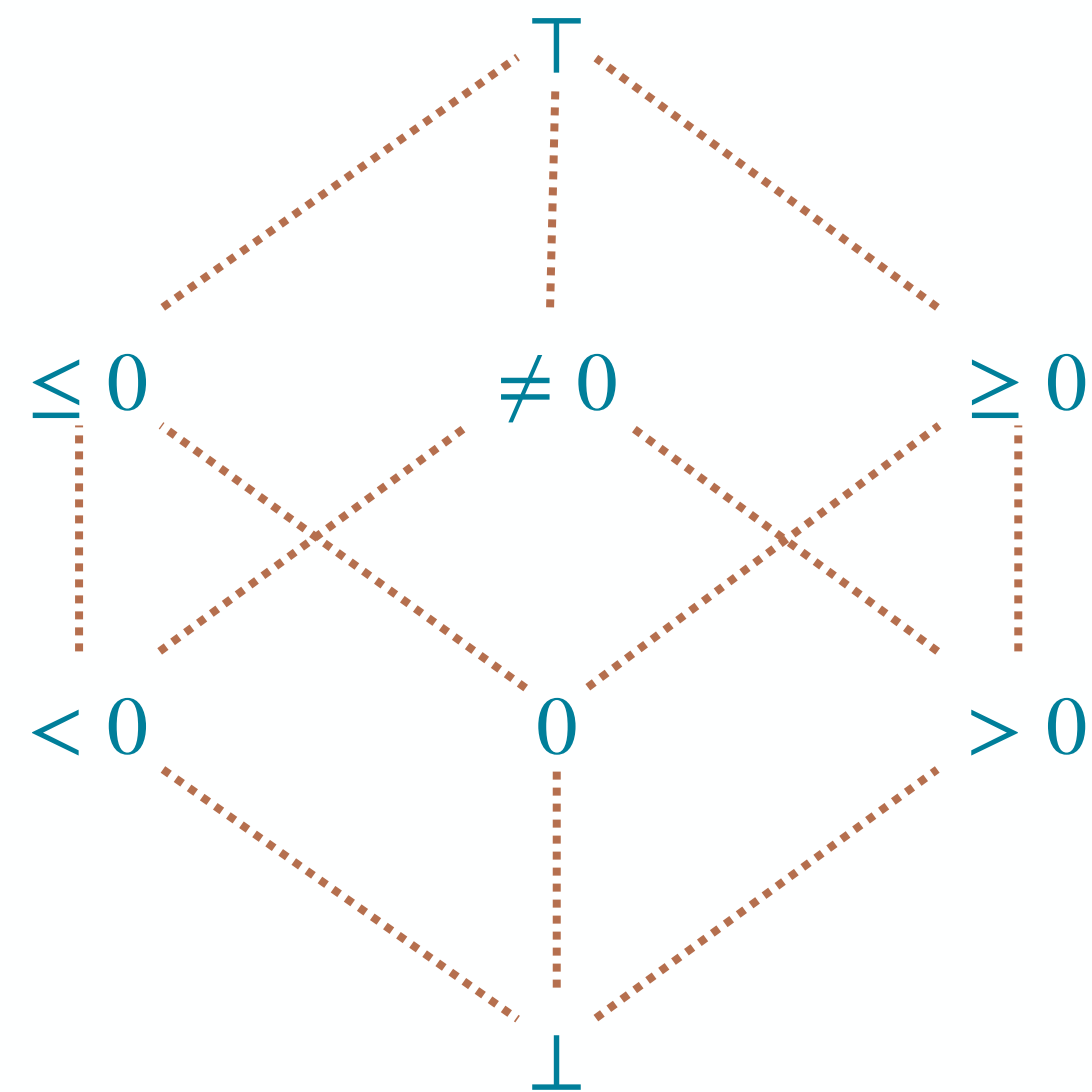
$$\gamma_{\text{Sign}}(\top) \stackrel{\text{def}}{=} \mathbb{Z}$$

$$\gamma' : A \rightarrow \mathcal{P}(\mathcal{E}) \quad \gamma'(a) \stackrel{\text{def}}{=} \{\rho \in \mathcal{E} \mid \forall x \in \mathbb{X} : \rho(x) \in \gamma_{\text{Sign}}(a(x))\}$$

$$\gamma : \mathcal{L} \times A \rightarrow \mathcal{P}(\Sigma) \quad \gamma((\ell, a)) \stackrel{\text{def}}{=} \{(\ell, \rho) \in \Sigma \mid \rho(x) \in \gamma'(a)\}$$

Sign Abstract Domain

$A: \mathbb{X} \rightarrow \text{Sign}$ maps **variables** to their **sign**



$\sqsubseteq_A$ defined by the diagram

$\sqcup_A$ defined by the diagram

$\sqcap_A$ defined by the diagram

$\text{ASSIGN}_A[[X \leftarrow e]]a$ maps X to e evaluated according to the **sign rules**

$$\begin{aligned} &> 0 + > 0 = > 0 - < 0 = > 0 \\ &< 0 + < 0 = < 0 - > 0 = < 0 \\ &\vdots \end{aligned}$$

$\text{FILTER}_A[[e \bowtie 0]]a$ modifies a to satisfy $e \bowtie 0$

Sign Static Analysis

$$S \stackrel{\text{def}}{=} \{ \langle \ell, \rho \rangle \in \Sigma \mid \ell \in \mathcal{L}, \rho \in \mathcal{E}, \rho(r) \geq 0 \}$$

$$\gamma(\mathcal{R}^{\sharp}(I)) \not\subseteq S$$



```

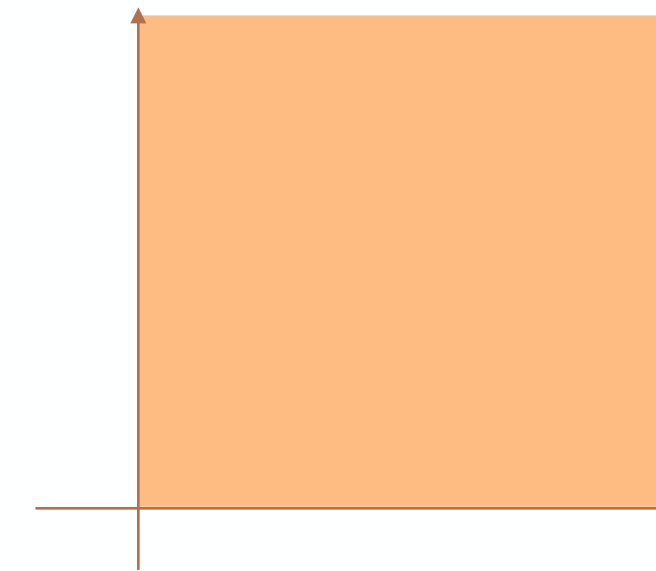
1
a ← [0, +∞]
2
b ← [0, +∞]
3
q ← 0
4
r ← a
5
while 6(r ≥ b) do
7
  r ← r - b
8
  q ← q + 1
9
done
10

```

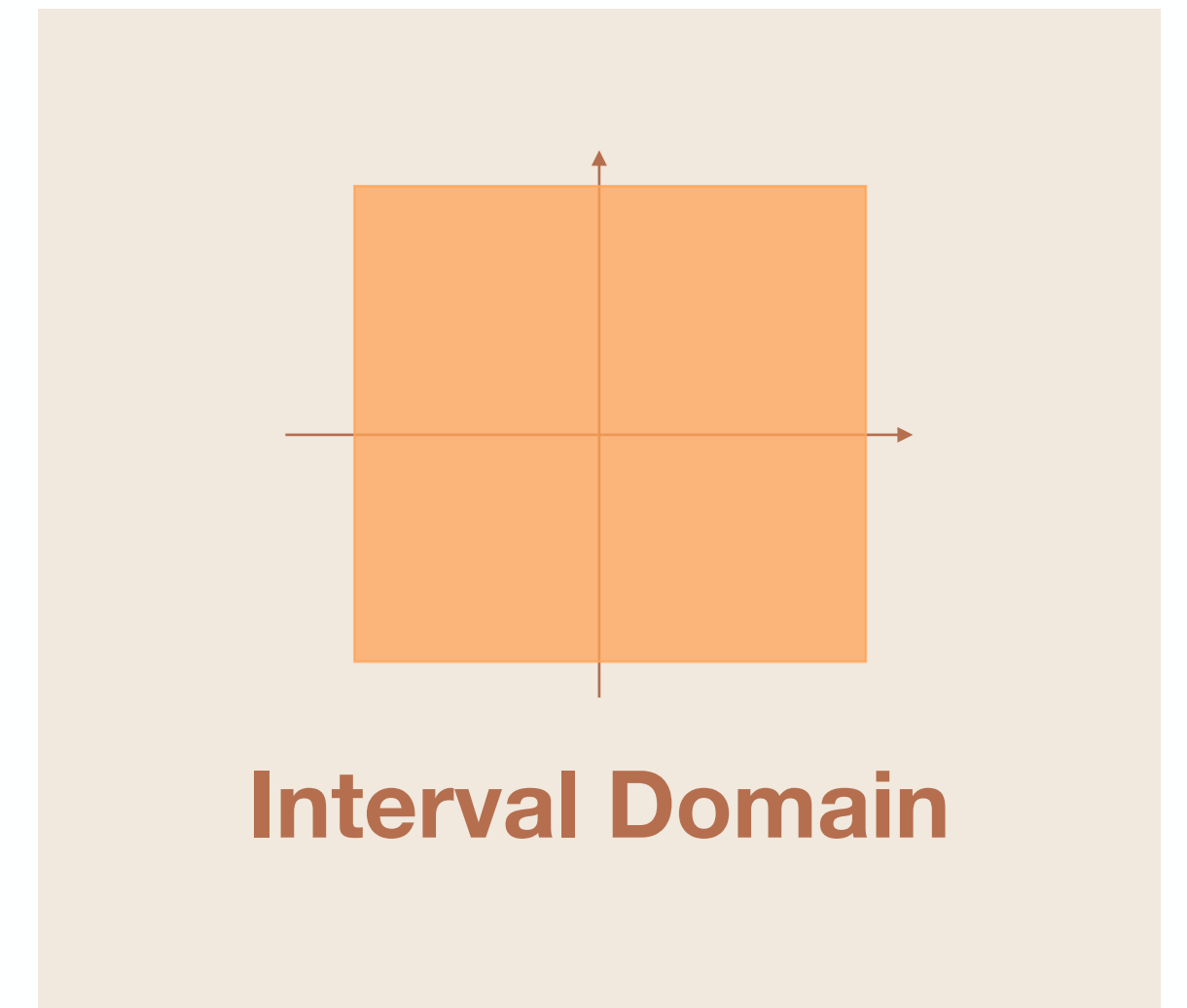
3:	$a \mapsto \geq 0$	$b \mapsto \geq 0$		
4:	$a \mapsto \geq 0$	$b \mapsto \geq 0$	$q \mapsto 0$	
5:	$a \mapsto \geq 0$	$b \mapsto \geq 0$	$q \mapsto 0$	$r \mapsto \geq 0$
6:	$a \mapsto \geq 0$	$b \mapsto \geq 0$	$q \mapsto 0$	$r \mapsto \geq 0$
7:	$a \mapsto \geq 0$	$b \mapsto \geq 0$	$q \mapsto 0$	$r \mapsto \geq 0$
8:	$a \mapsto \geq 0$	$b \mapsto \geq 0$	$q \mapsto 0$	$r \mapsto \top$
9:	$a \mapsto \geq 0$	$b \mapsto \geq 0$	$q \mapsto > 0$	$r \mapsto \top$
6:	$a \mapsto \geq 0$	$b \mapsto \geq 0$	$q \mapsto \geq 0$	$r \mapsto \top$
7:	$a \mapsto \geq 0$	$b \mapsto \geq 0$	$q \mapsto \geq 0$	$r \mapsto \geq 0$
8:	$a \mapsto \geq 0$	$b \mapsto \geq 0$	$q \mapsto \geq 0$	$r \mapsto \top$
9:	$a \mapsto \geq 0$	$b \mapsto \geq 0$	$q \mapsto > 0$	$r \mapsto \top$
6:	$a \mapsto \geq 0$	$b \mapsto \geq 0$	$q \mapsto \geq 0$	$r \mapsto \top$
10:	$a \mapsto \geq 0$	$b \mapsto \geq 0$	$q \mapsto \geq 0$	$r \mapsto \top$

Numerical Abstract Domains

Non-Relational Domains

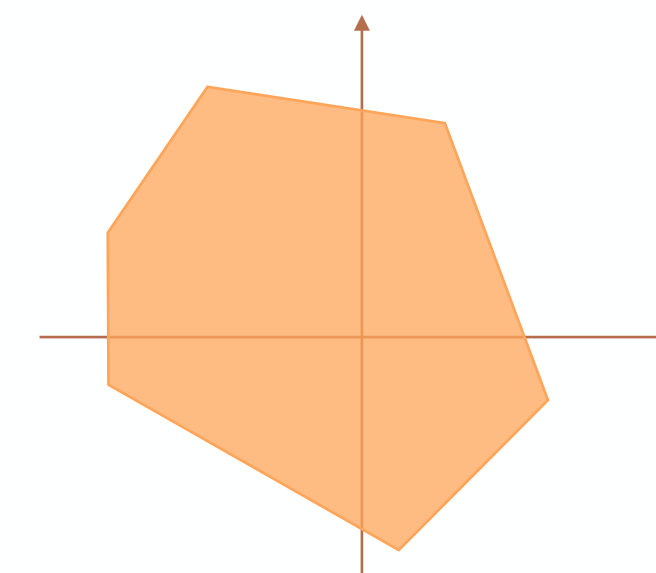


Sign Domain



Interval Domain

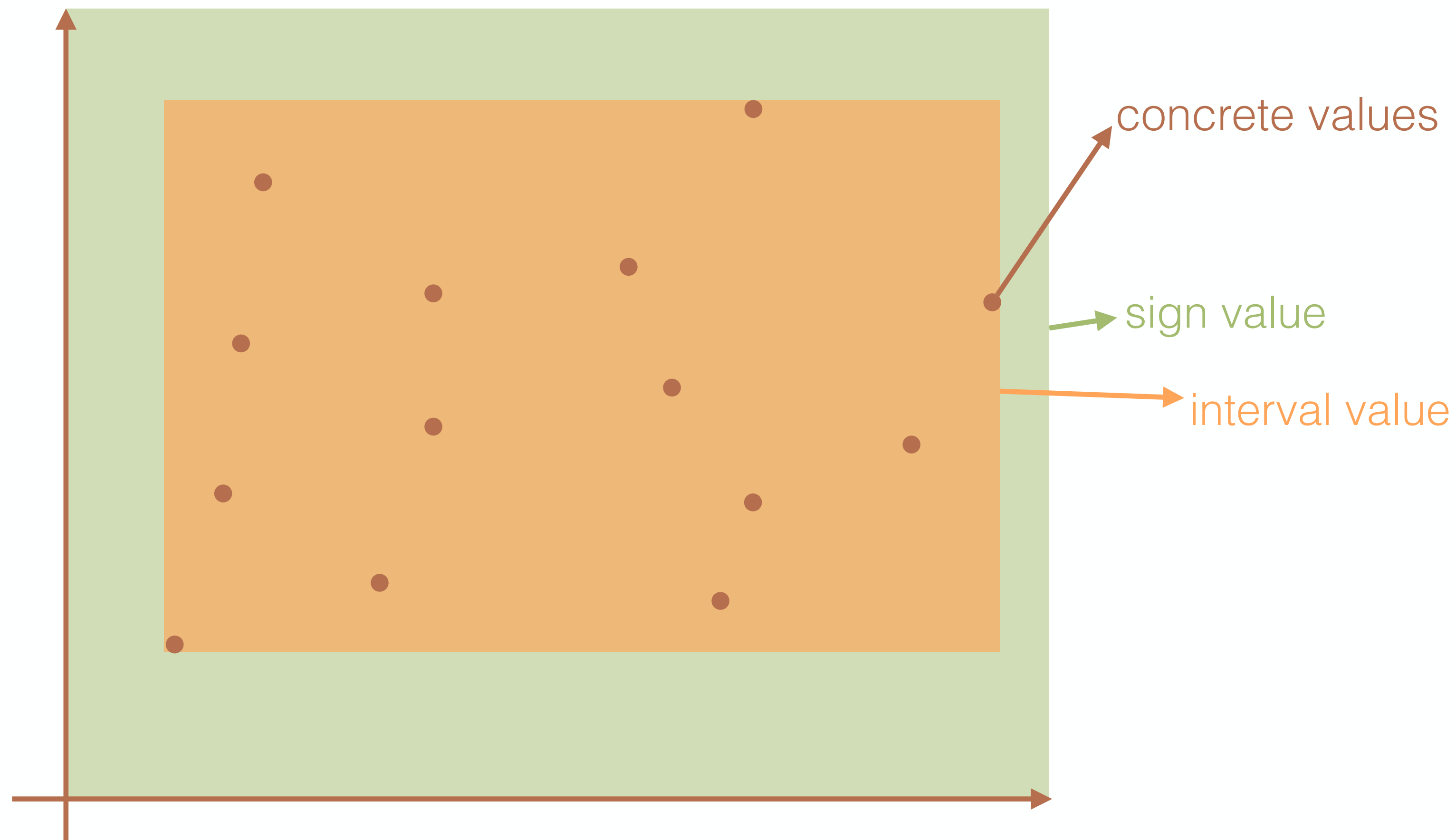
Relational Domains



Polyhedra Domain

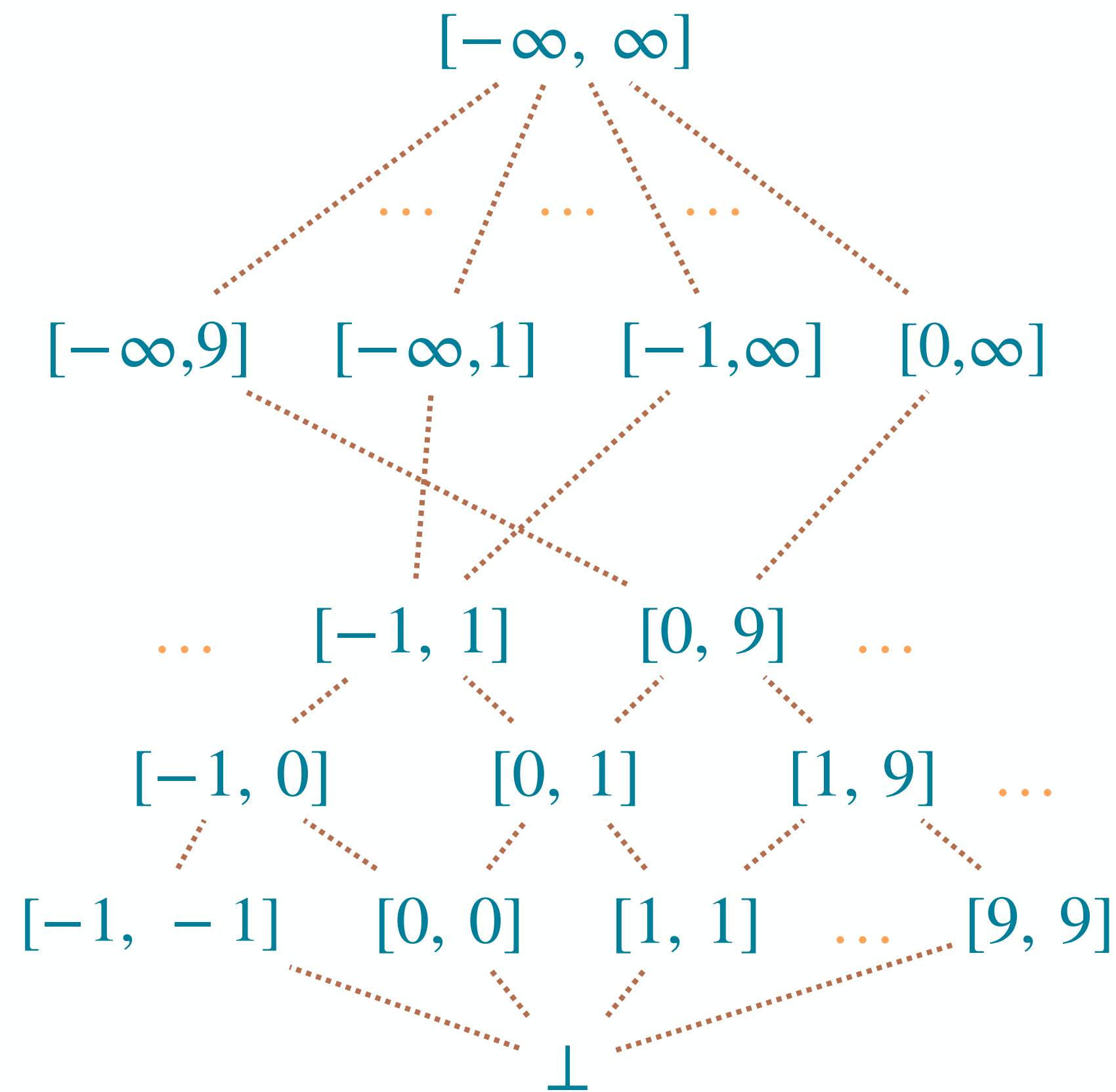
Interval Abstract Domain

Concrete Value are Replaced with Range Values



Interval Abstract Domain

$A: \mathbb{X} \rightarrow \text{Itv}$ maps **variables** to their **lower and upper bounds**



$$\gamma_{\text{Itv}}: \text{Itv} \rightarrow \mathcal{P}(\mathbb{Z})$$

$$\gamma_{\text{Itv}}(\perp) \stackrel{\text{def}}{=} \emptyset$$

$$\gamma_{\text{Itv}}([a, b]) \stackrel{\text{def}}{=} \{x \in \mathbb{Z} \mid a \leq x \leq b\}$$

$\vdots$

$$\gamma_{\text{Itv}}([\infty, \infty]) \stackrel{\text{def}}{=} \mathbb{Z}$$

$$\gamma': A \rightarrow \mathcal{P}(\mathcal{E})$$

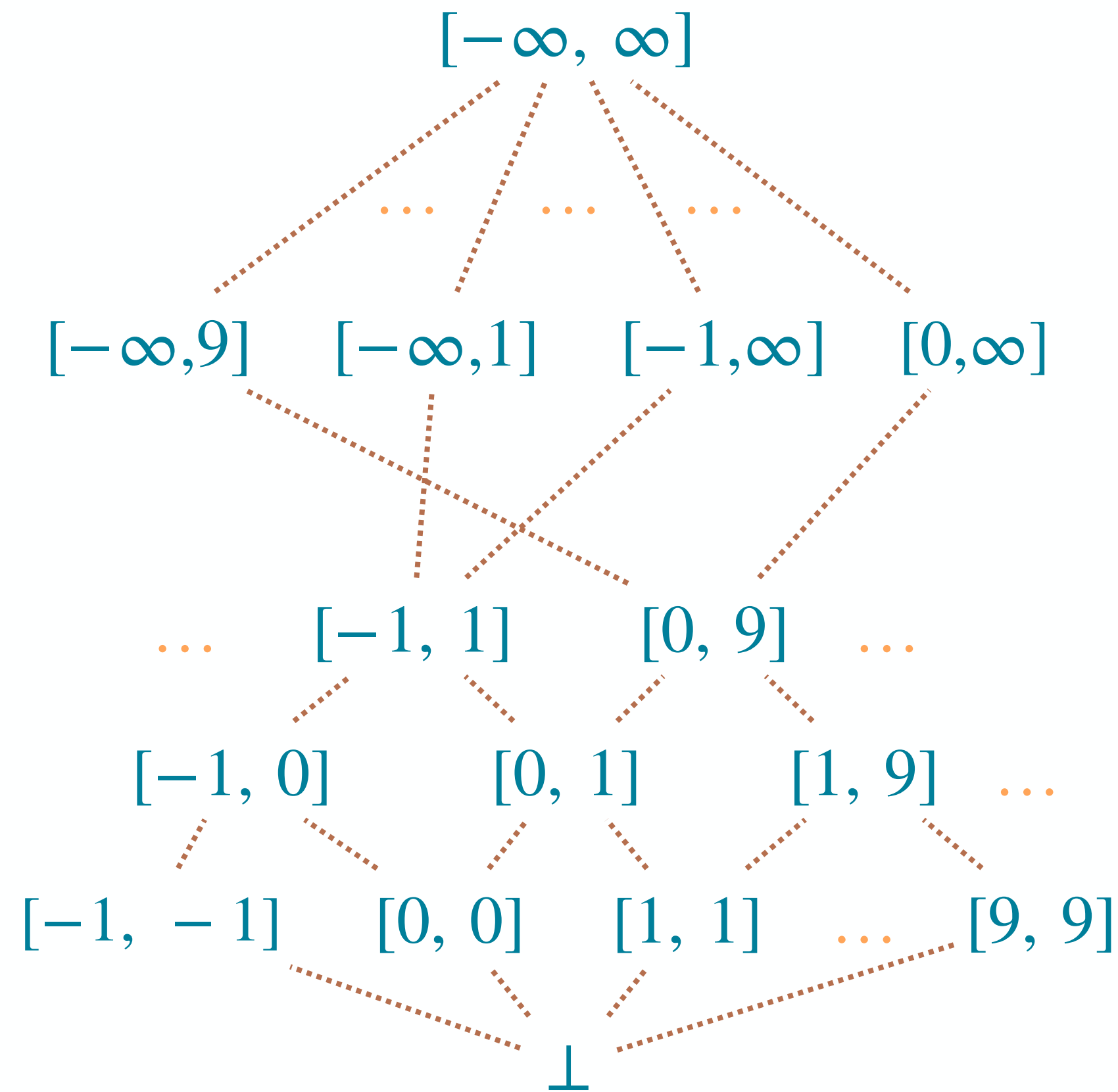
$$\gamma'(a) \stackrel{\text{def}}{=} \{\rho \in \mathcal{E} \mid \forall x \in \mathbb{X}: \rho(x) \in \gamma_{\text{Itv}}(a(x))\}$$

$$\gamma: \mathcal{L} \times A \rightarrow \mathcal{P}(\Sigma)$$

$$\gamma((\ell, a)) \stackrel{\text{def}}{=} \{(\ell, \rho) \in \Sigma \mid \rho(x) \in \gamma'(a)\}$$

Interval Abstract Domain

$A: \mathbb{X} \rightarrow \text{Itv}$ maps **variables** to their **lower and upper bounds**



$\sqsubseteq_A$ defined by the diagram

$\sqcup_A$ defined by the diagram

$\sqcap_A$ defined by the diagram

$\text{ASSIGN}_A[[X \leftarrow e]]a$ maps X to e evaluated with **interval arithmetic**

$\text{FILTER}_A[[e \bowtie 0]]a$ modifies a to satisfy $e \bowtie 0$

∇_A loosens the unstable bounds

$$[0, 1] \nabla_A [0, 2] = [0, \infty]$$

$$[0, 1] \nabla_A [-1, 1] = [-\infty, 1]$$

Interval Static Analysis

```

1
a ← [0, +∞]
2
b ← [0, +∞]
3
q ← 0
4
r ← a
5
while 6(r ≥ b) do
7
  r ← r - b
8
  q ← q + 1
9
done
10

```

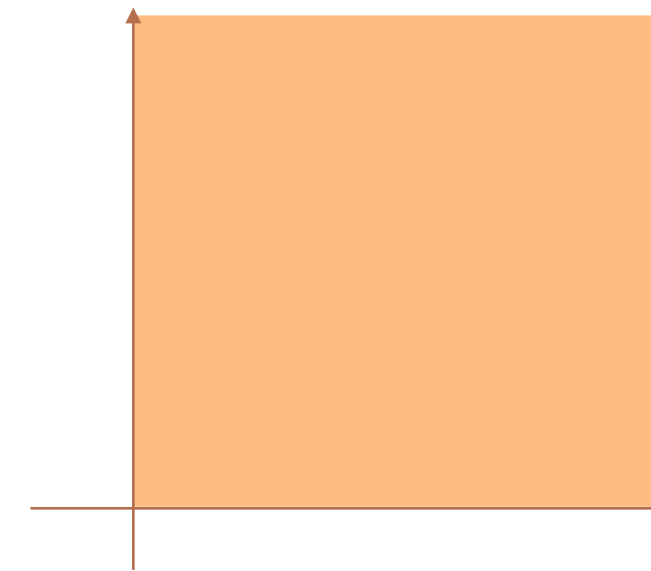
³ :	$a \mapsto [0, \infty]$	$b \mapsto [0, \infty]$
⁴ :	$a \mapsto [0, \infty]$	$b \mapsto [0, \infty]$
⁵ :	$a \mapsto [0, \infty]$	$b \mapsto [0, \infty]$
⁶ :	$a \mapsto [0, \infty]$	$b \mapsto [0, \infty]$
⁷ :	$a \mapsto [0, \infty]$	$b \mapsto [0, \infty]$
⁸ :	$a \mapsto [0, \infty]$	$b \mapsto [0, \infty]$
⁹ :	$a \mapsto [0, \infty]$	$b \mapsto [0, \infty]$
⁶ :	$a \mapsto [0, \infty]$	$b \mapsto [0, \infty]$
⁷ :	$a \mapsto [0, \infty]$	$b \mapsto [0, \infty]$
⁸ :	$a \mapsto [0, \infty]$	$b \mapsto [0, \infty]$
⁹ :	$a \mapsto [0, \infty]$	$b \mapsto [0, \infty]$
⁶ :	$a \mapsto [0, \infty]$	$b \mapsto [0, \infty]$
$\vdots$		
¹⁰ :	$a \mapsto [0, \infty]$	$b \mapsto [0, \infty]$

$$S \stackrel{\text{def}}{=} \{ \langle \ell, \rho \rangle \in \Sigma \mid \ell \in \mathcal{L}, \rho \in \mathcal{E}, \rho(r) \geq 0 \}$$

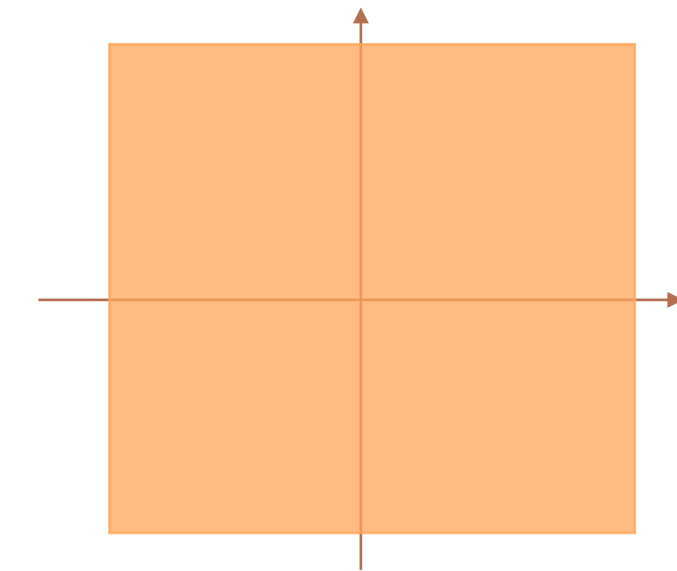
$q \mapsto [0, 0]$		$\gamma(\mathcal{R}^\sharp(I)) \not\subseteq S$
$q \mapsto [0, 0]$	$r \mapsto [0, \infty]$	 FALSE ALARM
$q \mapsto [0, 0]$	$r \mapsto [0, \infty]$	
$q \mapsto [0, 0]$	$r \mapsto [0, \infty]$	
$q \mapsto [0, 0]$	$r \mapsto [-\infty, \infty]$	
$q \mapsto [1, 1]$	$r \mapsto [-\infty, \infty]$	
$q \mapsto [0, 1]$	$r \mapsto [-\infty, \infty]$	
$q \mapsto [0, 1]$	$r \mapsto [0, \infty]$	
$q \mapsto [0, 1]$	$r \mapsto [-\infty, \infty]$	
$q \mapsto [1, 2]$	$r \mapsto [-\infty, \infty]$	
$q \mapsto [0, \infty]$	$r \mapsto [-\infty, \infty]$	
$q \mapsto [0, \infty]$	$r \mapsto [-\infty, \infty]$	

Numerical Abstract Domains

Non-Relational Domains

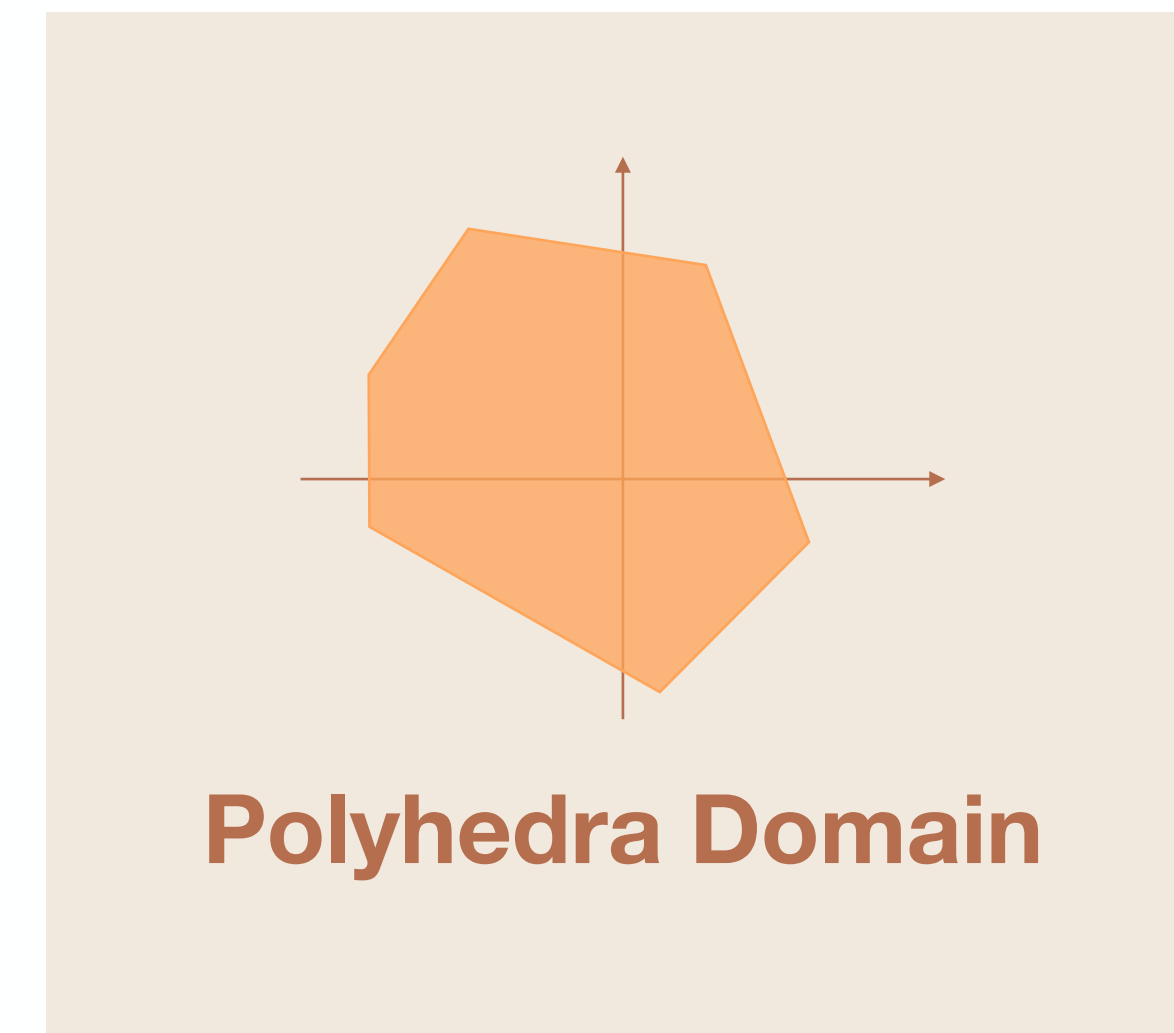


Sign Domain



Interval Domain

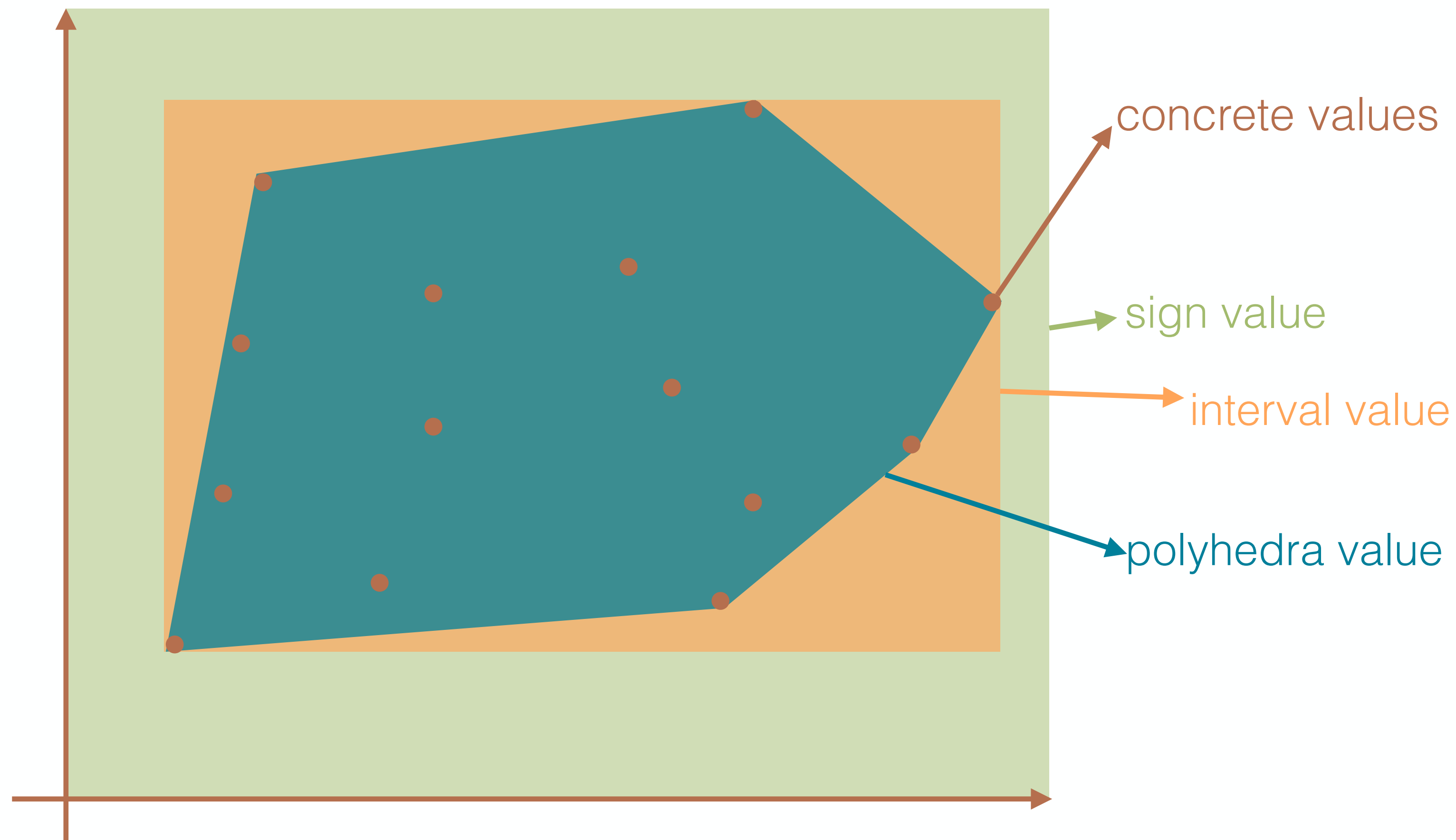
Relational Domains



Polyhedra Domain

Polyhedra Abstract Domain

Concrete Value are Replaced with Conjunctions of Linear Inequalities



Polyhedra Abstract Domain

A set of **convex polyhedra**

$\gamma': A \rightarrow \mathcal{P}(\mathcal{E})$ set of memory states within the polyhedra

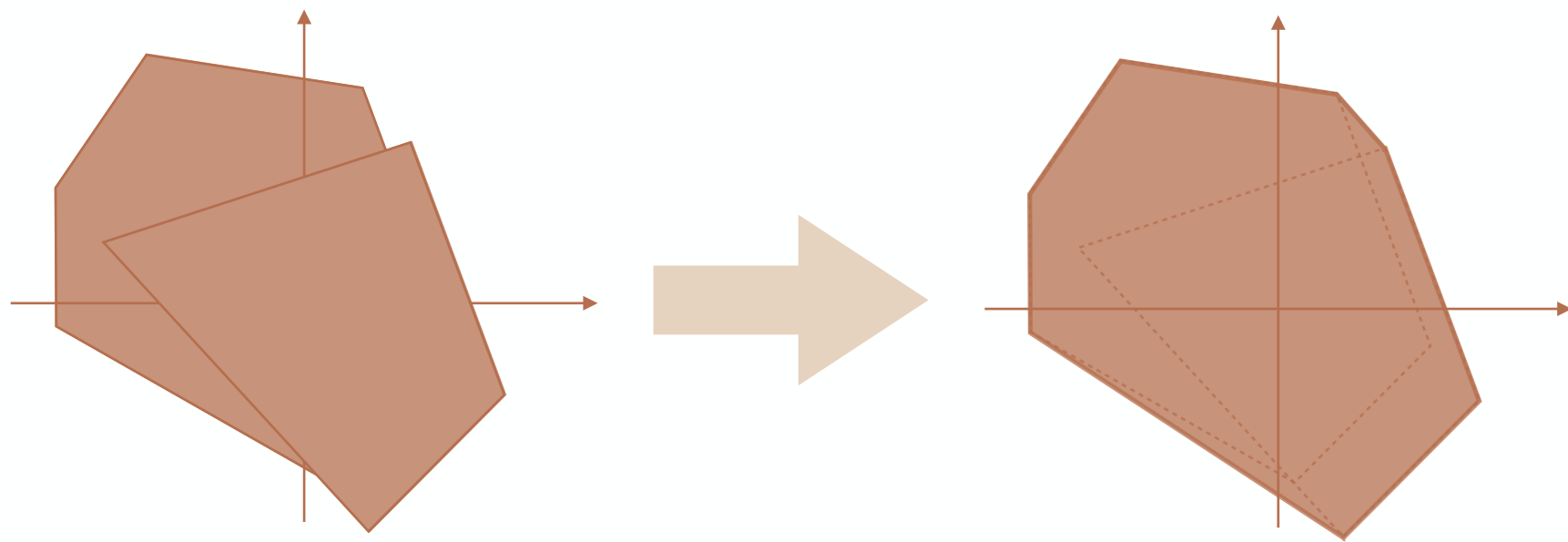
$\gamma: \mathcal{L} \times A \rightarrow \mathcal{P}(\Sigma)$ $\gamma((\ell, a)) \stackrel{\text{def}}{=} \{(\ell, \rho) \in \Sigma \mid \rho(x) \in \gamma'(a)\}$

Polyhedra Abstract Domain

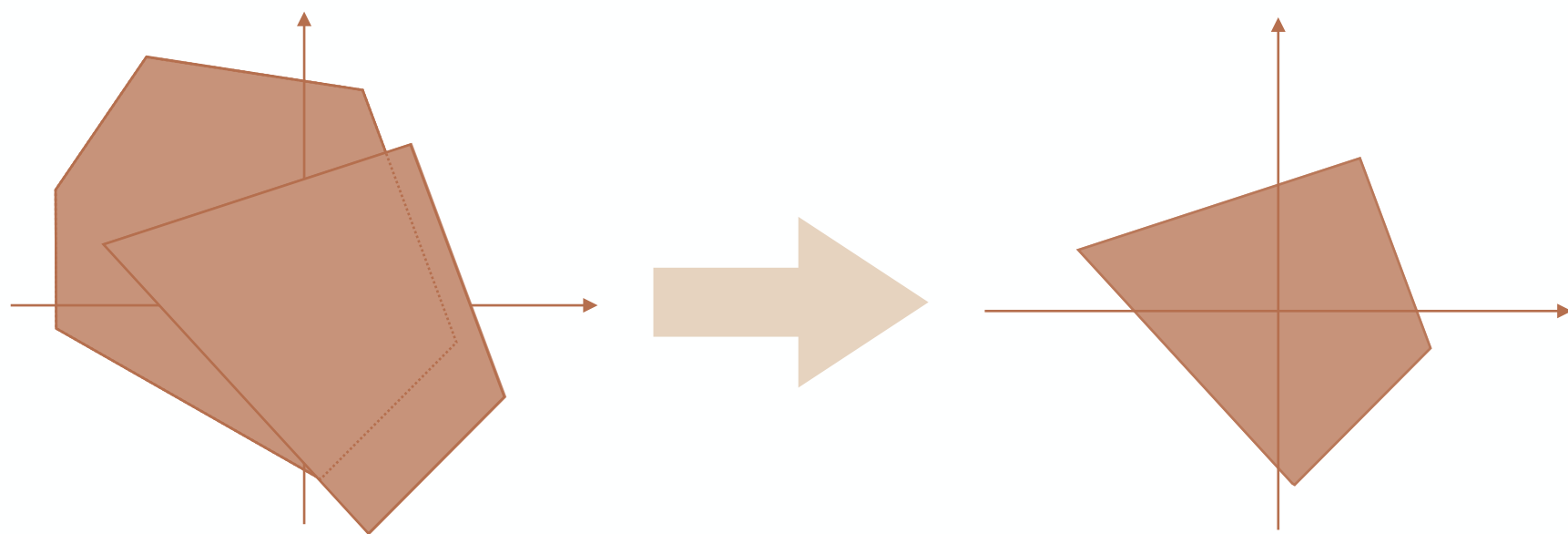
A set of **convex polyhedra**

$\sqsubseteq_A$ inclusion check

$\sqcup_A$ convex hull



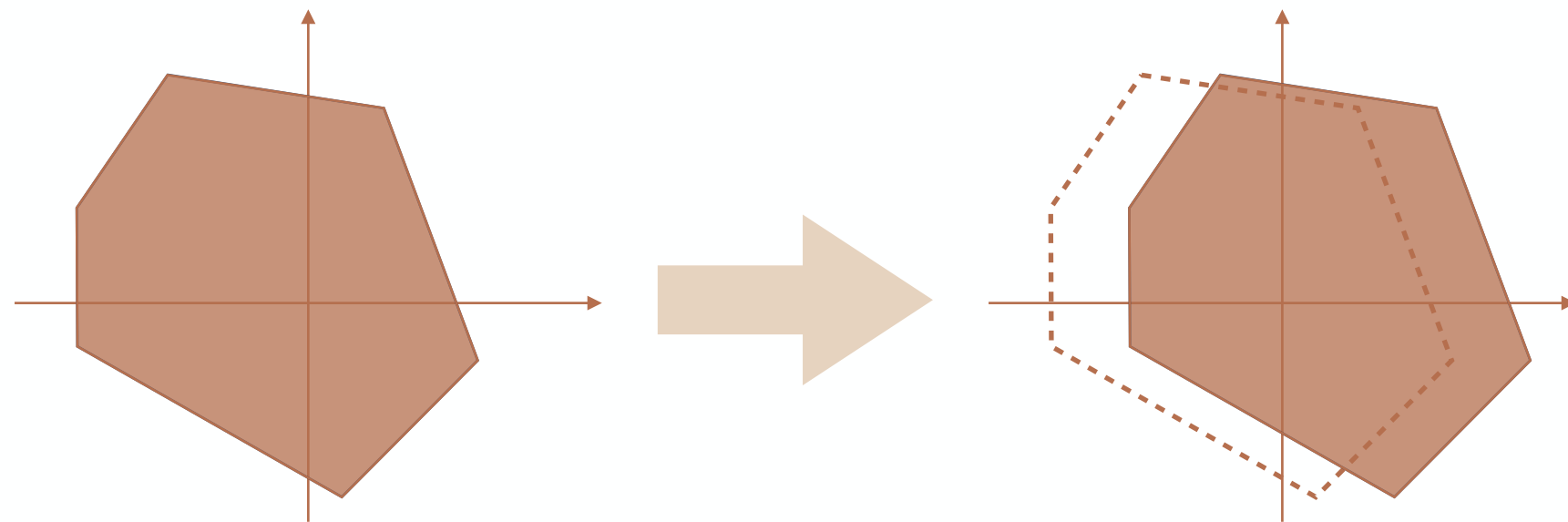
$\sqcap_A$ intersection



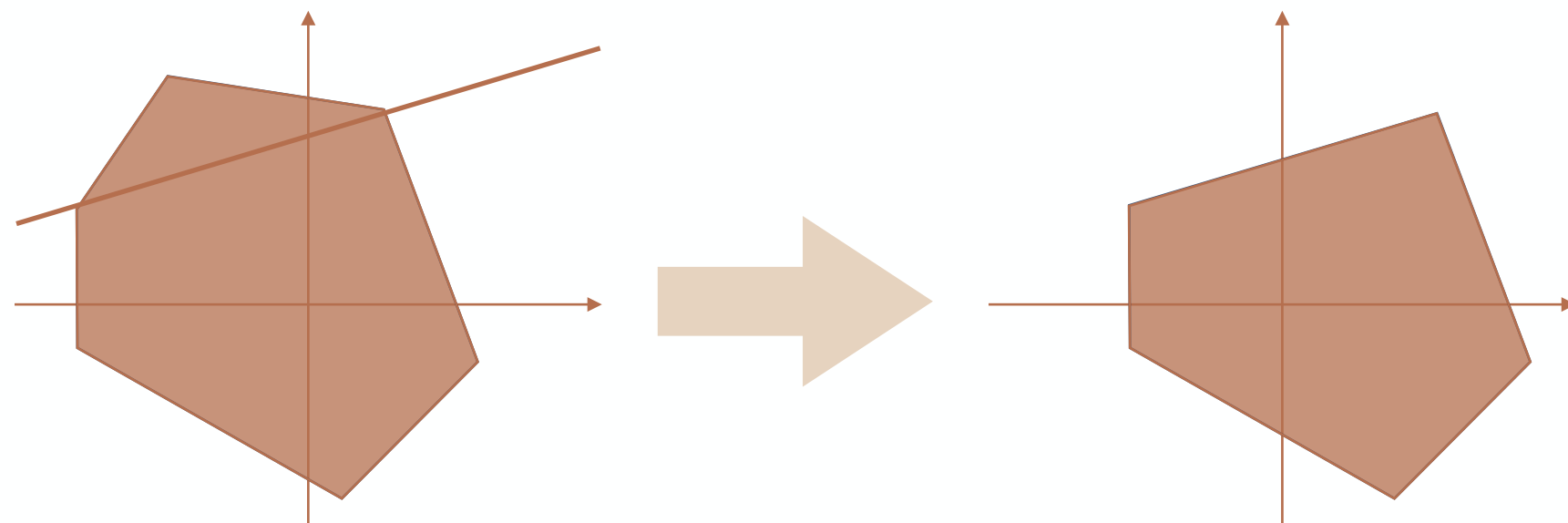
$\text{ASSIGN}_A[[X \leftarrow e]]a$ affine transformation

Example

$\text{ASSIGN}_A[[X \leftarrow X + 1]]a$



$\text{FILTER}_A[[e]]a$ intersection with e



∇_A remove unstable constraints

Polyhedra Static Analysis

$$S \stackrel{\text{def}}{=} \{ \langle \ell, \rho \rangle \in \Sigma \mid \ell \in \mathcal{L}, \rho \in \mathcal{E}, \rho(r) \geq 0 \}$$

$$\mathcal{R}(I) \subseteq \gamma(\mathcal{R}^\sharp(I)) \subseteq S$$



```

1
a ← [0, +∞]
2
b ← [0, +∞]
3
q ← 0
4
r ← a
5
while 6(r ≥ b) do
7
  r ← r - b
8
  q ← q + 1
9
done
10

```

```

3: a ≥ 0 ∧ b ≥ 0
4: a ≥ 0 ∧ b ≥ 0 ∧ q = 0
5: a ≥ 0 ∧ b ≥ 0 ∧ q = 0 ∧ r = a
6: a ≥ 0 ∧ b ≥ 0 ∧ q = 0 ∧ r = a
7: a ≥ 0 ∧ b ≥ 0 ∧ q = 0 ∧ r ≥ b
8: a ≥ 0 ∧ b ≥ 0 ∧ q = 0 ∧ r ≥ 0
9: a ≥ 0 ∧ b ≥ 0 ∧ q = 1 ∧ r ≥ 0
6: a ≥ 0 ∧ b ≥ 0 ∧ 0 ≤ q ≤ 1 ∧ r ≥ 0
7: a ≥ 0 ∧ b ≥ 0 ∧ 0 ≤ q ≤ 1 ∧ r ≥ b
8: a ≥ 0 ∧ b ≥ 0 ∧ 0 ≤ q ≤ 1 ∧ r ≥ 0
9: a ≥ 0 ∧ b ≥ 0 ∧ 1 ≤ q ≤ 2 ∧ r ≥ 0
6: a ≥ 0 ∧ b ≥ 0 ∧ q ≥ 0 ∧ r ≥ 0
⋮
10: a ≥ 0 ∧ b ≥ 0 ∧ q ≥ 0 ∧ b > r ≥ 0

```

Static Analysis by Abstract Interpretation

Static Analyzers

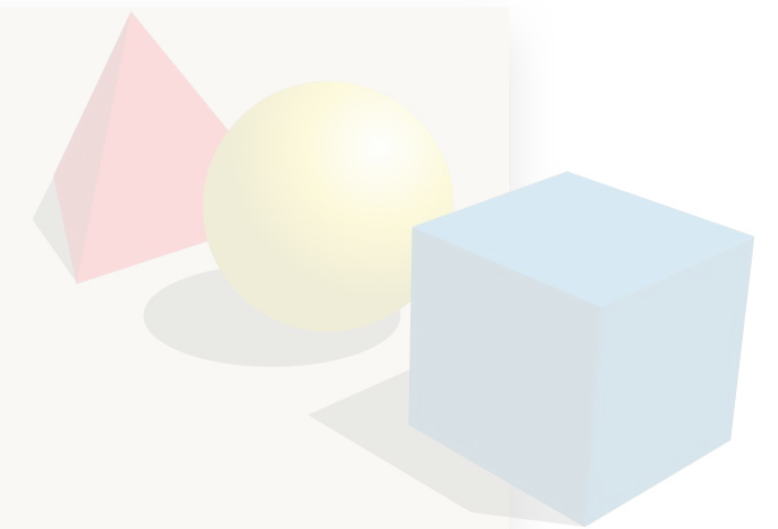
practical tools

targeting specific programs



abstract semantics, abstract domains

algorithmic approaches to decide program properties



concrete semantics

mathematical models of the program behavior



Astrée

ENS / AbsInt



Who uses Astrée?

Since 2003, Airbus France has been using Astrée in the development of safety-critical software for various aircraft series, including the A380.

In 2018, Bosch Automotive Steering replaced their legacy tools with Astrée and RuleChecker, resulting in significant savings thanks to faster analyses, higher accuracy, and optimized licensing and support costs.

Framatome employs Astrée for verification of their safety-critical TELEPERM XS platform that is used for engineering, testing, commissioning, operating and troubleshooting nuclear reactors.

The global automotive supplier Helbako in Germany is using Astrée to guarantee that no runtime errors can occur in their electronic control software and to demonstrate MISRA compliance of the code.

In 2008, Astrée proved the absence of any runtime errors in a C version of the automatic docking software of the Jules Verne Automated Transfer Vehicle, enabling ESA to transport payloads to the International Space Station.

A world leader in motors and ventilators for air-conditioning and refrigeration systems, ebm-papst is using Astrée for fully automatic continuous verification of safety-critical interrupt-driven control software for commutating high-efficiency EC motors for ventilator systems.

AIRBUS

BOSCH

framatome

HELBAKO

esa

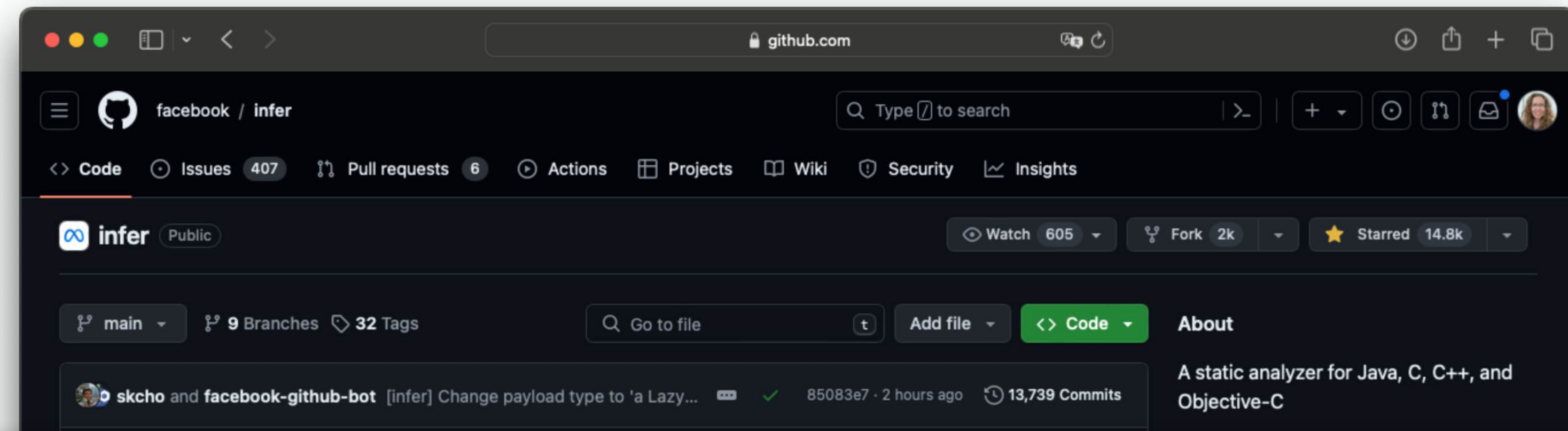
ebmpapst

Exploitation license of Astrée

Starting Dec. 2009, Astrée is available from AbsInt Angewandte Informatik **AbsInt** (www.absint.de/astree/).

Infer

Facebook / Meta



Who Uses Infer?

AdaCore



facebook



moz://a



sonatype



UBER



CodeAI

JD.com

Marks and
Spencer

Money Lover

Netcetera

OLA

Sky

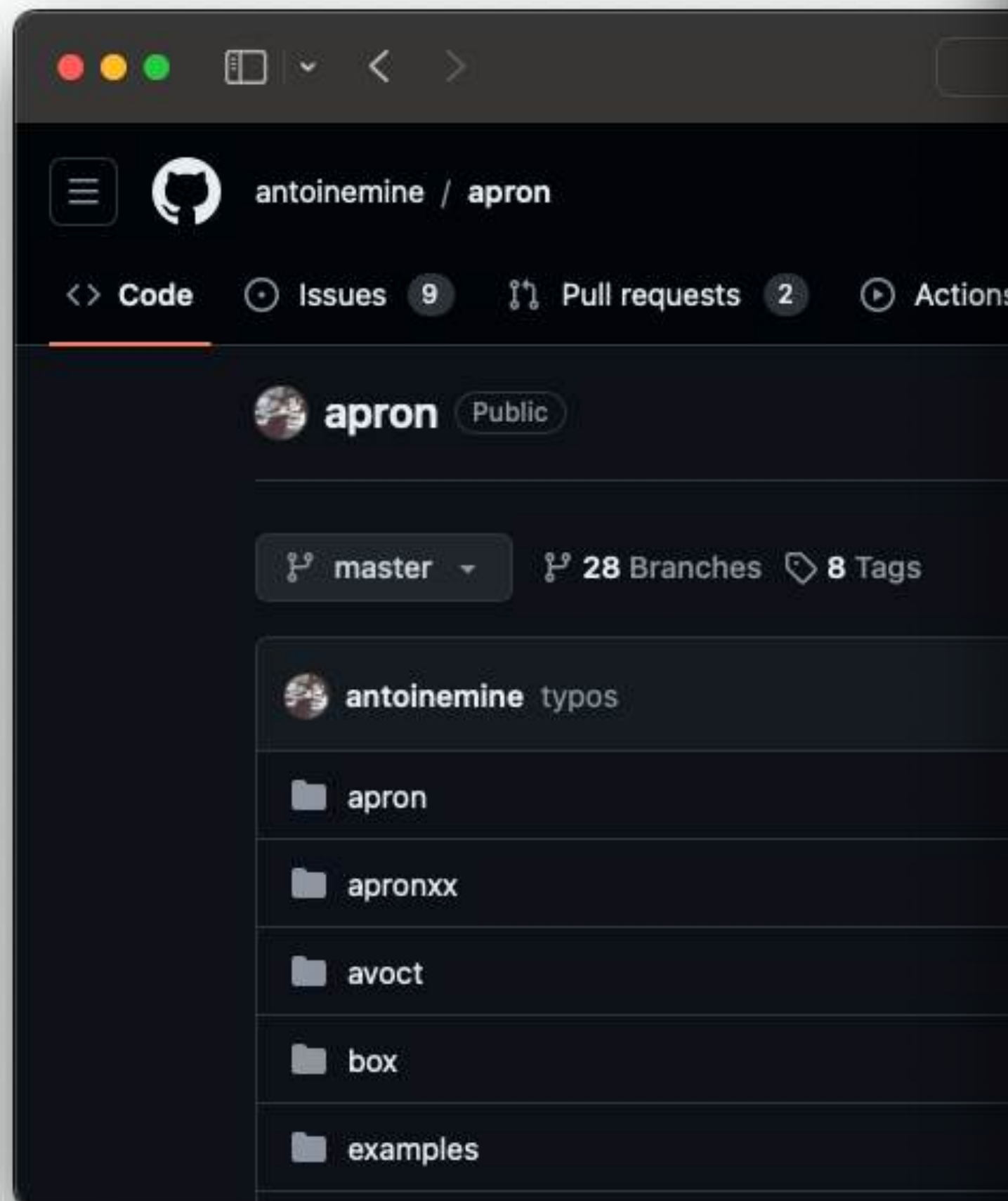
Tile

Vuo

wolfSSL

Apron

Library



APRON

Numerical Abstract Domain Library

Introduction | Contents | **Code** | **API Documentation** | Resources

The diagram illustrates the Apron architecture. On the left, a 'Typical Static Analyser Program' box contains a 'Front-end' box, which points to 'Semantic Equations'. 'Semantic Equations' points to a 'Solver' box, which in turn points to an 'Abstract Domain' box. On the right, a large box labeled 'Underlying libraries & abstract domains' contains three sub-boxes: 'box' (with 'intervals' and 'octagons'), 'NewPolka' (with 'convex polyhedra' and 'linear equalities'), and 'PPL + Wrapper' (with 'convex polyhedra' and 'linear congruences'). Arrows from these sub-boxes point to a 'Developer interface' box, which contains a table of datatypes and their semantics. The 'Developer interface' box points to a 'User interface' box.

Datatypes	Semantics: $A \xrightarrow{\gamma} \rho(\mathbb{R}^n)$
Coefficients	dimensions and space dimensionality
Expressions	
Constraints	Variables and Environments
Generators	Semantics: $A \xrightarrow{\gamma} (V \rightarrow \mathbb{R})$
Abs. values	

Introduction

Apron is a library to represent properties of numeric variables, such as variable bounds or linear relations between variables, and to manipulate these properties through semantic operations, such as variable assignments, tests, conjunctions, entailment.

Apron is intended to be used in static program analyzers, in order to infer invariants of numeric variables, i.e., properties that hold for all executions of a program. It is based on the theory of Abstract Interpretation.

The library is open-source, and [hosted on GitHub](#).

Static Analysis of Liveness Properties

The Art of Losing Precision

No Surprises, Please

What Could
Possibly Go Right?

It's Complicated

Trace Properties

$$T \in \mathcal{P}(\Sigma^\infty)$$

Liveness Properties = “Something Good Eventually Happens”

Example

- Termination: $T \stackrel{\text{def}}{=} \Sigma^*$

Liveness Property Verification

- T cannot be **verified** by **testing**



$$\mathcal{M} \subseteq T$$

- falsifying T requires finding **an infinite execution not in T**

Liveness Properties

- **Guarantee Properties**
“something good eventually happens at least once”
 - Example: Program Termination
- **Recurrence Properties**
“something good eventually happens infinitely often”
 - Example: Starvation Freedom



Zohar Manna

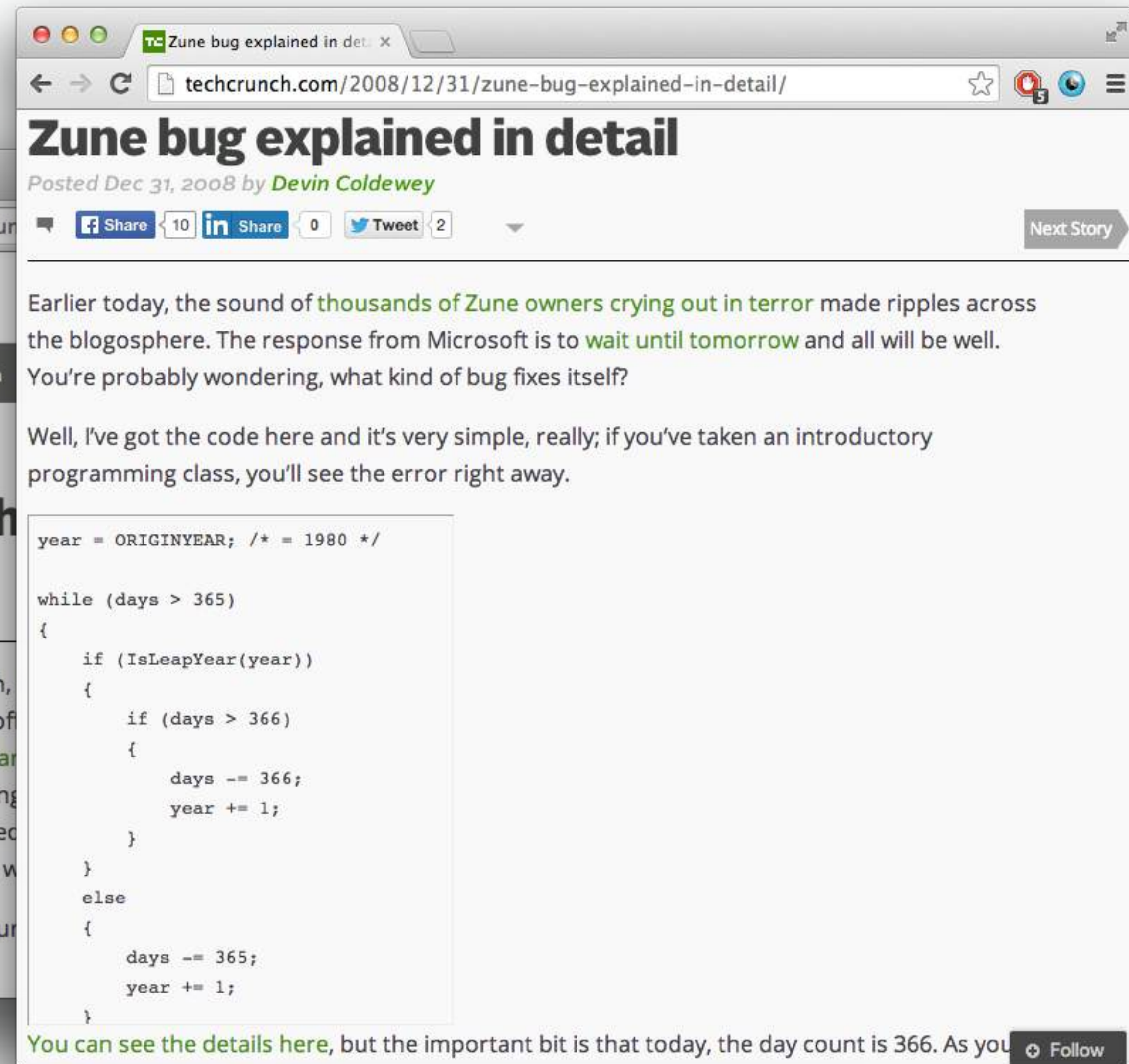
Amir Pnueli

Program Termination

The Zune Bug

31 December 2008

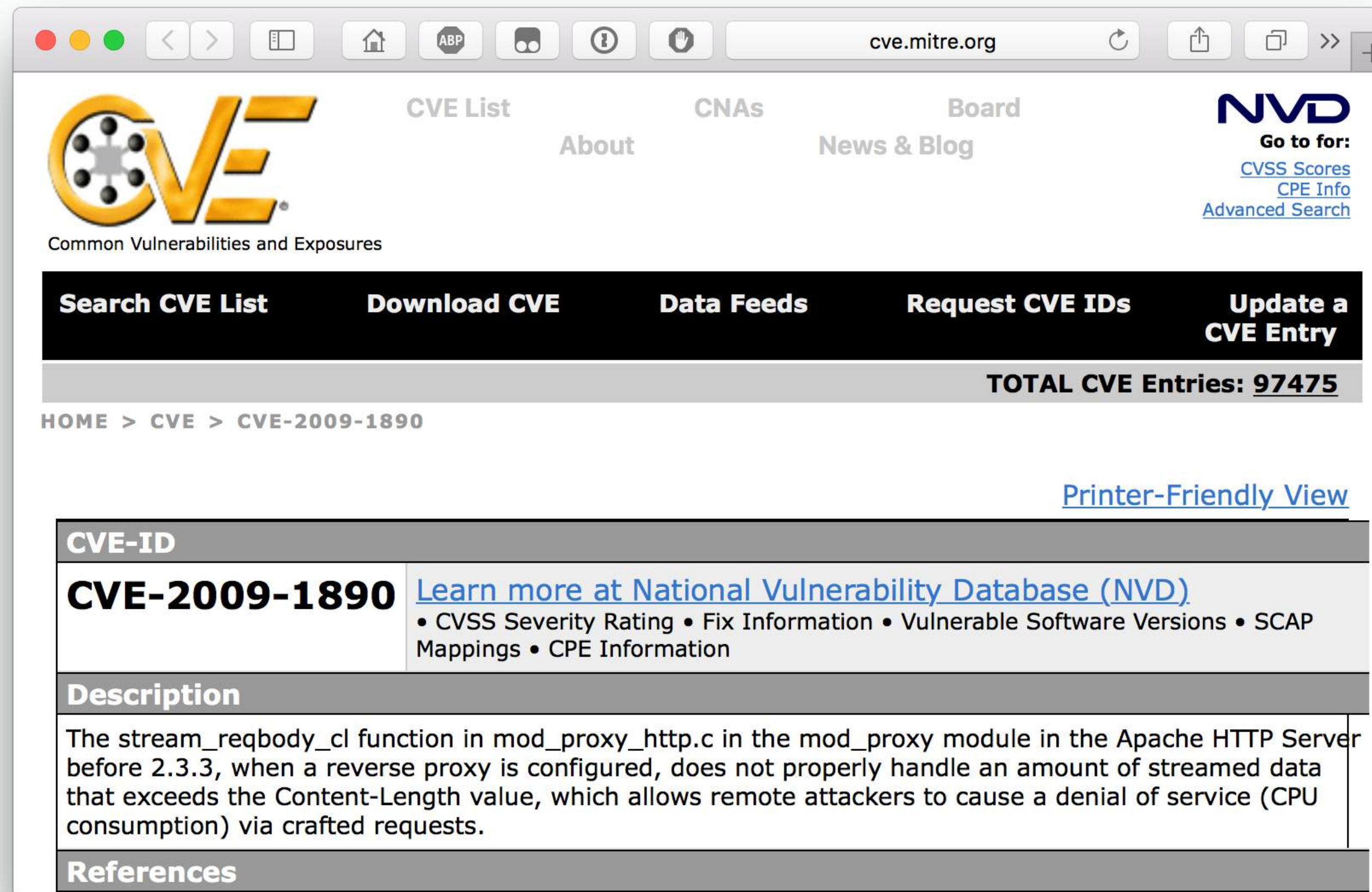
unresponsive
systems



Apache HTTP Server

Versions <2.3.3

denial-of-service
attacks



The screenshot shows the CVE MITRE website interface. At the top, there's a navigation bar with links like 'CVE List', 'CNAs', 'Board', 'About', and 'News & Blog'. The main header features the CVE logo and the text 'Common Vulnerabilities and Exposures'. Below this, there's a black bar with white text for 'Search CVE List', 'Download CVE', 'Data Feeds', 'Request CVE IDs', and 'Update a CVE Entry'. A status bar indicates 'TOTAL CVE Entries: 97475'. The breadcrumb trail shows 'HOME > CVE > CVE-2009-1890'. A 'Printer-Friendly View' link is present. The main content area is divided into sections: 'CVE-ID' with the entry 'CVE-2009-1890' and a link to 'Learn more at National Vulnerability Database (NVD)', 'Description' with a detailed text about a denial of service vulnerability in the Apache HTTP Server, and 'References'.

CVE-ID

CVE-2009-1890 [Learn more at National Vulnerability Database \(NVD\)](#)

- CVSS Severity Rating • Fix Information • Vulnerable Software Versions • SCAP Mappings • CPE Information

Description

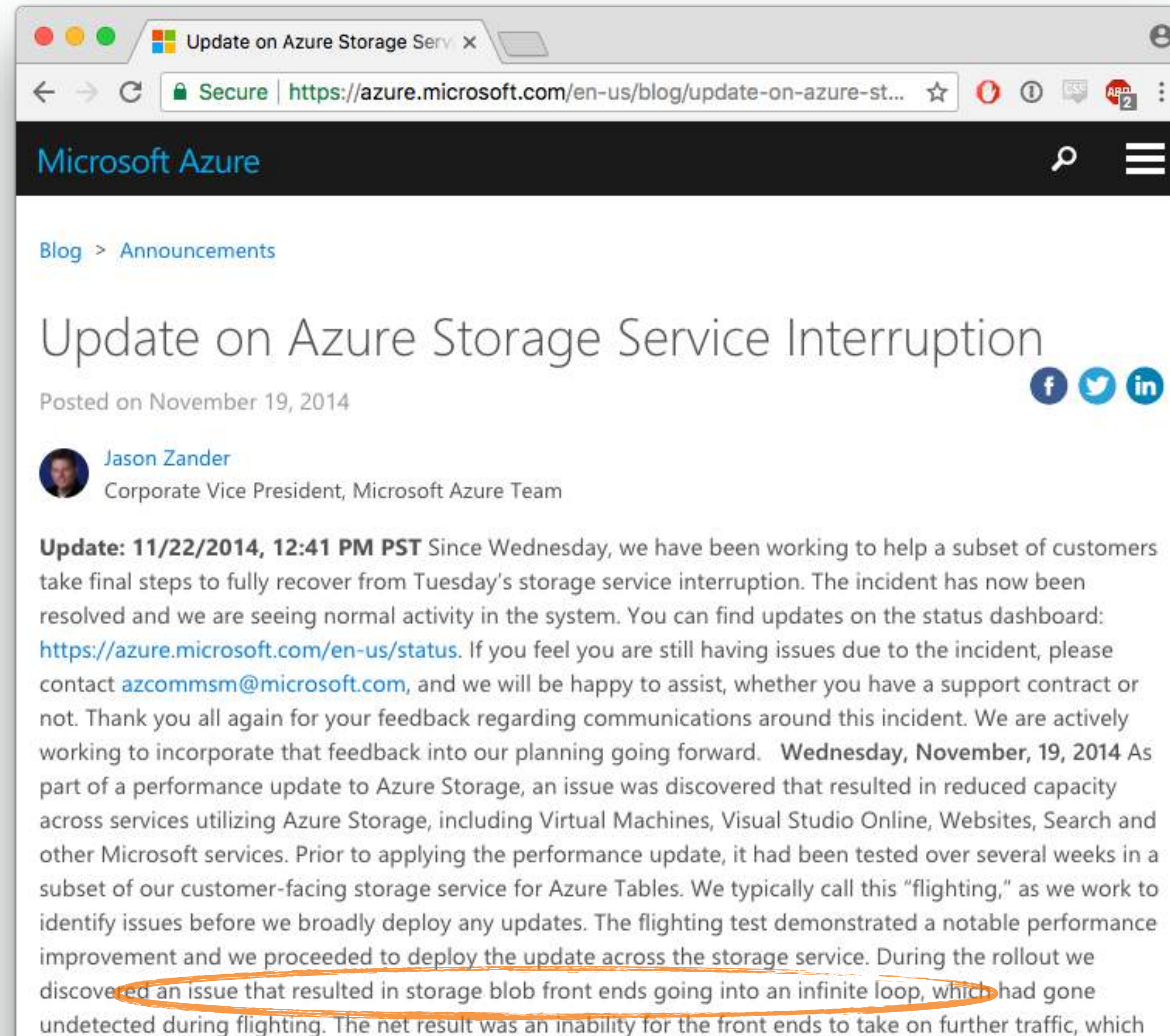
The stream_reqbody_cl function in mod_proxy_http.c in the mod_proxy module in the Apache HTTP Server before 2.3.3, when a reverse proxy is configured, does not properly handle an amount of streamed data that exceeds the Content-Length value, which allows remote attackers to cause a denial of service (CPU consumption) via crafted requests.

References

Azure Storage Service

19 November 2014

service
interruptions



The screenshot shows a web browser window with the title 'Update on Azure Storage Serv' and the URL 'https://azure.microsoft.com/en-us/blog/update-on-azure-st...'. The page is from the Microsoft Azure blog, specifically the 'Announcements' section. The post is titled 'Update on Azure Storage Service Interruption' and was posted on November 19, 2014, by Jason Zander, Corporate Vice President of the Microsoft Azure Team. The post content includes an update from 11/22/2014 at 12:41 PM PST, stating that the storage service interruption has been resolved. It also mentions a performance update to Azure Storage that caused the issue, which was discovered during a 'flighting' test. The text describes how the update resulted in storage blob front ends going into an infinite loop, which was undetected during the test. The phrase 'an issue that resulted in storage blob front ends going into an infinite loop, which' is circled in orange in the original image.

Microsoft Azure

Blog > Announcements

Update on Azure Storage Service Interruption

Posted on November 19, 2014

 **Jason Zander**
Corporate Vice President, Microsoft Azure Team

Update: 11/22/2014, 12:41 PM PST Since Wednesday, we have been working to help a subset of customers take final steps to fully recover from Tuesday's storage service interruption. The incident has now been resolved and we are seeing normal activity in the system. You can find updates on the status dashboard: <https://azure.microsoft.com/en-us/status>. If you feel you are still having issues due to the incident, please contact azcommsm@microsoft.com, and we will be happy to assist, whether you have a support contract or not. Thank you all again for your feedback regarding communications around this incident. We are actively working to incorporate that feedback into our planning going forward. **Wednesday, November 19, 2014** As part of a performance update to Azure Storage, an issue was discovered that resulted in reduced capacity across services utilizing Azure Storage, including Virtual Machines, Visual Studio Online, Websites, Search and other Microsoft services. Prior to applying the performance update, it had been tested over several weeks in a subset of our customer-facing storage service for Azure Tables. We typically call this "flighting," as we work to identify issues before we broadly deploy any updates. The flighting test demonstrated a notable performance improvement and we proceeded to deploy the update across the storage service. During the rollout we discovered an issue that resulted in storage blob front ends going into an infinite loop, which had gone undetected during flighting. The net result was an inability for the front ends to take on further traffic, which

Potential and Definite Termination

Potential Termination

Definition

A program with trace semantics
 $\mathcal{M} \in \mathcal{P}(\Sigma^\infty)$ **may terminate**
if and only if $\mathcal{M} \cap \Sigma^* \neq \emptyset$

Definite Termination

Definition

A program with trace semantics
 $\mathcal{M} \in \mathcal{P}(\Sigma^\infty)$ **must terminate**
if and only if $\mathcal{M} \subseteq \Sigma^*$

In absence of non-determinism, potential and definite termination coincide

Definite Termination

Ranking Functions



Alan Turing



Robert W. Floyd

Definition

Given a transition system $\langle \Sigma, \tau \rangle$, a **ranking function** is a partial function $f: \Sigma \rightarrow \mathcal{W}$ from the set of program states Σ into a well-ordered set $\langle \mathcal{W}, \leq \rangle$ whose value *strictly decreases* through transitions between states, that is,
$$\forall \sigma, \sigma' \in \text{dom}(f): (\sigma, \sigma') \in \tau \Rightarrow f(\sigma') < f(\sigma)$$

The best known *well-ordered sets* are **naturals** $\langle \mathbb{N}, \leq \rangle$ and **ordinals** $\langle \mathbb{O}, \leq \rangle$

Ranking Functions

Example

```
1  $x \leftarrow [-\infty, +\infty]$   
while 2  $(1 - x < 0)$  do  
    3  $x \leftarrow x - 1$   
done4
```

$\Sigma \stackrel{\text{def}}{=} \{\mathbf{1}, \mathbf{2}, \mathbf{3}, \mathbf{4}\} \times \mathcal{E}$

$\tau \stackrel{\text{def}}{=} \{((\mathbf{1}, \rho), (\mathbf{2}, \rho[X \mapsto v])) \mid \rho \in \mathcal{E}, v \in \mathbb{Z}\}$
 $\cup \{((\mathbf{2}, \rho), (\mathbf{3}, \rho)) \mid \mid \rho \in \mathcal{E}, \exists v \in E[[1 - x]]\rho : v < 0\}$
 $\cup \{((\mathbf{3}, \rho), (\mathbf{2}, \rho[X \mapsto v])) \mid \rho \in \mathcal{E}, v \in E[[x - 1]]\rho\}$
 $\cup \{((\mathbf{2}, \rho), (\mathbf{4}, \rho)) \mid \mid \rho \in \mathcal{E}, \exists v \in E[[1 - x]]\rho : v \not< 0\}$

Ranking Functions

Example

```
1x ← [-∞, +∞]  
while 2(1 - x < 0) do  
    3x ← x - 1  
done4
```

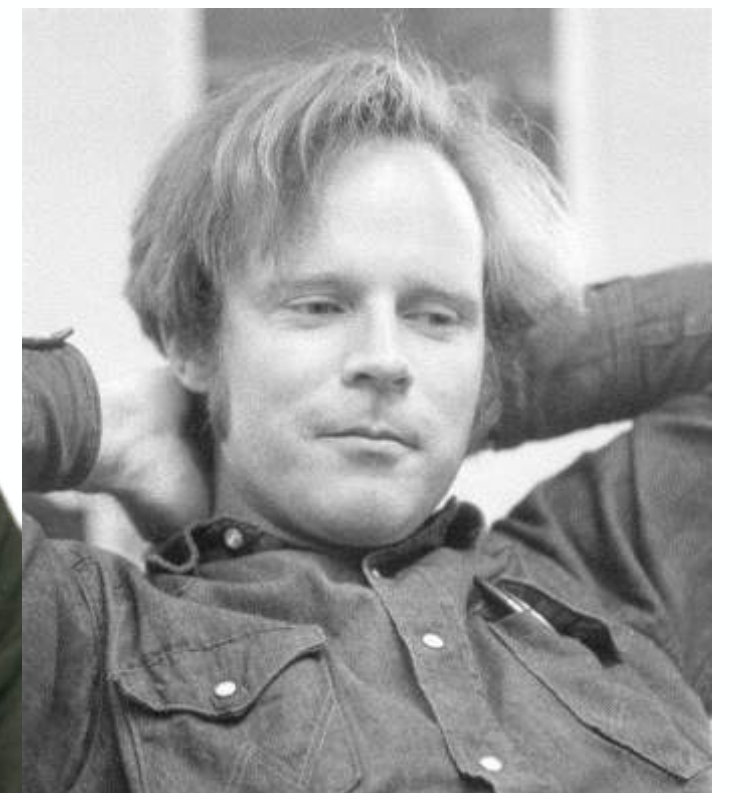
Most obvious ranking function:
a mapping $f: \Sigma \rightarrow \mathbb{O}$ from each program state to
(an upper bound on) the number of steps until termination

We define $f: \Sigma \rightarrow \mathbb{O}$ by partitioning with respect to the program control points, i.e., $f: \mathcal{L} \rightarrow (\mathcal{E} \rightarrow \mathbb{O})$

$$\begin{aligned} f(\mathbf{4}) &\stackrel{\text{def}}{=} \lambda\rho.0 \\ f(\mathbf{2}) &\stackrel{\text{def}}{=} \lambda\rho. \begin{cases} 1 & 1 - \rho(x) \not< 0 \\ 2\rho(x) - 1 & 1 - \rho(x) < 0 \end{cases} \\ f(\mathbf{3}) &\stackrel{\text{def}}{=} \lambda\rho. \begin{cases} 2 & 2 - \rho(x) \not< 0 \\ 2\rho(x) - 2 & 2 - \rho(x) < 0 \end{cases} \\ f(\mathbf{1}) &\stackrel{\text{def}}{=} \lambda\rho.\omega \end{aligned}$$



Alan Turing



Robert W. Floyd

Static Termination Analysis

3-Step Recipe

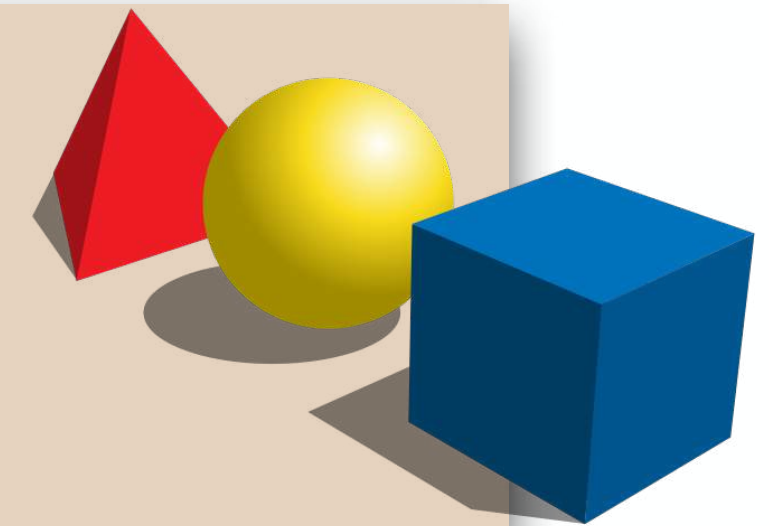
practical tools

targeting specific programs



abstract semantics, abstract domains

algorithmic approaches to decide program properties



concrete semantics

mathematical models of the program behavior



Static Termination Analysis

Program Termination Semantics

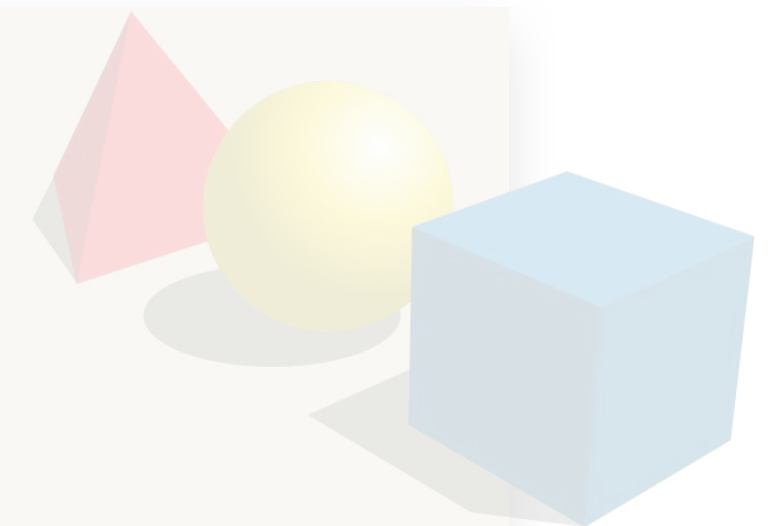
practical tools

targeting specific programs



abstract semantics, abstract domains

algorithmic approaches to decide program properties

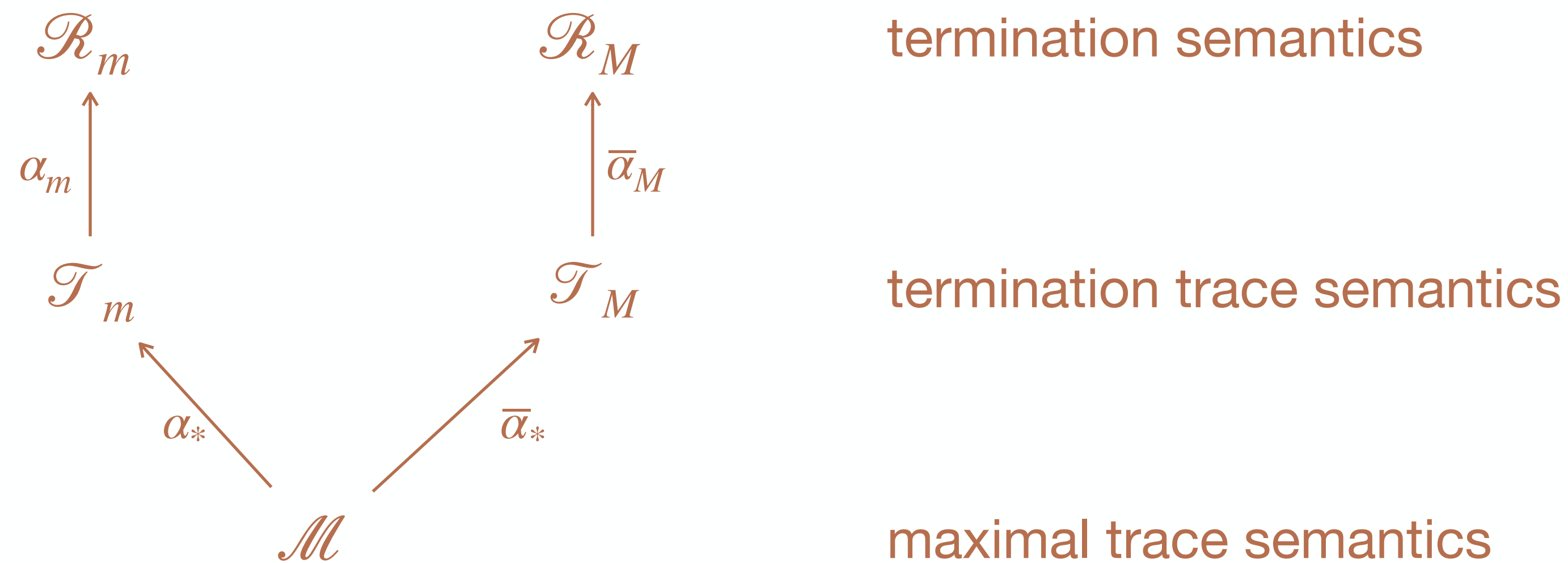


concrete semantics

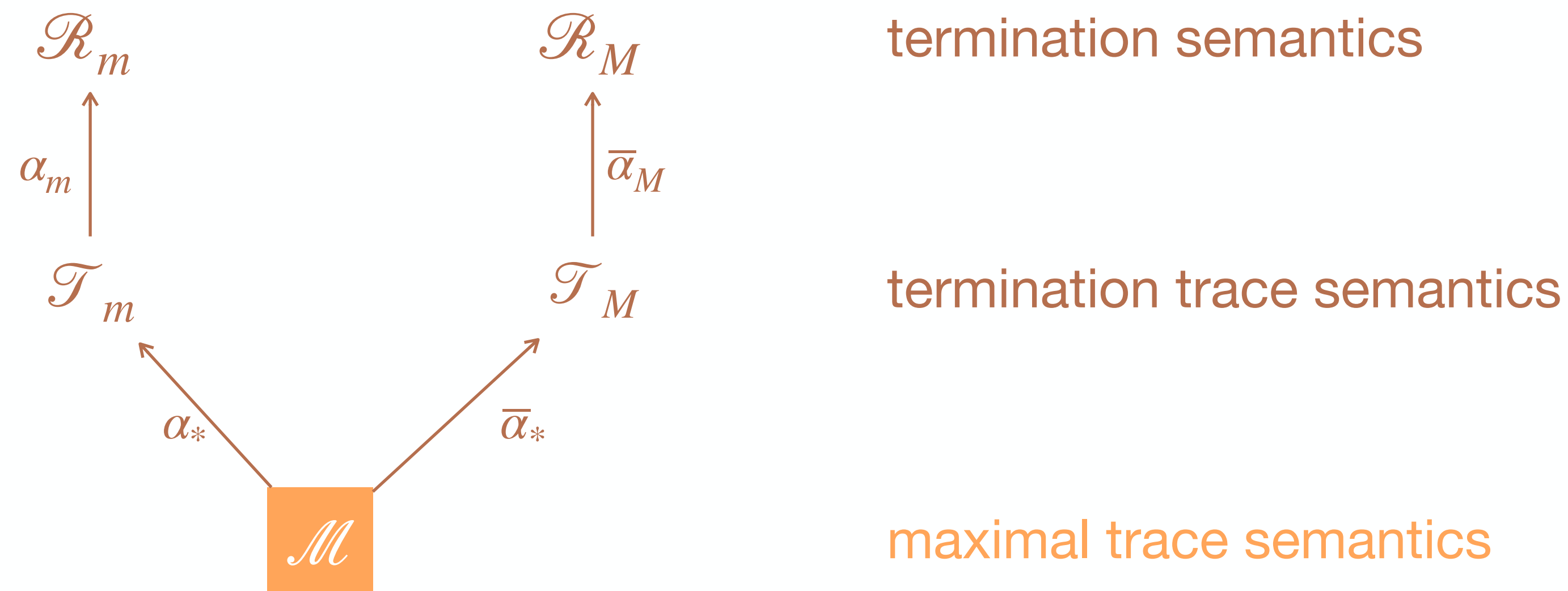
mathematical models of the program behavior



(Yet Another) Hierarchy of Semantics



(Yet Another) Hierarchy of Semantics

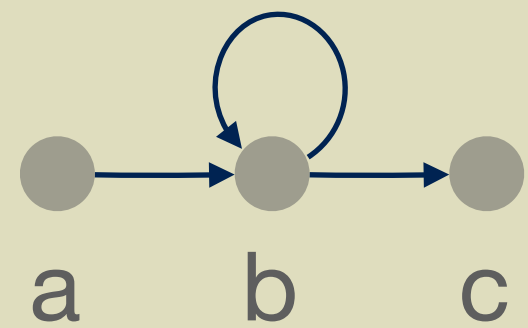


Maximal Trace Semantics

Least Fixpoint Formulation

$$\mathcal{M} = \text{lfp}_{\Sigma^\omega} F$$

$$F(T) \stackrel{\text{def}}{=} \mathcal{B} \cup \tau; T$$



- $F^0(\emptyset) = \Sigma^\omega$
- $F^1(F^0) = \{c\} \cup \{ab\Sigma^\omega, bb\Sigma^\omega, bc\Sigma^\omega\}$
- $F^2(F^1) = \{bc, c\} \cup \{abb\Sigma^\omega, bbb\Sigma^\omega, abc\Sigma^\omega, bbc\Sigma^\omega\}$
- $F_p^3(F_p^2) = \{abc, bbc, bc, c\} \cup \{abbb\Sigma^\omega, bbbb\Sigma^\omega, abbc\Sigma^\omega, bbbc\Sigma^\omega\}$

$$\mathcal{M} = \{ab^i c, b^i c, c \mid i \geq 1\} \cup \{ab^\omega, b^\omega\}$$

Maximal Trace Semantics

Example

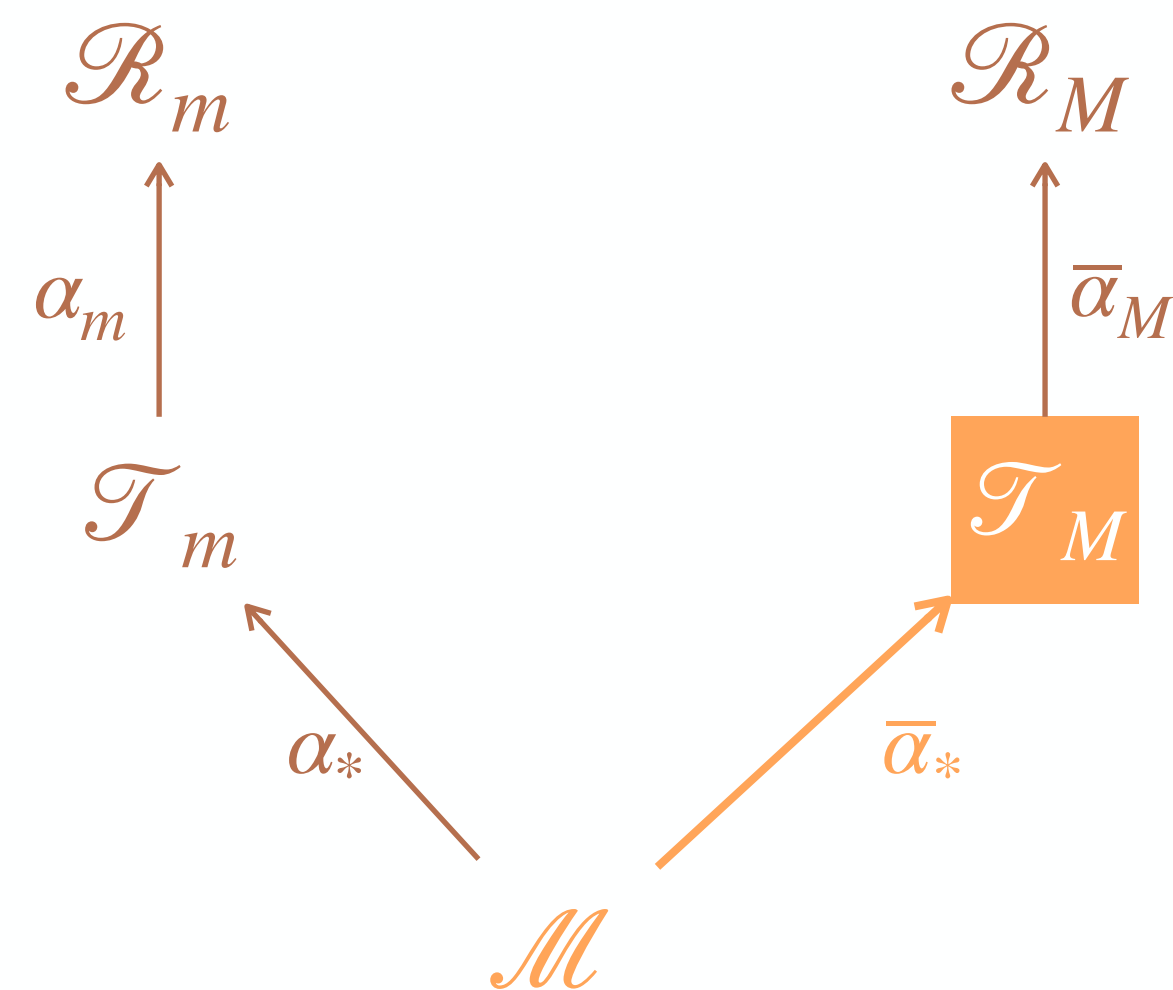
while ¹ $([-\infty, +\infty] \neq 0)$ do
 ²skip
done³

$$\Sigma \stackrel{\text{def}}{=} \{\mathbf{1}, \mathbf{2}, \mathbf{3}\} \times \mathcal{E}$$

$$\tau \stackrel{\text{def}}{=} \{((\mathbf{1}, \rho), (\mathbf{2}, \rho)) \mid \rho \in \mathcal{E}\} \\ \cup \{((\mathbf{2}, \rho), (\mathbf{1}, \rho)) \mid \rho \in \mathcal{E}\} \\ \cup \{((\mathbf{1}, \rho), (\mathbf{3}, \rho)) \mid \rho \in \mathcal{E}\}$$

$$\mathcal{M} \stackrel{\text{def}}{=} \{(\mathbf{1}, \rho)(\mathbf{2}, \rho)^*(\mathbf{3}, \rho) \mid \rho \in \mathcal{E}\} \cup \{(\mathbf{1}, \rho)(\mathbf{2}, \rho)^\omega \mid \rho \in \mathcal{E}\}$$

(Yet Another) Hierarchy of Semantics



termination semantics

definite termination trace semantics

maximal trace semantics

Definite Termination Trace Semantics

Definite Termination Abstraction

$$\langle \mathcal{P}(\Sigma^\infty), \sqsubseteq \rangle \xrightarrow{\bar{\alpha}_*} \langle \mathcal{P}(\Sigma^*), \subseteq \rangle$$

$$\bar{\alpha}_*(T) \stackrel{\text{def}}{=} \{t \in T \cap \Sigma^* \mid \text{nhdb}(t, T \cap \Sigma^\omega) = \emptyset\}$$

$$\text{where } \text{nhdb}(t, T) \stackrel{\text{def}}{=} \{t' \in T \mid \text{pf}(t) \cap \text{pf}(t') \neq \emptyset\}$$

$$\text{pf}(t) \stackrel{\text{def}}{=} \{t' \in \Sigma^\infty \setminus \{\epsilon\} \mid \exists t'' \in \Sigma^\infty : t = t' \cdot t''\}$$

Example:

$$\alpha_*(\{ab, aba, bb, ba^\omega\}) = \{ab, aba\} \text{ since } \text{pf}(bb) \cap \text{pf}(ba^\omega) = \{b\} \neq \emptyset$$

Definite Termination Trace Semantics

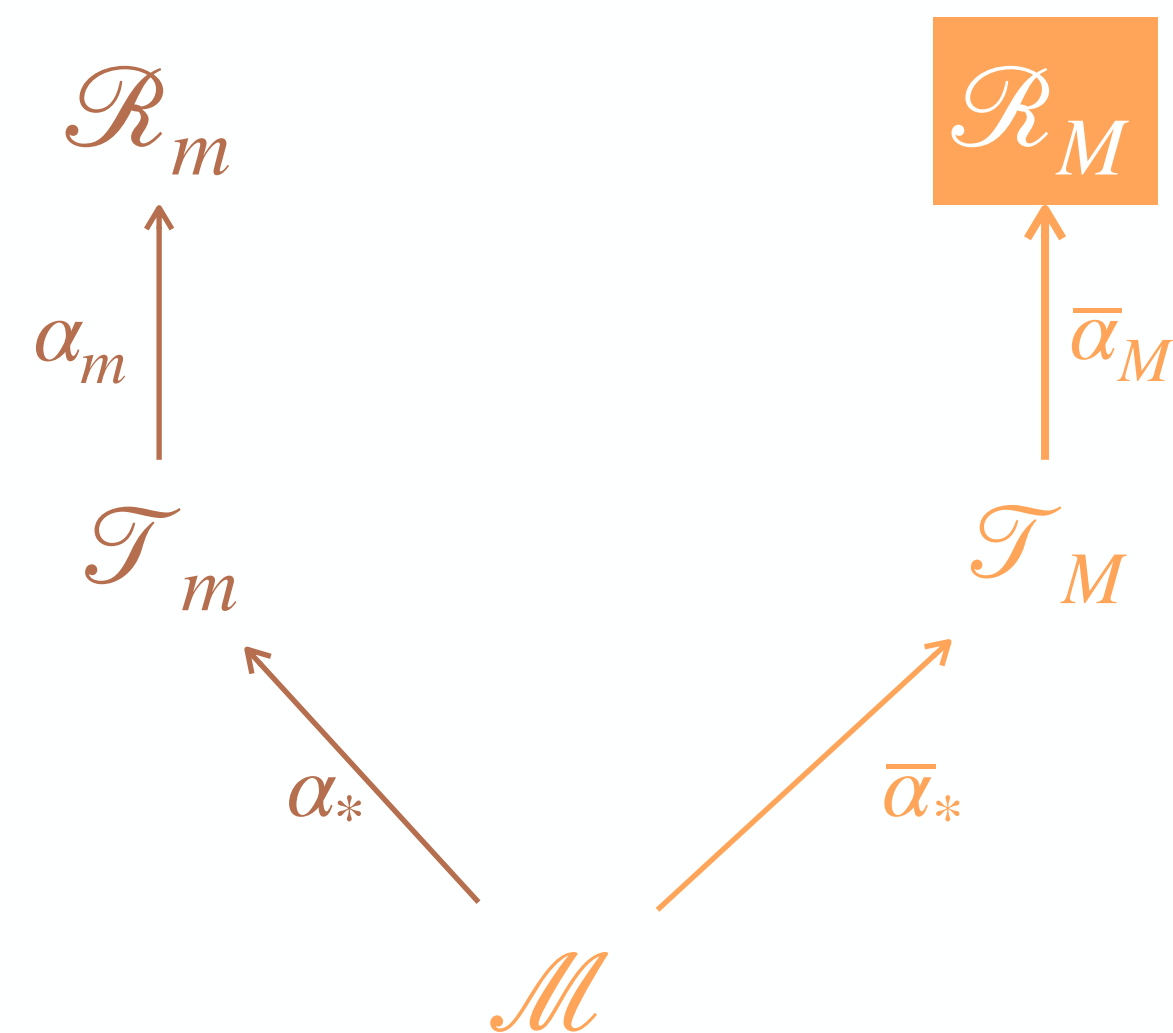
Example

while ¹ $([-\infty, +\infty] \neq 0)$ **do**
 ²**skip**
done³

$$\mathcal{M} \stackrel{\text{def}}{=} \{(\mathbf{1}, \rho)(\mathbf{2}, \rho)^*(\mathbf{3}, \rho) \mid \rho \in \mathcal{E}\} \cup \{(\mathbf{1}, \rho)(\mathbf{2}, \rho)^\omega \mid \rho \in \mathcal{E}\}$$

$$\mathcal{T}_M \stackrel{\text{def}}{=} \emptyset$$

(Yet Another) Hierarchy of Semantics



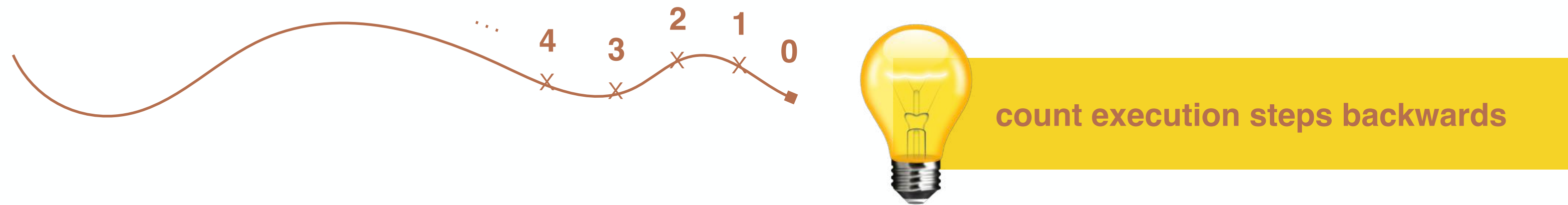
definite termination semantics

termination trace semantics

maximal trace semantics

Definite Termination Semantics

Definite Ranking Abstraction



$$\langle \mathcal{P}(\Sigma^*), \subseteq \rangle$$

$$\langle \Sigma \rightarrow \mathbb{O}, \leq \rangle$$

α_M

$$f_1 \leq f_2 \stackrel{\text{def}}{=} \text{dom}(f_1) \subseteq \text{dom}(f_2) \wedge \forall x \in \text{dom}(f_1): f_1(x) \leq f_2(x)$$

$$\bar{\alpha}_M(T) \stackrel{\text{def}}{=} \bar{\alpha}_V(\vec{\alpha}(T))$$

$$\text{where } \bar{\alpha}_V(\emptyset) \stackrel{\text{def}}{=} \emptyset$$

$$\bar{\alpha}_V(r)\sigma \stackrel{\text{def}}{=} \begin{cases} 0 & \forall \sigma' \in \Sigma: (\sigma, \sigma') \notin r \\ \sup\{\bar{\alpha}_V(r)\sigma' + 1 \mid \sigma' \in \text{dom}(\bar{\alpha}_V(r)) \wedge (\sigma, \sigma') \in r\} & \text{otherwise} \end{cases}$$

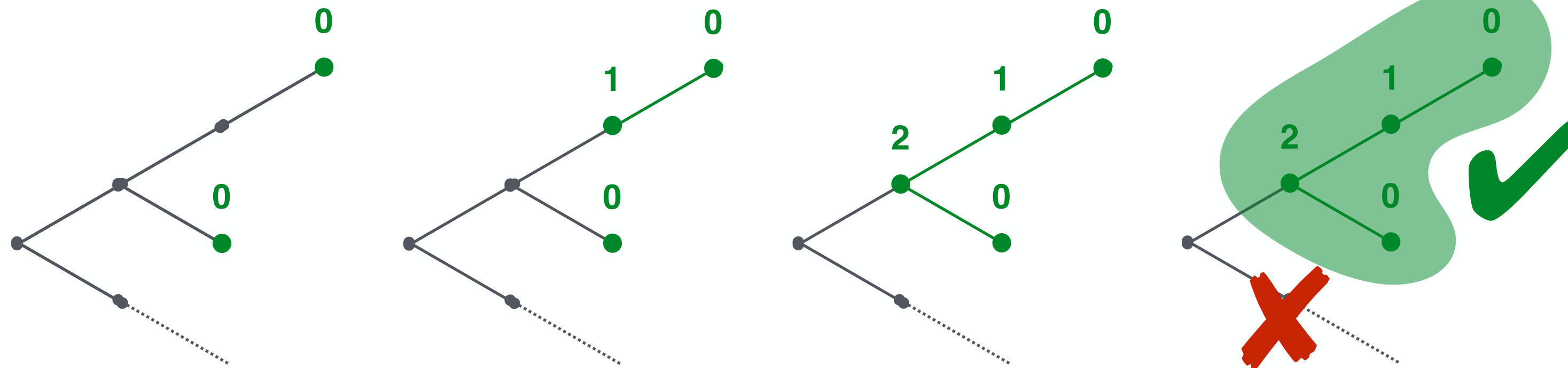
$$\vec{\alpha}(T) \stackrel{\text{def}}{=} \{(\sigma, \sigma') \in \Sigma \times \Sigma \mid \exists t \in \Sigma^*, t' \in \Sigma^\infty: t\sigma\sigma't' \in T\}$$

Definite Termination Semantics

Least Fixpoint Formulation

$$\mathcal{R}_M \stackrel{\text{def}}{=} \bar{\alpha}_M(\mathcal{T}_M) = \text{lfp}^{\leq} \bar{F}_M$$

$$\bar{F}_M(f)\sigma \stackrel{\text{def}}{=} \begin{cases} 0 & \sigma \in \mathcal{B} \\ \sup\{f(\sigma') + 1 \mid (\sigma, \sigma') \in \tau\} & \sigma \in \tilde{\text{pre}}_{\tau}(\text{dom}(f)) \\ \text{undefined} & \text{otherwise} \end{cases}$$



Theorem

A program **must terminate** for traces starting from a set of initial state I if and only if $I \subseteq \text{dom}(\mathcal{R}_M)$

Definite Termination Semantics

Denotational Formulation

We define the $\mathcal{R}_M: \Sigma \rightarrow \mathbb{O}$ by partitioning with respect to $\mathcal{L}$, i.e., $\mathcal{R}_M: \mathcal{L} \rightarrow (\mathcal{E} \rightarrow \mathbb{O})$.

Thus, for each program instruction *stmt*, we define a transformer $\mathcal{R}_M[\text{stmt}]: (\mathcal{E} \rightarrow \mathbb{O}) \rightarrow (\mathcal{E} \rightarrow \mathbb{O})$:

- $\mathcal{R}_M[X \leftarrow e]$
- $\mathcal{R}_M[\text{if } e \bowtie 0 \text{ then } s \text{ end}]$
- $\mathcal{R}_M[\text{while } e \bowtie 0 \text{ do } s \text{ done}]$
- $\mathcal{R}_M[s_1; s_2]$

```
stmt ::=  ${}^\ell X \leftarrow \text{expr}^\ell$   
        |  $\text{if } {}^\ell \text{expr} \bowtie 0 \text{ then } \text{stmt} \text{ end}^\ell$   
        |  $\text{while } {}^\ell \text{expr} \bowtie 0 \text{ do } \text{stmt} \text{ done}^\ell$   
        |  $\text{stmt}; \text{stmt}$ 
```

Definite Termination Semantics

$$\mathcal{R}_M[X \leftarrow e]$$

$$\mathcal{R}_M[X \leftarrow e]f \stackrel{\text{def}}{=} \lambda\rho. \begin{cases} \sup\{f(\rho[X \mapsto v]) + 1 \mid v \in E[e]\rho\} & \exists \bar{v} \in E[e]\rho \wedge \forall v \in E[e]\rho: \rho[X \mapsto v] \in \text{dom}(f) \\ \text{undefined} & \text{otherwise} \end{cases}$$

Example:

Let $\mathbb{V} = \{x\}$ and $f: \mathcal{E} \rightarrow \mathbb{O}$ defined as follows:

$$f(\rho) \stackrel{\text{def}}{=} \begin{cases} 2 & \rho(x) = 1 \\ 3 & \rho(x) = 2 \\ \text{undefined} & \text{otherwise} \end{cases}$$

We have

$$\mathcal{R}_M[x \leftarrow x + [1,2]]f \stackrel{\text{def}}{=} \lambda\rho. \begin{cases} 4 & \rho(x) = 0 \\ \text{undefined} & \text{otherwise} \end{cases}$$

Definite Termination Semantics

$\mathcal{R}_M[\text{if } e \bowtie 0 \text{ then } s \text{ end}]$

$$\mathcal{R}_M[\text{if } e \bowtie 0 \text{ then } s \text{ end}]f \stackrel{\text{def}}{=} \lambda\rho. \begin{cases} \textcircled{1} \\ \textcircled{2} \\ \textcircled{3} \\ \text{undefined} \quad \text{otherwise} \end{cases}$$

$$\textcircled{1} \quad \sup\{\mathcal{R}_M[s]f(\rho) + 1, f(\rho) + 1\} \quad \rho \in \text{dom}(\mathcal{R}_M[s]f) \cap \text{dom}(f) \wedge \exists v_1, v_2 \in E[e]\rho: v_1 \bowtie 0 \wedge v_2 \not\bowtie 0$$

$$\textcircled{2} \quad \mathcal{R}_M[s]f(\rho) + 1 \quad \rho \in \text{dom}(\mathcal{R}_M[s]f) \wedge \forall v \in E[e]\rho: v \bowtie 0$$

$$\textcircled{3} \quad f(\rho) + 1 \quad \rho \in \text{dom}(f) \wedge \forall v \in E[e]\rho: v \not\bowtie 0$$

Definite Termination Semantics

$\mathcal{R}_M[\text{if } e \bowtie 0 \text{ then } s \text{ end}]$ (continue)

Example:

Let $\mathbb{V} = \{x\}$ and $f: \mathcal{E} \rightarrow \mathbb{Q}$, and $\mathcal{R}_M[\llbracket s \rrbracket]f$ defined as follows:

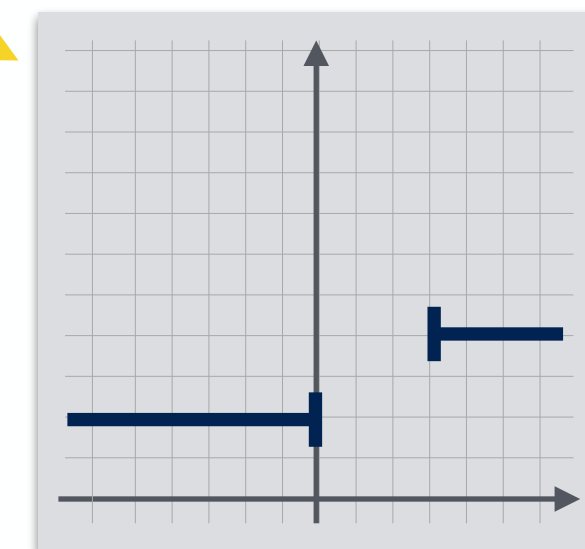
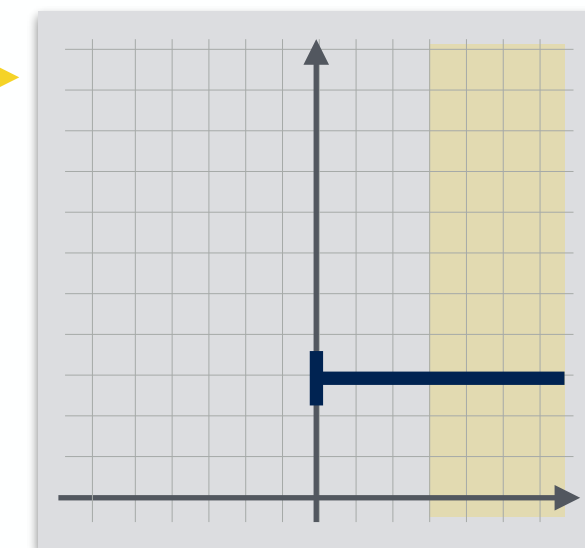
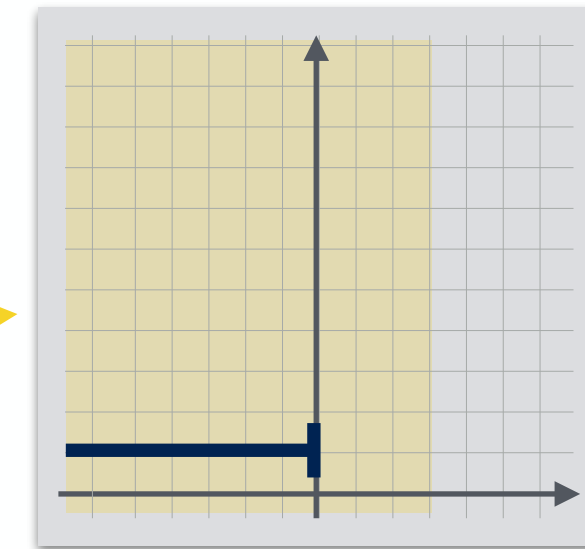
$$f \stackrel{\text{def}}{=} \lambda \rho . \begin{cases} 1 & \rho(x) \leq 0 \\ \text{undefined} & \text{otherwise} \end{cases}$$

$$\mathcal{R}_M[\llbracket s \rrbracket]f \stackrel{\text{def}}{=} \lambda \rho . \begin{cases} 3 & 0 \leq \rho(x) \\ \text{undefined} & \text{otherwise} \end{cases}$$

We have

$$\mathcal{R}_M[\llbracket \text{if } 3 - x < 0 \text{ then } s \rrbracket]f \stackrel{\text{def}}{=} \lambda \rho . \begin{cases} 2 & \rho(x) \leq 0 \\ 4 & 3 < \rho(x) \\ \text{undefined} & \text{otherwise} \end{cases}$$

$$\text{and } \mathcal{R}_M[\llbracket \text{if } [-\infty, +\infty] \neq 0 \text{ then } s \rrbracket]f \stackrel{\text{def}}{=} \lambda \rho . \begin{cases} 4 & \rho(x) = 0 \\ \text{undefined} & \text{otherwise} \end{cases}$$



Definite Termination Semantics

$\mathcal{R}_M[\text{while } e \bowtie 0 \text{ do } s \text{ done}]$

$\mathcal{R}_M[\text{while } e \bowtie 0 \text{ do } s \text{ done}]f \stackrel{\text{def}}{=} \text{lfp}_{\emptyset}^{\preceq} \bar{F}_M$

where $F_M(x) \stackrel{\text{def}}{=} \lambda \rho . \begin{cases} \textcircled{1} \\ \textcircled{2} \\ \textcircled{3} \\ \text{undefined} & \text{otherwise} \end{cases}$

$\textcircled{1} \quad \sup\{\mathcal{R}_M[s]x(\rho) + 1, f(\rho) + 1\} \quad \rho \in \text{dom}(\mathcal{R}_M[s]x) \cap \text{dom}(f) \wedge \exists v_1, v_2 \in E[e]\rho: v_1 \bowtie 0 \wedge v_2 \nbowtie 0$

$\textcircled{2} \quad \mathcal{R}_M[s]x(\rho) + 1 \quad \rho \in \text{dom}(\mathcal{R}_M[s]x) \wedge \forall v \in E[e]\rho: v \bowtie 0$

$\textcircled{3} \quad f(\rho) + 1 \quad \rho \in \text{dom}(f) \wedge \forall v \in E[e]\rho: v \nbowtie 0$

Definite Termination Semantics

$\mathcal{R}_M[[s_1; s_2]]$

$\mathcal{R}_M[[s_1; s_2]]f \stackrel{\text{def}}{=} \mathcal{R}_M[[s_1]](\mathcal{R}_M[[s_2]]f)$

Definite Termination Semantics

Denotational Formulation

Definition

The **definite termination semantics** $\mathcal{R}_M[[s^\ell]]: \mathcal{E} \rightarrow \mathbb{O}$ of a program s^ℓ is:

$$\mathcal{R}_M[[s^\ell]] \stackrel{\text{def}}{=} \mathcal{R}_M[[s]](\lambda\rho.0)$$

where $\mathcal{R}_M[[s]]: (\mathcal{E} \rightarrow \mathbb{O}) \rightarrow (\mathcal{E} \rightarrow \mathbb{O})$ is the definite termination semantics of each instruction s

Theorem

A program s^ℓ **must terminate** starting from a set of initial states I if and only if $I \subseteq \text{dom}(\mathcal{R}_M[[s^\ell]])$

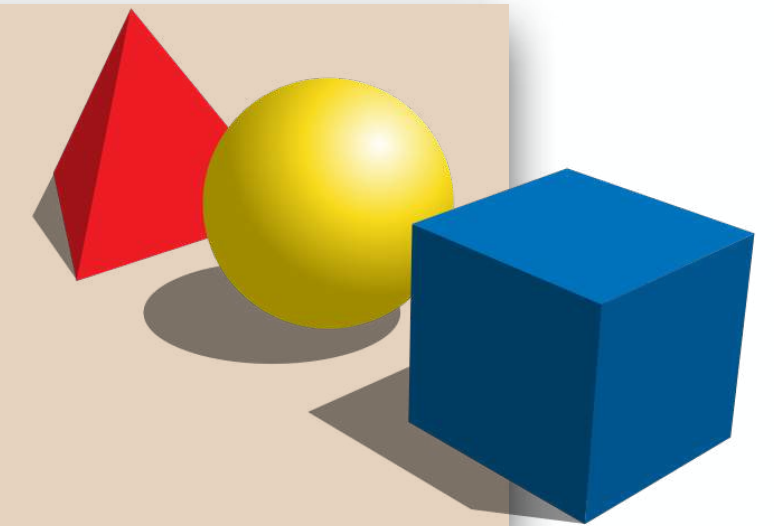
Termination Static Analysis

Abstract Program Termination Semantics

practical tools
targeting specific programs



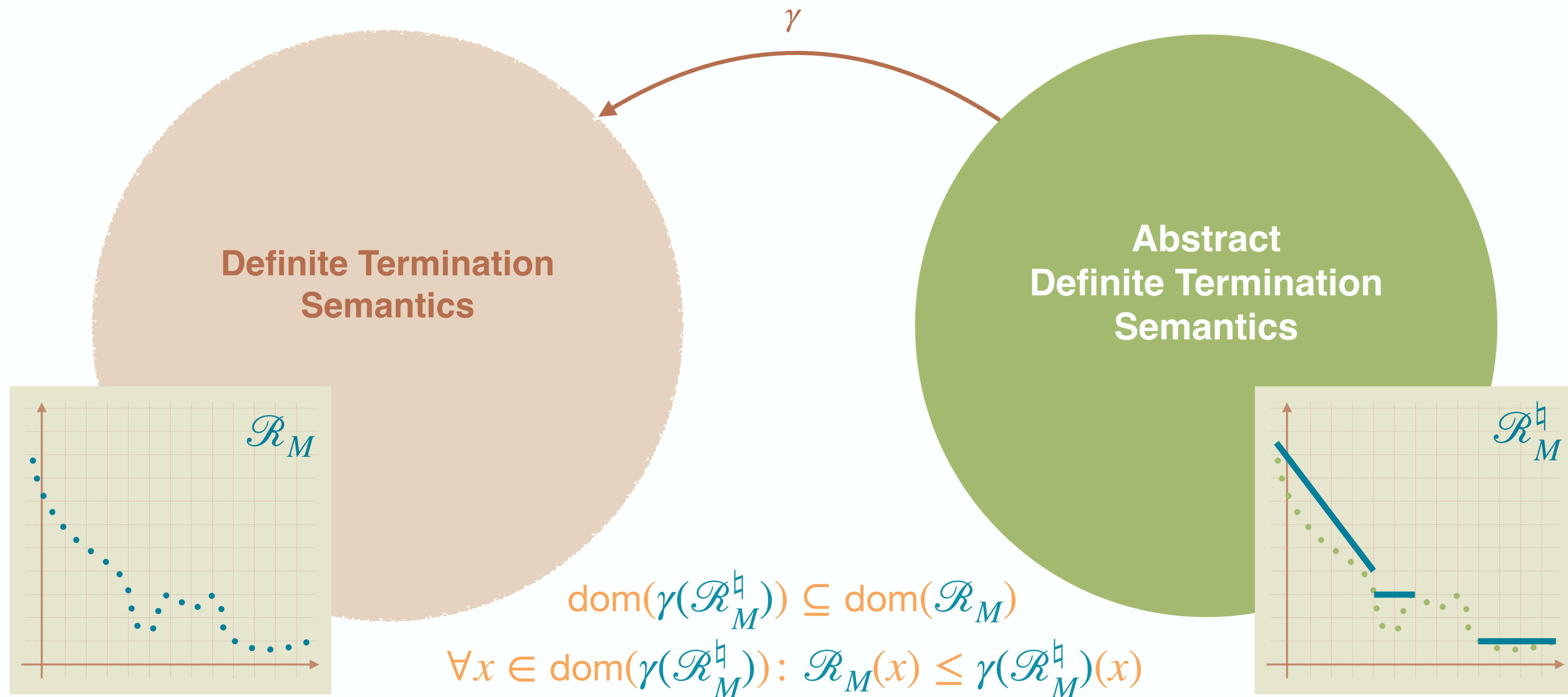
abstract semantics, abstract domains
algorithmic approaches to decide program properties



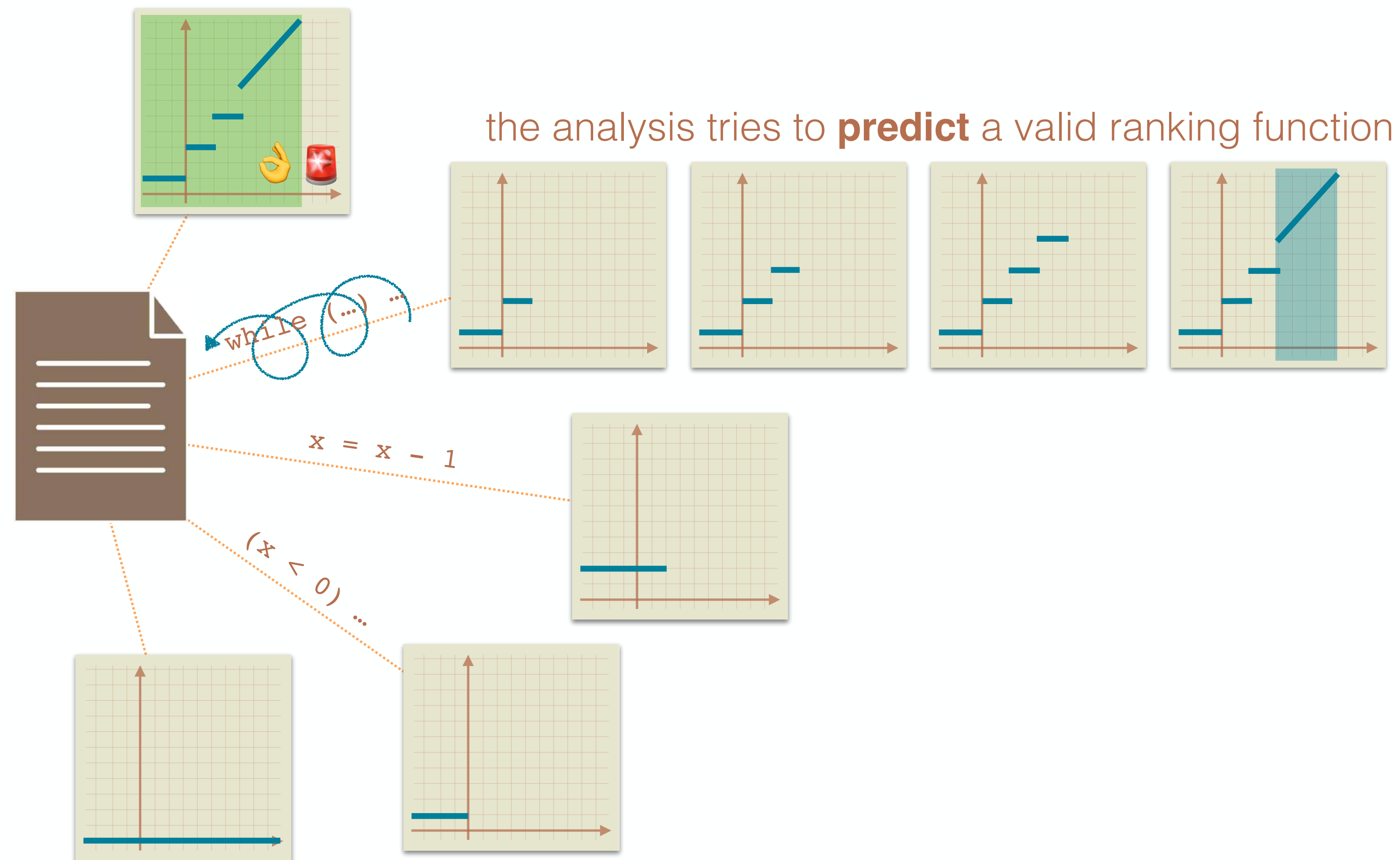
concrete semantics
mathematical models of the program behavior



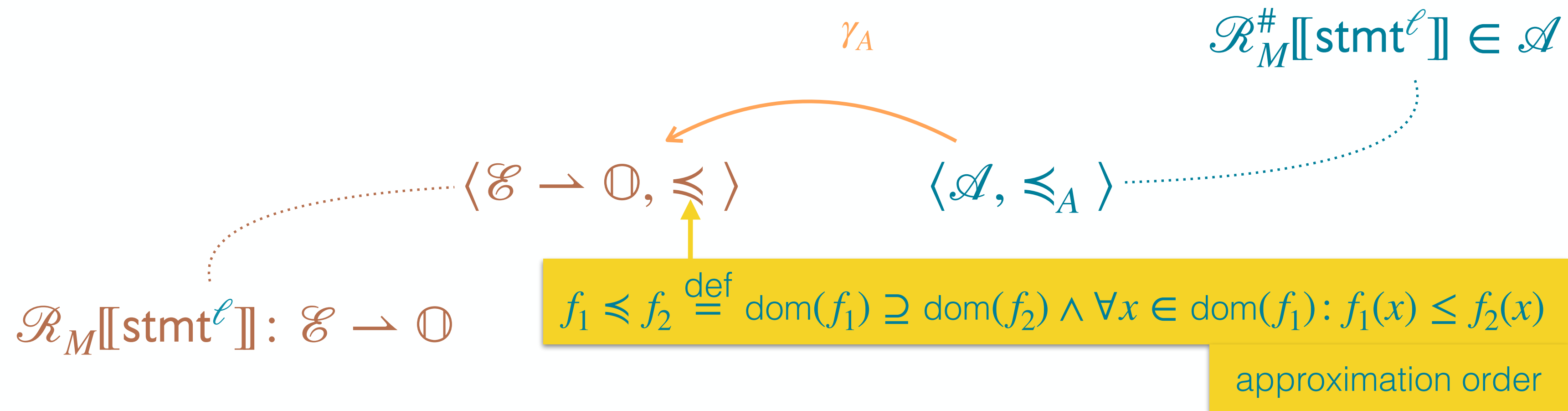
Piecewise-Defined Ranking Functions



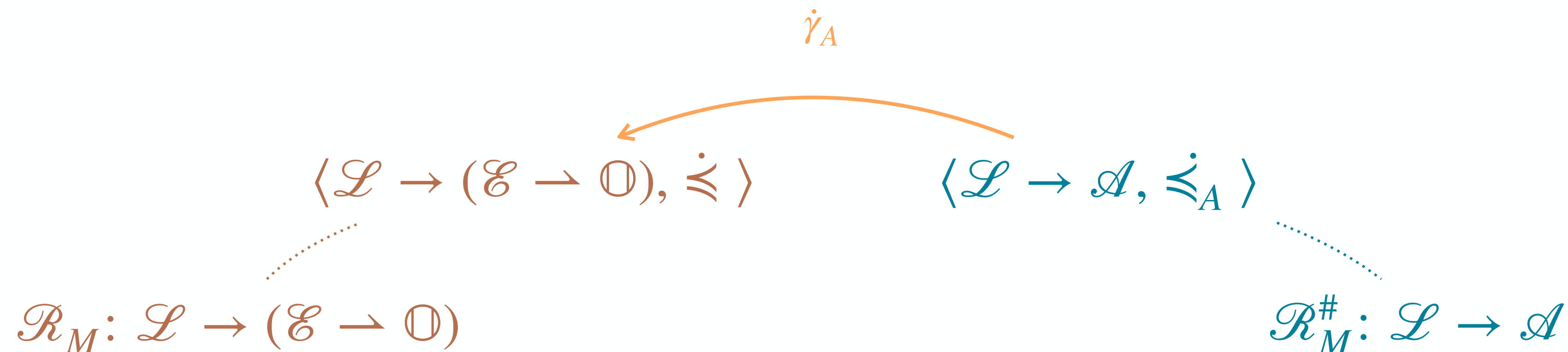
Termination Static Analysis



Piecewise-Defined Function Abstraction



By *pointwise lifiting* we obtain an abstraction $\mathcal{R}_M^\#$ of $\mathcal{R}_M$:

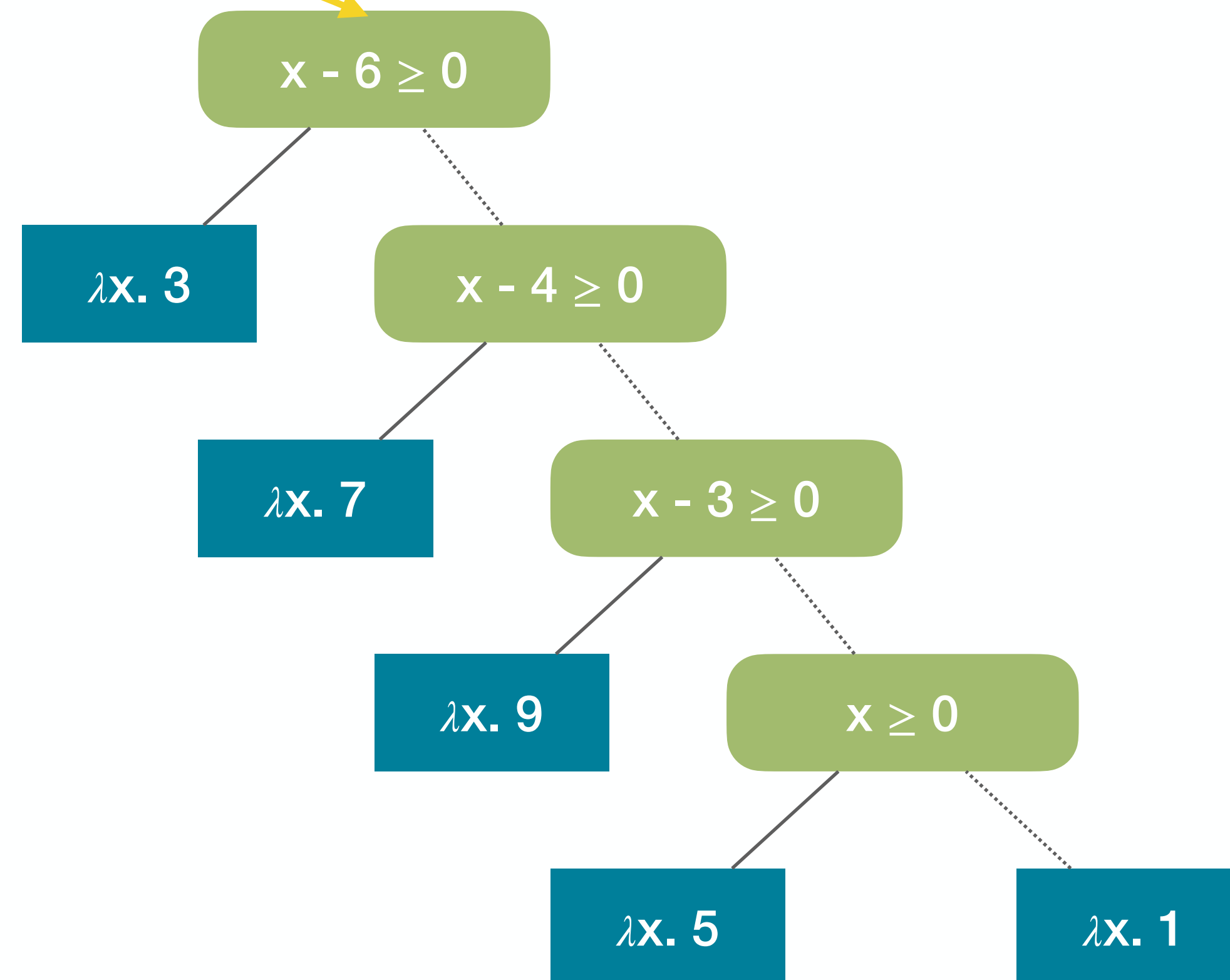
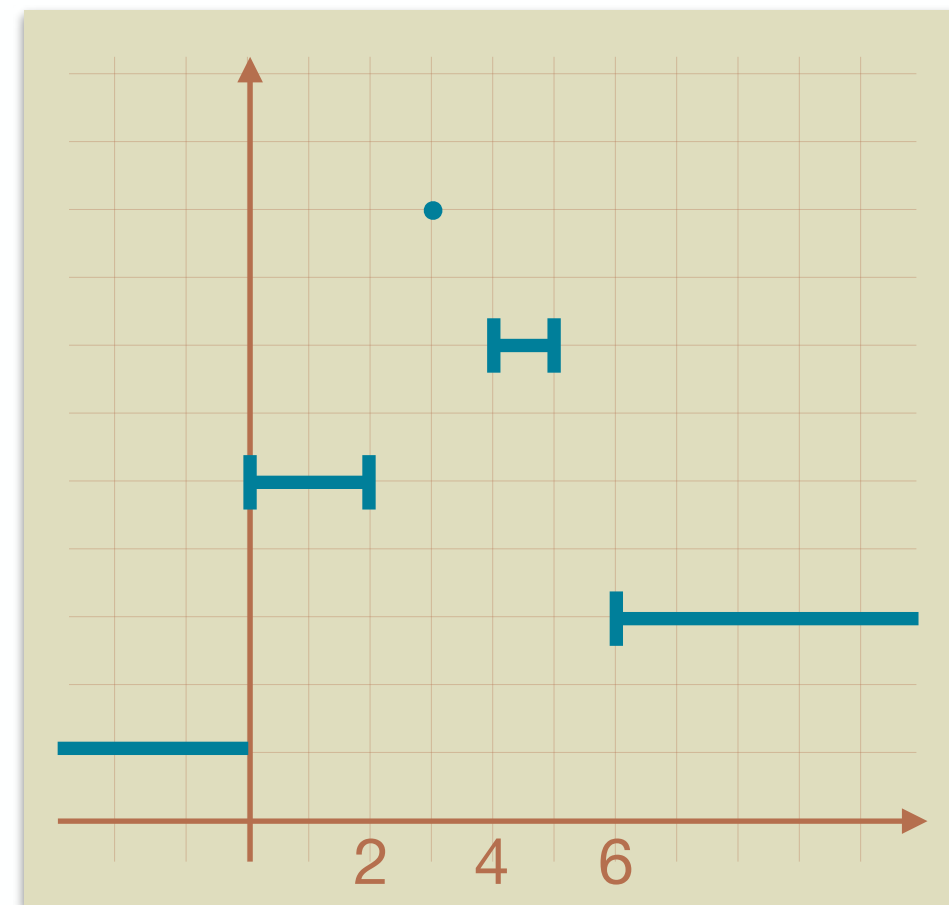


Piecewise-Defined Function Domain

$\langle \mathcal{A}, \preceq_A \rangle$

Example

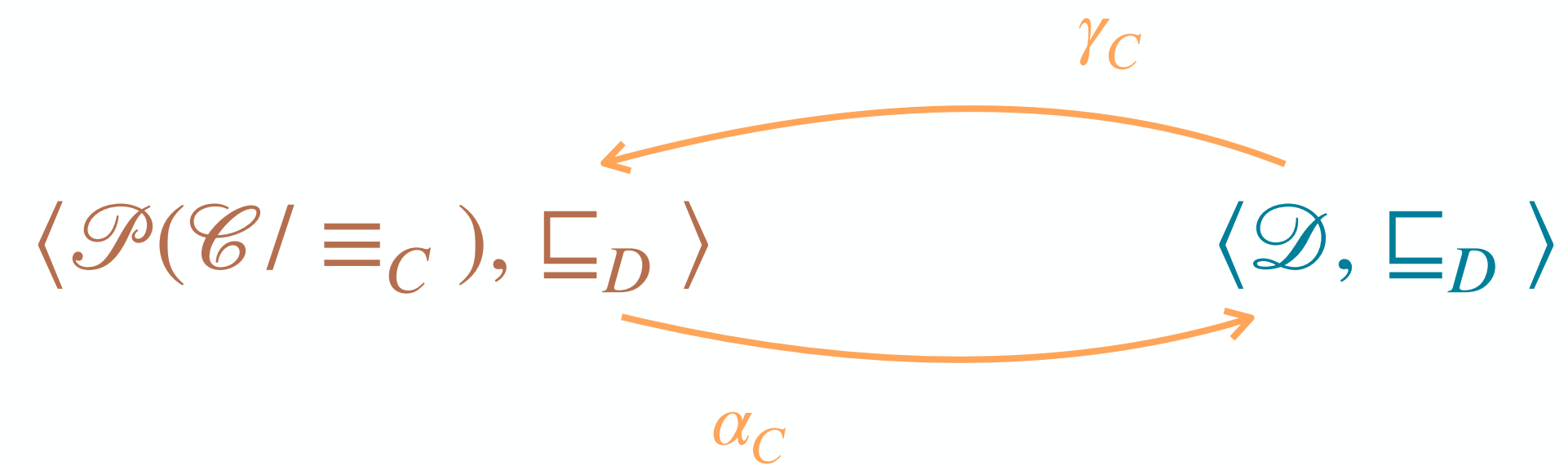
$^1x \leftarrow [-\infty, +\infty]$
while $^2(x \geq 0)$ **do**
 $^3x \leftarrow -2 \cdot x + 10$
done 4



Piecewise-Defined Function Domain

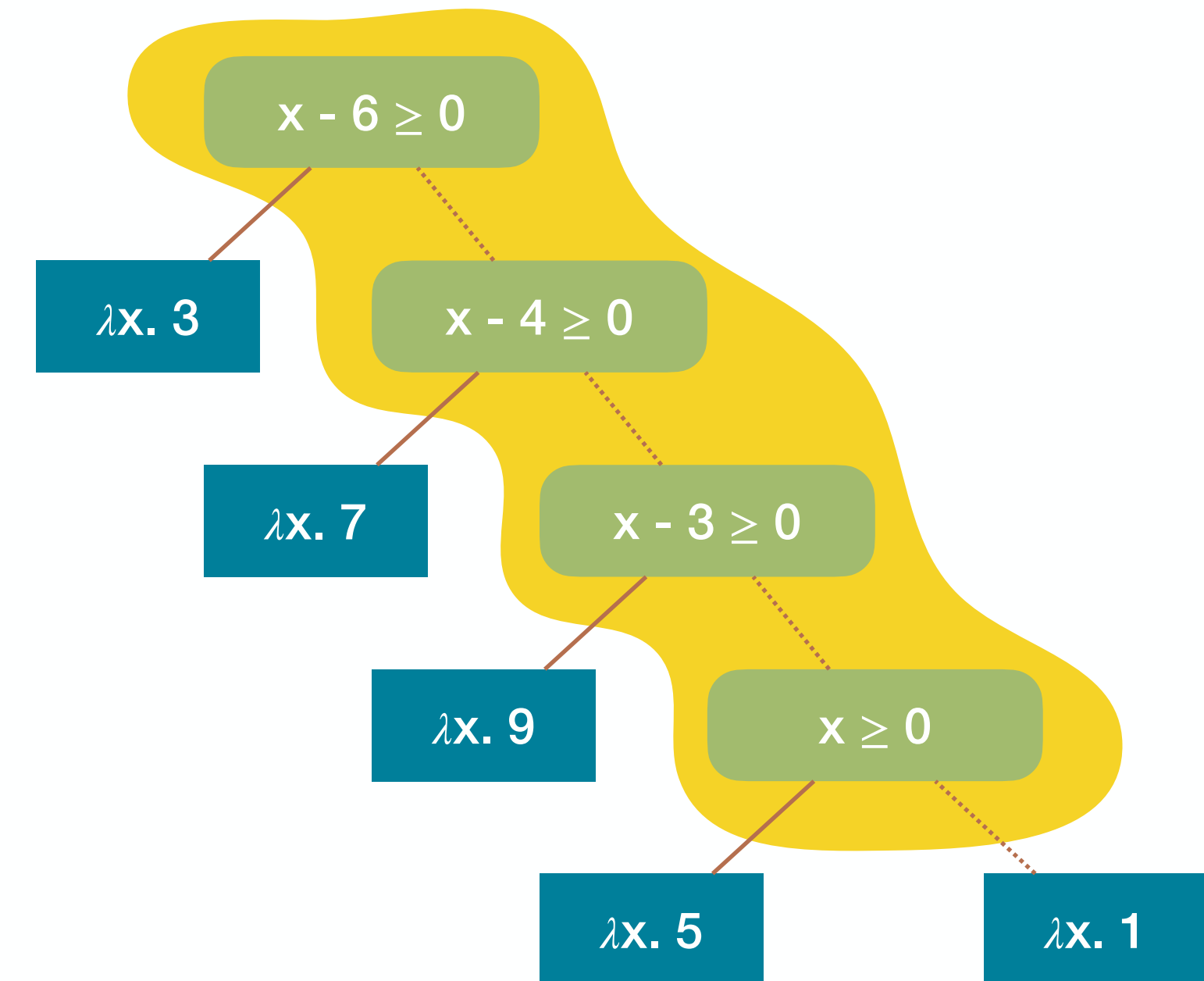
Linear Constraints Auxiliary Abstract Domain

- Parameterized by an *underlying numerical abstract domain* $\langle \mathcal{D}, \sqsubseteq_D \rangle$ (e.g., intervals, polyhedra):



Example:

$$X \rightarrow [-\infty, 3], Y \rightarrow [0, \infty] \xrightarrow{\gamma_C} \{3 - X \geq 0, Y \geq 0\}$$



- $\mathcal{C}$ is a set of linear constraints *in canonical form*, equipped with a total order $\leq_C$:

$$\mathcal{C} \stackrel{\text{def}}{=} \{c_1 \cdot X_1 + c_k \cdot X_k + c_{k+1} \geq 0 \mid X_1, \dots, X_k \in \mathbb{V} \wedge c_1, \dots, c_{k+1} \in \mathbb{Z} \wedge \gcd(|c_1|, \dots, |c_{k+1}|) = 1\}$$

Natural-Valued Ranking Functions

Piecewise-Defined Function Domain

Functions Auxiliary Abstract Domain

- Parameterized by an *underlying numerical abstract domain* $\langle \mathcal{D}, \sqsubseteq_D \rangle$

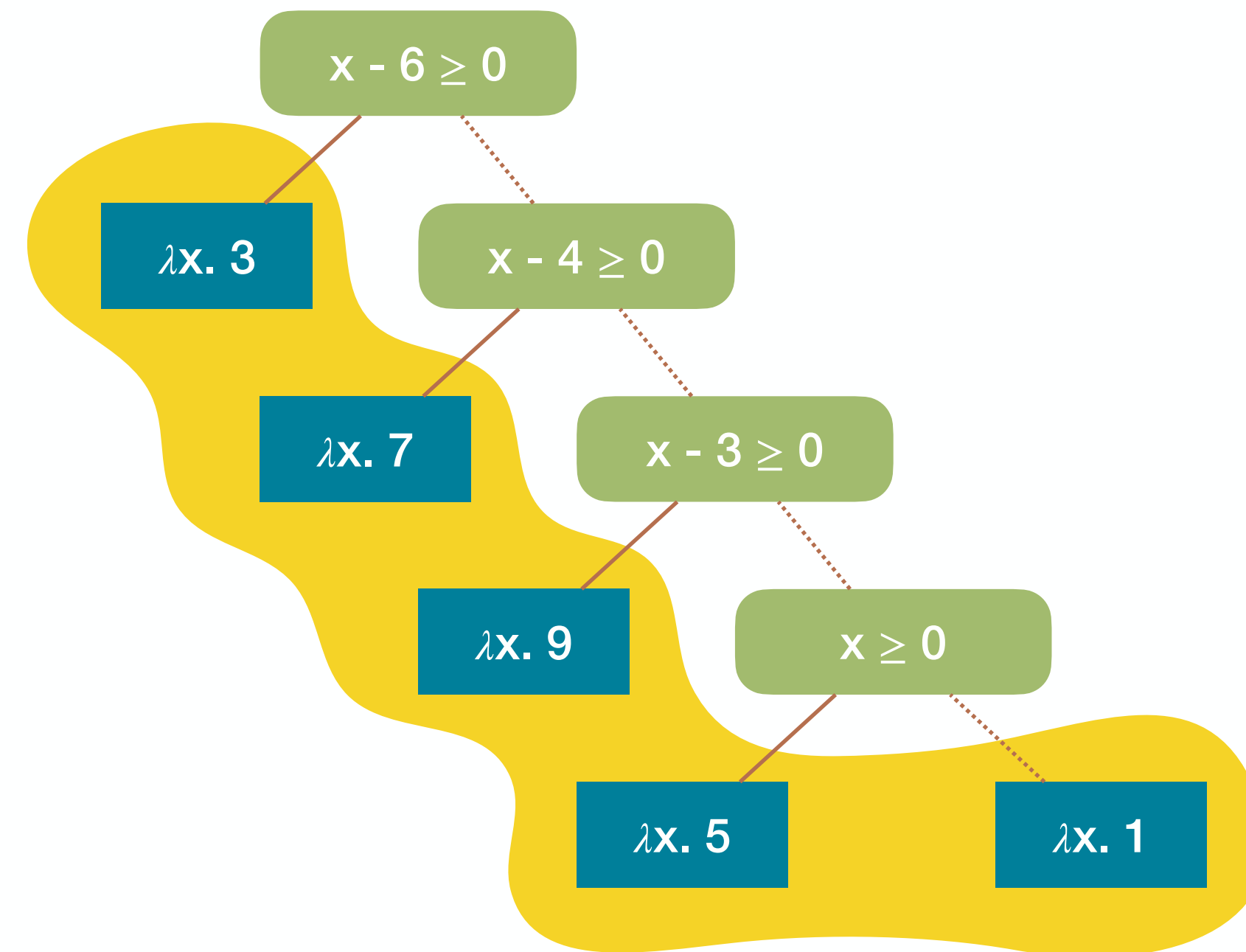
- $\mathcal{F} \stackrel{\text{def}}{=} \{ \perp_F \} \cup (\mathbb{Z}^{|V|} \rightarrow \mathbb{N}) \cup \{ \top_F \}$

We consider **affine functions**:

$$\mathcal{F}_A \stackrel{\text{def}}{=} \{ \perp_F \} \cup \{ f: \mathbb{Z}^{|V|} \rightarrow \mathbb{N} \mid$$

$$f(X_1, \dots, X_k) = \sum_{i=1}^k m_i \cdot X_i + q$$

$$\} \cup \{ \top_F \}$$



Piecewise-Defined Function Domain

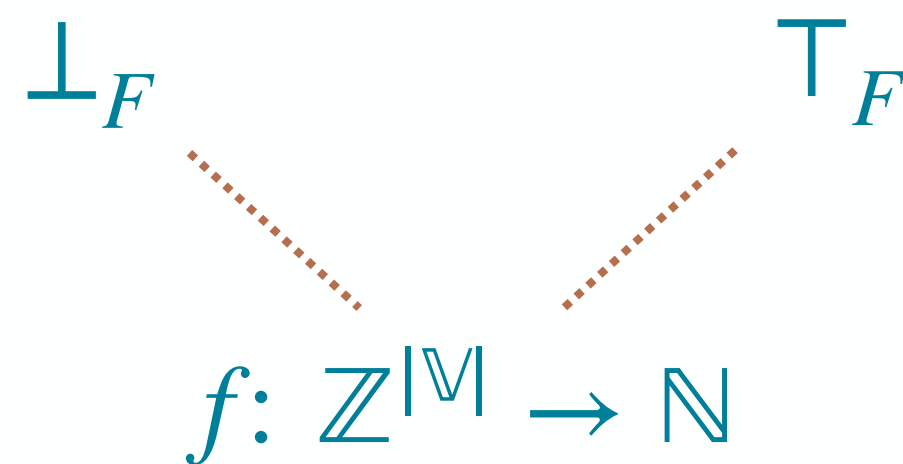
Functions Auxiliary Abstract Domain (continue)

- approximation order $\leqslant_F[D]$, where $D \in \mathcal{D}$:

- between defined leaf nodes:

$$f_1 \leqslant_F[D] f_2 \stackrel{\text{def}}{=} \forall \rho \in \gamma_D(D): f_1(\dots, \rho(X_i), \dots) \leq f_2(\dots, \rho(X_i), \dots)$$

- otherwise (i.e., when one or both leaf nodes are undefined):



Piecewise-Defined Function Domain

Functions Auxiliary Abstract Domain (continue)

- **computational order** $\sqsubseteq_F[D]$, where $D \in \mathcal{D}$:

- between defined leaf nodes:

$$f_1 \preceq_F[D] f_2 \stackrel{\text{def}}{=} \forall \rho \in \gamma_D(D): f_1(\dots, \rho(X_i), \dots) \leq f_2(\dots, \rho(X_i), \dots)$$

- otherwise (i.e., when one or both leaf nodes are undefined):

$$\begin{array}{c} \top_F \\ \vdots \\ f: \mathbb{Z}^{\mathbb{M}} \rightarrow \mathbb{N} \\ \vdots \\ \perp_F \end{array}$$

Piecewise-Defined Functions Domain

$$\mathcal{A} \stackrel{\text{def}}{=} \{\text{LEAF}: f \mid f \in \mathcal{F}\} \cup \{\text{NODE}\{c\}: t_1; t_2 \mid c \in \mathcal{C} \wedge t_1, t_2 \in \mathcal{A}\}$$

- **concretization function** $\gamma_A: \mathcal{A} \rightarrow (\mathcal{E} \rightarrow \mathbb{O})$:

$$\gamma_A(t) \stackrel{\text{def}}{=} \bar{\gamma}_A[\emptyset](t)$$

where $\bar{\gamma}_A: \mathcal{P}(\mathcal{C} / \equiv_C) \rightarrow \mathcal{A} \rightarrow (\mathcal{E} \rightarrow \mathbb{O})$:

$$\begin{aligned} \bar{\gamma}_A[C](\text{LEAF}: f) &\stackrel{\text{def}}{=} \gamma_F[\alpha_C(C)](f) \\ \bar{\gamma}_A[C](\text{NODE}\{c\}: t_1; t_2) &\stackrel{\text{def}}{=} \bar{\gamma}_A[C \cup \{c\}](t_1) \dot{\cup} \bar{\gamma}_A[C \cup \{\neg c\}](t_2) \end{aligned}$$

and $\gamma_F: \mathcal{D} \rightarrow \mathcal{F} \rightarrow (\mathcal{E} \rightarrow \mathbb{O})$:

$$\begin{aligned} \gamma_F[D](\perp_F) &\stackrel{\text{def}}{=} \emptyset \\ \gamma_F[D](f) &\stackrel{\text{def}}{=} \lambda \rho \in \gamma_D(D): f(\dots, \rho(X_i), \dots) \\ \gamma_F[D](\top_F) &\stackrel{\text{def}}{=} \emptyset \end{aligned}$$